

A Reflexive Root-Polytope Model for Preorder h -Polynomials

Simplicial-polytopal realization, palindromicity, and unimodality

With Part II (added 28 September 2026): height-two preorders — matching supports, nonreal-root families, and crown obstructions

With Part III (added 29 September 2026): cactus rigidity for matching-support determinants

With Part IV (added 29 September 2026): gamma-positivity for every finite preorder

Research manuscript prepared for Vladimir Reshetnikov
20 September 2026

Abstract

For a finite preorder τ on $[n]$, let $h_\tau(t)$ count the integer points of its preorder polytope by the number of nonzero coordinates. We give a proposed general proof that this polynomial is the h -polynomial of an n -dimensional simplicial polytope, establishing Conjecture 5.1(c), and hence (a) and (b), of Athanasiadis and Chapoton. The principal construction is the explicit n -dimensional lattice polytope

$$C_\tau = \text{conv}(\{\pm \mathbf{e}_i : 1 \leq i \leq n\} \cup \{\mathbf{e}_i - \mathbf{e}_j : j \leq_\tau i, i \neq j\}).$$

We prove $h^*(C_\tau, t) = h_\tau(t)$ by an integer-fiber decomposition of an augmented bipartite root polytope. Every boundary pulling triangulation of C_τ , coned from the origin, is unimodular; its boundary is combinatorially polytopal. We also prove reflexivity, an exact formula for the gauge of C_τ , and a Gorenstein translation formula for the larger root polytope. The higher-dimensional root-polytope identity and preorder duality are already results of Dai, Hou, Liu, Thawinrak, and Wang, and are not claimed as new here. A separate hypertree argument reconstructs that known identity. Exact verification code and complete small-case data accompany the manuscript.

Addition of 28 September 2026 (batch 39). Part II, from a later manuscript, proves that for a poset of height at most two with bipartite comparability graph G the support polynomial is $(1+t)^n p_G(t/(1+t)^2)$, where p_G counts perfectly matchable vertex sets; this gives γ -positivity (Conjecture 5.2) in height two. It records that real-rootedness (Conjecture 5.3) fails, crediting the first counterexample, an eight-element poset, to a posting by Shivam Patel, and proves an exact real-root count for an infinite family extending it. It shows that an induced crown on $2r \geq 6$ elements forces a minimal nonface of size r in every lattice triangulation of C_τ ; Conjecture 5.1(d) itself remains open.

Addition of 29 September 2026 (batch 42). Part III, from a further manuscript, proves that a bipartite graph carries a complex matrix on its edges whose square minors have squared modulus exactly the perfect-matchability indicator if and only if it is a cactus, and that this is strictly stronger than real stability of the signed matching-support polynomial; noncacti miss the exact model by the sharp universal defect $(\sqrt{17} - 3)/2$.

Addition of 29 September 2026 (batch 44). Part IV, from two further manuscripts, proves Conjecture 5.2 for every finite preorder: the k th γ -coefficient of h_τ counts ordered pairs of disjoint k -element sets joined by a τ -increasing bijection. It also shows that for a reflexive relation the doubled matching-support polynomial is palindromic exactly when the relation is transitive.

Scope and status. The intended new result is the general simplicial-polytopal realization, not the already established duality theorem. The proof uses explicitly identified results on support-enumerators and pulling triangulations. All steps specific to the construction are proved below; the external results are identified explicitly. The manuscript has not been independently refereed or checked by a proof assistant. The finite computations test the constructions; they do not replace the proof. Flag realizability, γ -positivity, and real-rootedness are not proved. [Added 28 September 2026, batch 39: this sentence describes Part I. Part II proves γ -positivity for posets of height at most two, records counterexamples to real-rootedness (the first credited to prior work), and obstructs flag lattice triangulations of C_τ ; see Section 11.3.] [Added 29 September 2026, batch 44: Part IV proves γ -positivity for every finite preorder; see Section 46.3.]

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Reading route. Sections 1–5 contain the main argument. Section 6 gives a second geometric check. Sections 8–10 provide examples, exact experiments, and limitations. Appendix A gives an alternative proof of the known higher-dimensional identity; Appendix B records the proof dependencies and the main normalization checks. Part II (Sections 11–25), added on 28 September 2026, treats height-two preorders; it precedes the appendices of Part I. Part III (Sections 26–45), added on 29 September 2026, answers Part II’s question on exact cactus-style matrices; it follows Part II. Part IV (Sections 46–65), added on 29 September 2026, proves γ -positivity for every finite preorder; it follows Part III.

1 The problem and the result

A *preorder* is a reflexive, transitive binary relation; antisymmetry is not required. For a preorder τ on $E = [n]$, write $\mathcal{J}(\tau)$ for its order ideals, including $\emptyset$ and E . For a vector x and a subset $A \subseteq E$, write $x(A) = \sum_{i \in A} x_i$. The preorder polytope and its support polynomial are

$$\mathcal{Q}_\tau = \{x \in \mathbb{R}_{\geq 0}^n : x(I) \leq |I| \text{ for every } I \in \mathcal{J}(\tau)\}, \quad (1.1)$$

$$P_\tau = \mathcal{Q}_\tau \cap \mathbb{Z}^n, \quad h_\tau(t) = \sum_{x \in P_\tau} t^{|\text{supp}(x)|} = \sum_{k=0}^n h_k(\tau) t^k. \quad (1.1)$$

Here $\text{supp}(x) = \{i : x_i \neq 0\}$. These are the objects of Athanasiadis and Chapoton [1]. Their Conjecture 5.1(c) asks whether $(h_0(\tau), \dots, h_n(\tau))$ is the h -vector of an n -dimensional simplicial polytope. Parts (a) and (b) ask, respectively, for palindromicity and palindromic unimodality. The stronger part (d) requires a flag simplicial polytope.

1.1 What was already known when this problem was selected

The arXiv version of [1] is dated 26 May 2026. A subsequent paper by Dai, Hou, Liu, Thawinrak, and Wang [2], in its 27 August 2026 revision, proves that the support-enumerator of a bipartite graph is the Ehrhart numerator of its augmented root polytope. Its Theorem 1.3 explicitly settles Conjecture 5.4 of [1], the identity $h_\tau = h_{\tau^*}$ under reversal of the preorder. These facts must therefore be treated as prior results, not as open problems. Problem 5.3 of [2] still discusses the coefficient-shape conjectures for preorders. Menon's August 2026 paper [6] gives further special cases of γ -positivity for arbors.

The target of the present manuscript is the *general* assertion in Conjecture 5.1(c). The advance proposed here is a dimension-reducing lattice fiber decomposition and the resulting explicit root-polytope realization. The literature check establishes these version-specific statements; it does not certify absolute priority against every unpublished or unindexed argument.

Theorem 1.1 (Main theorem). *Let τ be any preorder on $[n]$, where $n \geq 1$, and put*

$$C_\tau = \text{conv}(\{\pm \mathbf{e}_i : i \in [n]\} \cup \{\mathbf{e}_i - \mathbf{e}_j : j \leq_\tau i, i \neq j\}) \subset \mathbb{R}^n. \quad (1.2)$$

Then C_τ is reflexive and

$$h^*(C_\tau, t) = h_\tau(t). \quad (1.3)$$

Moreover, any pulling triangulation of ∂C_τ is combinatorially the boundary of an n -dimensional simplicial polytope S_τ satisfying

$$h(S_\tau, t) = h_\tau(t).$$

Consequently $h_k(\tau) = h_{n-k}(\tau)$ for all k , and

$$1 = h_0(\tau) \leq h_1(\tau) \leq \dots \leq h_{\lfloor n/2 \rfloor}(\tau).$$

In fact the full g -theorem conditions hold for $h(\tau)$.

A pulling triangulation is obtained by fixing an order on the vertices and recursively triangulating faces by cones from their first vertex. The actual root polytope C_τ need not be simplicial. The theorem asserts that its triangulated boundary has a simplicial convex realization with the desired h -vector. These two assertions should not be conflated.

The empty preorder is harmless: its support polynomial is 1, and both root models degenerate to a point. We treat $n \geq 1$ below to avoid unnecessary zero-dimensional boundary conventions.

1.2 A few useful sanity checks

Since E is an ideal, every $x \in P_\tau$ has $x(E) \leq n$. Hence P_τ is finite, $h_0 = 1$, and $h_n = 1$: the only point with all coordinates positive is $\mathbf{1}$. For a point supported just at i , the largest allowed value of its coordinate is $|\downarrow i|$, where $\downarrow i = \{j : j \leq_\tau i\}$. Therefore

$$h_1(\tau) = \sum_{i=1}^n |\downarrow i| = |\{(j, i) : j \leq_\tau i\}|. \quad (1.4)$$

Elements in the same equivalence class of the preorder remain separate coordinates throughout. Replacing a preorder by its quotient poset without retaining class sizes would change the problem.

2 The augmented bipartite root polytope

Let G_τ have left vertices $1_L, \dots, n_L$, right vertices $1_R, \dots, n_R$, and an edge $i_L j_R$ precisely when $j \leq_\tau i$. Adjoin 0_L and 0_R , with 0_L adjacent to every right vertex and 0_R adjacent to every left vertex. Denote the resulting graph by $\widehat{G}_\tau$. In particular $0_L 0_R$ is an edge. Its root polytope is

$$B_\tau = \text{conv}\{\mathbf{e}_i + \mathbf{f}_j : i_L j_R \in E(\widehat{G}_\tau)\} \subset \mathbb{R}^{n+1} \oplus \mathbb{R}^{n+1}. \quad (2.1)$$

The two coordinate blocks will be denoted (u, v) . We use the plus-sign convention for bipartite roots. Negating the second block converts this into the difference convention and preserves the lattice.

The affine span of B_τ is

$$H_1 = \{(u, v) : \sum_{i=1}^n u_i = \sum_{j=1}^n v_j = 1\}. \quad (2.2)$$

Indeed, the edges $i_L 0_R$ and $0_L j_R$ form a spanning tree, whose $2n + 1$ root vertices are affinely independent. Thus $\dim B_\tau = 2n$. At dilation m , the lattice is the full integer lattice in $H_m = \{\sum u_i = \sum v_j = m\}$, not a smaller lattice generated by selected coordinates.

2.1 The known support-enumerator theorem

For a bipartite graph with left neighborhoods $D_1, \dots, D_n$ in an n -element right shore, its draconian vectors are the nonnegative integer vectors x satisfying

$$x(A) \leq \left| \bigcup_{i \in A} D_i \right| \quad (A \subseteq [n]). \quad (2.3)$$

Theorem 1.2 of [2] identifies their support-enumerator with the h^* -polynomial of the graph augmented by the two universal vertices just used in (2.1).

Lemma 2.1. *For $D_i = \downarrow i$, the draconian vectors in (2.3) are exactly P_τ .*

Proof. If $x \in P_\tau$ and $A \subseteq E$, transitivity makes $\downarrow A$ an ideal. Nonnegativity gives

$$x(A) \leq x(\downarrow A) \leq |\downarrow A| = \left| \bigcup_{i \in A} D_i \right|.$$

Conversely, if (2.3) holds and I is an ideal, reflexivity and downward closure give $\bigcup_{i \in I} D_i = I$. Taking $A = I$ gives $x(I) \leq |I|$. $\square$

Proposition 2.2 (Known higher-dimensional identity). *For every preorder on $[n]$,*

$$\text{Ehr}_{B_\tau}(t) := \sum_{m \geq 0} |mB_\tau \cap \mathbb{Z}^{2n+2}| t^m = \frac{h_\tau(t)}{(1-t)^{2n+1}}. \quad (2.4)$$

Proof. Apply the support-enumerator theorem of [2] and Lemma 2.1. $\square$

Appendix A supplies another proof of this proposition using the Kálmán–Postnikov theorem [3]. Its elementary matching construction also explains why a support statistic, rather than a sum-of-coordinates statistic, occurs. Neither proof makes the already known proposition a novelty claim of this manuscript.

3 A complete inequality description

The special feature of a preorder is that all relevant neighborhood sets can be reduced to ideals. We need the reduction over the reals, not only for integer degree vectors.

Lemma 3.1 (Weighted Hall criterion). *Let G be a bipartite graph with shores L, R . Nonnegative margins u and v with common total m are the row and column sums of a nonnegative matrix supported on the edges of G if and only if*

$$u(A) \leq v(N(A)) \quad (A \subseteq L). \quad (3.1)$$

If the margins are integral, the matrix can be chosen integral.

Proof. Necessity follows by summing the entries in the rows of A . For sufficiency, make a network with source-to-left capacities u_i , right-to-sink capacities v_j , and capacity $m + 1$ on each allowed left-to-right edge. A cut of capacity at most m cannot cut an allowed left-to-right edge. If its source side contains the left set A , it must therefore contain $N(A)$ on the right; its capacity is at least $m - u(A) + v(N(A)) \geq m$. A flow of value m follows from the max-flow/min-cut argument. For completeness, that argument chooses a maximum feasible flow, which exists by compactness; if its residual graph contains a source-to-sink path, a positive augmentation contradicts maximality. Otherwise the vertices reachable from the source give a cut with capacity equal to the flow value. Integral capacities allow integral augmentations from the zero flow, so an integral maximum flow exists in the integral case. $\square$

Proposition 3.2 (Ideal inequalities for B_τ). *For every real $m \geq 0$, a pair $(u, v) \in H_m$ belongs to mB_τ exactly when all its coordinates are nonnegative and*

$$u(I) - v(I) \leq v_0 \quad (I \in \mathcal{J}(\tau)). \quad (3.2)$$

Here $u(I), v(I)$ involve only the coordinates indexed by $[n]$.

Proof. A point of mB_τ is precisely the pair of margins of a nonnegative matrix of total m supported on $\widehat{G}_\tau$. For a left subset containing 0_L , its neighborhood is the entire right shore, so (3.1) follows from nonnegativity and the total-sum condition. For $A \subseteq [n]$,

$$N(A) = \{0_R\} \cup (\downarrow A)_R.$$

Thus the remaining Hall conditions are $u(A) \leq v_0 + v(\downarrow A)$. If (3.2) holds, then

$$u(A) \leq u(\downarrow A) \leq v_0 + v(\downarrow A),$$

so all Hall conditions hold. Conversely, taking $A = I$ for an ideal gives $\downarrow I = I$ and recovers (3.2). The case $m = 0$ is immediate. $\square$

The empty-ideal inequality is just $v_0 \geq 0$, and the full-ideal inequality is equivalent to $u_0 \geq 0$. They may be redundant, but retaining them makes later formulas uniform. We do not assert that every inequality in this presentation defines a distinct facet.

4 The dimension-reducing lattice fibers

Define the projection

$$\pi(u, v) = (u_i - v_i)_{i=1}^n. \quad (4.1)$$

On the root vertices in (2.1), it has values 0, $\pm \mathbf{e}_i$, and $\mathbf{e}_i - \mathbf{e}_j$ for $j \leq_\tau i$. Therefore

$$\pi(B_\tau) = C_\tau. \quad (4.2)$$

Its kernel on the direction space of H_1 is spanned by $\mathbf{e}_i + \mathbf{f}_i - \mathbf{e}_0 - \mathbf{f}_0$, $1 \leq i \leq n$. The dimension loss is exactly n . A linear projection alone does *not* usually preserve an Ehrhart numerator. The following explicit fibers are what make this one work.

For $y \in \mathbb{R}^n$, put

$$\begin{aligned} y_i^+ &= \max(y_i, 0), & y_i^- &= \max(-y_i, 0), \\ p(y) &= \sum_i y_i^+, & q(y) &= \sum_i y_i^-, \end{aligned} \quad (4.3)$$

$$a_\tau(y) = \max_{I \in \mathcal{J}(\tau)} y(I), \quad g_\tau(y) = q(y) + a_\tau(y). \quad (4.4)$$

The empty and full ideals imply $a_\tau(y) \geq 0$ and $a_\tau(y) \geq p(y) - q(y)$.

Theorem 4.1 (Exact fibers and an integral gauge). *For $y \in \mathbb{R}^n$, define*

$$\begin{aligned} u^{\min}(y) &= (a_\tau(y) + q(y) - p(y), y_1^+, \dots, y_n^+), \\ v^{\min}(y) &= (a_\tau(y), y_1^-, \dots, y_n^-). \end{aligned} \quad (4.5)$$

Every point of mB_τ projecting to y has a unique expression

$$(u, v) = (u^{\min}(y), v^{\min}(y)) + \sum_{i=0}^n w_i(\mathbf{e}_i + \mathbf{f}_i), \quad (4.6)$$

where $w_i \geq 0$ and

$$\sum_{i=0}^n w_i = m - g_\tau(y). \quad (4.7)$$

Consequently,

$$y \in mC_\tau \iff g_\tau(y) \leq m. \quad (4.8)$$

For integral y, m , all terms in (4.5) are integral, and the lattice points in the fiber correspond exactly to nonnegative integer vectors w satisfying (4.7).

Proof. The two vectors in (4.5) are nonnegative. Their sums are both $a_\tau(y) + q(y) = g_\tau(y)$, and their difference on $[n]$ is y . Moreover,

$$u^{\min}(I) - v^{\min}(I) = y(I) \leq a_\tau(y) = v_0^{\min}.$$

Proposition 3.2 puts this pair in $g_\tau(y)B_\tau$. Adding a nonnegative multiple of a diagonal root leaves every difference $u(I) - v(I)$ unchanged. For the root indexed by 0, the right side v_0 also

increases, so all inequalities remain valid. This proves that every expression in (4.6) with (4.7) is an admissible point.

Conversely, suppose $(u, v) \in mB_\tau$ projects to y . For $1 \leq i \leq n$, nonnegativity and $u_i - v_i = y_i$ imply uniquely

$$u_i = y_i^+ + w_i, \quad v_i = y_i^- + w_i, \quad w_i = \min(u_i, v_i) \geq 0.$$

The ideal inequalities give $v_0 \geq \max_I y(I) = a_\tau(y)$. Set $w_0 = v_0 - a_\tau(y) \geq 0$. The equality of the two total sums then forces

$$u_0 = v_0 + q(y) - p(y) = u_0^{\min}(y) + w_0.$$

All coordinates in (4.6) are now determined. Summing either block yields (4.7). Projection proves (4.8). If y is integral, the maximum defining $a_\tau(y)$ is an integer, which establishes the final assertion without any lattice-saturation assumption. $\square$

Corollary 4.2 (Equality of Ehrhart numerators). *For every finite preorder,*

$$\text{Ehr}_{B_\tau}(t) = \frac{\text{Ehr}_{C_\tau}(t)}{(1-t)^n}, \quad h^*(B_\tau, t) = h^*(C_\tau, t) = h_\tau(t). \quad (4.9)$$

Proof. For integral y , the least admissible integral dilation is $g_\tau(y)$. Theorem 4.1 gives

$$\text{Ehr}_{B_\tau}(t) = \frac{\sum_{y \in \mathbb{Z}^n} t^{g_\tau(y)}}{(1-t)^{n+1}}, \quad \text{Ehr}_{C_\tau}(t) = \frac{\sum_{y \in \mathbb{Z}^n} t^{g_\tau(y)}}{1-t}.$$

These are legitimate formal power series: each coefficient receives only finitely many contributions because C_τ is bounded. Dividing the two identities proves the first equality in (4.9). The dimensions are $2n$ and n , respectively, so their Ehrhart numerators coincide. Finally apply Proposition 2.2. $\square$

There is a useful coefficient-level form. The fiber over an integral y at dilation m contains

$$\binom{m - g_\tau(y) + n}{n} \quad \text{if } g_\tau(y) \leq m, \quad 0 \quad \text{otherwise.} \quad (4.10)$$

Thus the factor $(1-t)^{-n}$ is not a heuristic cancellation of dimensions; it comes from a concrete, uniquely parametrized family of integer fibers.

4.1 An order-theoretic interpretation of the gauge

The gauge can also be written

$$g_\tau(y) = \max_{I \in \mathcal{J}(\tau)} \left(\sum_{i \in I} y_i^+ + \sum_{i \notin I} y_i^- \right). \quad (4.11)$$

Thus testing membership in a dilation of C_τ is a maximum-weight-ideal problem. One need not enumerate ideals in a practical implementation: put source capacity y_i^+ at i , sink capacity y_i^- at i , and sufficiently large capacities on arcs $i \rightarrow j$ for $j \leq_\tau i$. A finite minimum cut has a downward-closed source side I , and its capacity is $p(y) - y(I)$. Hence $a_\tau(y)$, and therefore $g_\tau(y)$, can be obtained by a network-flow calculation. The accompanying small-case verifier deliberately uses exhaustive ideal enumeration instead, keeping the check independent of a flow library.

5 Reflexivity and simplicial-polytopal realization

We now finish the proof of Theorem 1.1. The required geometry is particularly simple because all nonzero vertices of C_τ are type- A roots with one coordinate deleted.

5.1 The polar is integral

Since C_τ contains $\text{conv}\{\pm \mathbf{e}_i\}$, it is full-dimensional and contains 0 in its interior. Its polar is

$$C_\tau^\circ = \{z \in \mathbb{R}^n : -1 \leq z_i \leq 1, \quad z_i - z_j \leq 1 \text{ whenever } j \leq_\tau i\}. \quad (5.1)$$

Lemma 5.1 (Root determinants). *Put $\mathbf{e}_0 = 0 \in \mathbb{Z}^n$. Any n linearly independent vectors of the form $\mathbf{e}_i - \mathbf{e}_j$, with $i, j \in \{0, \dots, n\}$ and $i \neq j$, form a basis of $\mathbb{Z}^n$.*

Proof. Associate the vector $\mathbf{e}_i - \mathbf{e}_j$ to a directed edge from j to i on $\{0, \dots, n\}$. A cycle in the underlying undirected graph would give a linear dependence, with signs determined by the edge directions. Consequently n independent vectors give a forest with n edges on $n + 1$ vertices, hence a spanning tree. The matrix of these vectors is a directed incidence matrix with the row indexed by 0 deleted. Remove a leaf different from 0; its row has one nonzero entry, equal to 1 or -1 . Expanding the determinant along that row and repeating reduces the determinant to 1 or -1 . $\square$

Proposition 5.2. *The polar (5.1) is a lattice polytope. In fact all its vertices have coordinates in $\{-1, 0, 1\}$. Thus C_τ is reflexive.*

Proof. At a vertex of the full-dimensional polar, choose n linearly independent active constraints. With $z_0 = 0$, each selected equality is of the form $z_i - z_j = 1$. Its coefficient row is a root as in Lemma 5.1. The resulting square matrix has determinant ± 1 , so its solution is integral. The bounds $-1 \leq z_i \leq 1$ restrict these integers to $-1, 0, 1$. Both the original polytope and its polar are therefore integral, with the origin in the interior, which is precisely reflexivity. $\square$

This proof also supplies a finite exact way to test lattice-point membership: one may evaluate against all ternary vectors satisfying (5.1), since that set contains every vertex of the polar. The verifier uses this method as a check independent of the gauge formula.

5.2 Unimodular cones over the boundary

Recall that a lattice simplex with vertices $0, r_1, \dots, r_n$ is unimodular when $r_1, \dots, r_n$ form a lattice basis. A triangulation is unimodular if all its full-dimensional simplices are unimodular.

Proposition 5.3. *Let Δ be any triangulation of ∂C_τ using only vertices of C_τ . Then the star triangulation $0 * \Delta$ of C_τ is unimodular.*

Proof. Every boundary simplex is contained in a supporting hyperplane not containing 0. The n vertices of a maximal boundary simplex are therefore linearly independent: an affine dependence is already excluded by being a simplex, and a linear dependence with nonzero coefficient sum would place 0 in that supporting hyperplane. All its vertices are roots $\mathbf{e}_i - \mathbf{e}_j$, allowing index 0. Lemma 5.1 shows that the cone simplex from 0 is unimodular. Since 0 is interior, these cones form a triangulation of all of C_τ . $\square$

5.3 Why the triangulated boundary is polytopal

We use the following standard pulling fact, stated as Lemma 2.1 in Athanasiadis [4]: a pulling (reverse lexicographic) triangulation of the boundary of a convex n -polytope is combinatorially the boundary of a simplicial convex n -polytope. The geometric mechanism is successive small outward perturbations of the ordered vertices; at each step one chooses the perturbation small enough not to change the already fixed incidences. The resulting convex boundary realizes the prescribed pulling subdivision. This is an actual polytopality assertion, not merely shellability or the assertion that the complex is a sphere.

Here is the Ehrhart connection, with its normalization made explicit. If K is a d -dimensional unimodular triangulation of a lattice polytope P , then

$$\text{Ehr}_P(t) = \sum_{\sigma \in K} \left(\frac{t}{1-t} \right)^{|\sigma|} = \frac{h(K, t)}{(1-t)^{d+1}}, \quad (5.2)$$

where the empty face contributes 1. Indeed, the relative interior of an r -dimensional unimodular simplex contributes $\binom{m-1}{r}$ lattice points at positive dilation m , with generating series $t^{r+1}/(1-t)^{r+1}$. Relative interiors partition the triangulation, giving the first equality. The second equality is the definition of its h -polynomial. For a cone, the face generating function gains a factor $1 + t/(1-t)$, while the denominator exponent increases by one; hence

$$h(0 * \Delta, t) = h(\Delta, t). \quad (5.3)$$

Completion of the proof of Theorem 1.1. Choose a pulling triangulation Δ of ∂C_τ . By the pulling fact, Δ is the boundary of an n -dimensional simplicial polytope S_τ . Proposition 5.3 and (5.2)–(5.3) give

$$h(S_\tau, t) = h(\Delta, t) = h^*(C_\tau, t) = h_\tau(t),$$

where the last equality is Corollary 4.2. Reflexivity was proved in Proposition 5.2. The Dehn–Sommerville relations for simplicial polytopes give palindromicity, and the g -theorem gives the stated inequalities and its stronger numerical conditions; see also the polytopal conclusions summarized in [4, Theorem 3.5]. $\square$

6 A second geometric check in dimension $2n$

The larger polytope B_τ supplies two independent checks on the preceding argument: an interior-point translation and an explicit special simplex. They are useful safeguards against a sign error or a mismatch of dimensions.

6.1 An exact interior translation

Let $\mathbf{c} = (1, \dots, 1; 1, \dots, 1) \in \mathbb{Z}^{2n+2}$. For every $m \geq n+1$,

$$(\text{relint}(mB_\tau)) \cap \mathbb{Z}^{2n+2} = \mathbf{c} + ((m-n-1)B_\tau \cap \mathbb{Z}^{2n+2}). \quad (6.1)$$

To see this, note that every coordinate inequality is nonconstant on B_τ , and each ideal inequality has strict slack at $\mathbf{e}_0 + \mathbf{f}_0$. Consequently relative interior is characterized by positive coordinates and strict versions of (3.2). At integer points these strict inequalities are

$$u(I) - v(I) \leq v_0 - 1.$$

Subtracting 1 from every coordinate reduces each block sum by $n + 1$. The terms $|I|$ cancel on the left, and the right side becomes the new v_0 . This is exactly the weak inequality description of $(m - n - 1)B_\tau$. Conversely, adding $\mathbf{c}$ makes every defining inequality strict. There are no interior integer points for $1 \leq m < n + 1$, since $n + 1$ positive coordinates in either block already require total at least $n + 1$.

In particular, B_τ has codegree $n + 1$, with unique interior integer point $\mathbf{c}$ in its first nonempty interior dilation. It is Gorenstein of index $n + 1$. Ehrhart reciprocity translates (6.1) into

$$t^{2n+1}h^*(B_\tau, t^{-1}) = t^{n+1}h^*(B_\tau, t),$$

which again gives $h_\tau(t) = t^n h_\tau(t^{-1})$. This symmetry proof does not use the g -theorem or polytopality of a sphere.

6.2 The diagonal special simplex

A simplex with r vertices inside a polytope is *special* if each facet contains exactly $r - 1$ of those vertices. Consider

$$\Sigma_\tau = \text{conv}\{d_i : 0 \leq i \leq n\}, \quad d_i = \mathbf{e}_i + \mathbf{f}_i. \quad (6.2)$$

Reflexivity of the preorder makes these root vertices available, and they are affinely independent. A coordinate hyperplane $u_i = 0$ or $v_i = 0$ contains exactly the diagonal vertices other than d_i . On an ideal hyperplane $u(I) - v(I) = v_0$, every d_i with $i > 0$ lies on the hyperplane and d_0 does not. Every facet is furnished by at least one nonredundant inequality of Proposition 3.2; thus Σ_τ is special.

Every triangulation of a bipartite root polytope using its vertices is unimodular. Indeed, affine dependence of edge roots is equivalent to a cycle in the supporting bipartite graph; a full-dimensional simplex therefore corresponds to a spanning tree. Leaf elimination in its incidence matrix shows that its edge vectors form a basis of the integer lattice $\{(u, v) : \sum u_i = \sum v_j\}$. Intersecting with the height-one hyperplane shows affine unimodularity. In particular every pulling ordering is compressed in the terminology of [4].

Athanasiadis's special-simplex theorem [4, Theorem 3.5], applied with dimension $2n$ and $n + 1$ special vertices, therefore realizes $h^*(B_\tau, t)$ as the h -polynomial of a simplicial polytope of dimension $2n - (n + 1) + 1 = n$. This recovers the main theorem by another route. The broader Gorenstein/unimodular-triangulation framework of Bruns and Römer [5] provides a further compatible interpretation, but is not needed for our main proof.

7 Consequences and quantitative formulas

7.1 Duality, with its prior attribution

Swapping the coordinate blocks identifies B_τ with B_{τ^*} . This is the already established duality mechanism of [2]. In the smaller model it becomes the especially transparent equality

$$C_{\tau^*} = -C_\tau. \quad (7.1)$$

Thus $h_{\tau^*} = h_\tau$. The gauge gives a third check. Ideals of τ^* are complements of ideals of τ , so

$$a_{\tau^*}(-y) = a_\tau(y) - y(E), \quad g_{\tau^*}(-y) = g_\tau(y).$$

This rederivation is included to connect the new lower-dimensional model with the known theorem, not to reclassify Conjecture 5.4 as open.

7.2 Ehrhart polynomials and volumes

The ordinary Ehrhart polynomial of C_τ is now explicit in terms of the preorder support counts:

$$|mC_\tau \cap \mathbb{Z}^n| = \sum_{k=0}^n h_k(\tau) \binom{m+n-k}{n}, \quad m \in \mathbb{Z}_{\geq 0}. \quad (7.2)$$

Similarly,

$$|mB_\tau \cap \mathbb{Z}^{2n+2}| = \sum_{k=0}^n h_k(\tau) \binom{m+2n-k}{2n}. \quad (7.3)$$

A binomial term with nonnegative upper argument smaller than its lower argument is zero. These formulas also extend as polynomial identities in m . Normalized volume, meaning dimension factorial times volume in the relevant lattice, satisfies

$$\text{Vol}_{\mathbb{Z}}(C_\tau) = \text{Vol}_{\mathbb{Z}}(B_\tau) = h_\tau(1) = |P_\tau|. \quad (7.4)$$

The two volumes in this equality live in different dimensions and are both *normalized* volumes.

Remark 5.6 of [1] considers $h_\tau(t)/(1-t)^{n+1}$. Equation (7.2) identifies its coefficients as the Ehrhart polynomial of C_τ . This gives that series a concrete geometric interpretation. It does not prove that its polynomial has all complex roots on a particular vertical line; no such root-location assertion follows just from reflexivity.

7.3 Vertices, elementary bounds, and the g -vector

Every vector displayed in (1.2) is a vertex. To expose $\mathbf{e}_i$, choose a linear functional taking value 1 at coordinate i and a fixed $\varepsilon \in (0, 1)$ at all other coordinates. The negative case is analogous. To expose $\mathbf{e}_i - \mathbf{e}_j$, give coordinates i, j the values 1, -1 and all others the value 0; this root is the unique displayed vector with value 2. Thus

$$f_0(C_\tau) = n + |\{(j, i) : j \leq_\tau i\}| = n + h_1(\tau). \quad (7.5)$$

Pulling adds no vertices, so the same number occurs in S_τ .

The inclusions

$$[0, 1]^n \subseteq Q_\tau \subseteq \{x \geq 0 : x(E) \leq n\}$$

give the useful coefficientwise bounds

$$\binom{n}{k} \leq h_k(\tau) \leq \binom{n}{k}^2 \quad (0 \leq k \leq n). \quad (7.6)$$

For the upper bound, choose the support in $\binom{n}{k}$ ways; the positive integer values on it with total at most n can be chosen in $\binom{n}{k}$ ways by stars and bars. The lower bound uses the corresponding 0–1 points.

For completeness, the full numerical strengthening supplied by polytopality can be stated as follows. Put $g_0 = 1$ and $g_i = h_i - h_{i-1}$ for $1 \leq i \leq \lfloor n/2 \rfloor$. The g -theorem says this sequence is an M -sequence. In particular $g_i \geq 0$ and $g_{i+1} \leq g_i^{\langle i \rangle}$ whenever both sides are relevant. Here, if

$$a = \binom{b_i}{i} + \binom{b_{i-1}}{i-1} + \cdots + \binom{b_j}{j}, \quad b_i > b_{i-1} > \cdots > b_j \geq j \geq 1,$$

is the Macaulay expansion, then $a^{\langle i \rangle} = \binom{b_{i+1}}{i+1} + \cdots + \binom{b_{j+1}}{j+1}$, with $0^{\langle i \rangle} = 0$. These inequalities are stronger than unimodality and follow here from an actual simplicial-polytope realization.

8 Examples and the role of transitivity

8.1 Antichains and a single equivalence class

For an antichain, $\mathcal{Q}_\tau = [0, 1]^n$ and

$$h_\tau(t) = (1 + t)^n, \quad C_\tau = \text{conv}\{\pm \mathbf{e}_1, \dots, \pm \mathbf{e}_n\}.$$

The gauge is $\sum_i |y_i|$. The smaller model is already the simplicial cross polytope, so no boundary refinement is needed.

For the indiscrete preorder, in which every two elements are equivalent, $\mathcal{Q}_\tau = \{x \geq 0 : x(E) \leq n\}$. The same stars-and-bars argument used above shows

$$h_\tau(t) = \sum_{k=0}^n \binom{n}{k}^2 t^k.$$

Here $B_\tau = \Delta_n \times \Delta_n$, and the lower-dimensional polytope contains all roots $\pm \mathbf{e}_i$ and $\mathbf{e}_i - \mathbf{e}_j$. Its gauge is $\max(p(y), q(y))$, since its only ideals are $\emptyset, E$. These two extreme examples check both the coefficient bounds and the change of Ehrhart denominator.

8.2 A non-self-dual four-element example

Let τ have one minimum below three incomparable maximal elements. Its polynomial, obtained directly from (1.1), is

$$h_\tau(t) = 1 + 7t + 12t^2 + 7t^3 + t^4.$$

For example, for supports of size two not containing the minimum, the three pairs of maximal elements each allow the values $(1, 1), (1, 2), (2, 1)$, giving nine points; the three supports containing the minimum each allow only $(1, 1)$, giving three more. The dual order is not isomorphic to the original, but its directed root polytope is the negative of the original one and its support polynomial agrees. This is a useful test that duality is not merely invariance under relabeling.

8.3 A preorder with a nontrivial equivalence class

Use labels a, b, c, d, e , with $a \sim c, a < e, c < e, b < e$, and $b < d$. No other comparisons hold beyond those forced by reflexivity and transitivity. The principal ideals are

$$\{a, c\}, \quad \{b\}, \quad \{a, c\}, \quad \{b, d\}, \quad \{a, b, c, e\}.$$

Direct enumeration gives

$$h_\tau(t) = 1 + 11t + 34t^2 + 34t^3 + 11t^4 + t^5. \tag{8.1}$$

This agrees with the running example of [1]. There are 92 points in P_τ , 22 vertices of B_τ , and 16 vertices of C_τ . The number 92 is also the normalized volume of either root model.

For $x = (0, 0, 1, 2, 2)$, the matching construction in Appendix A can assign the five demands to the five right vertices as follows: the demand at c uses c_R , the two demands at d use b_R, d_R , and the two demands at e use a_R, e_R . Attach every left vertex to 0_R , then attach each right vertex to its assigned left vertex. This gives the spanning-tree certificate included in the JSON data.

8.4 Selected coefficient vectors

Preorder	n	$(h_0, \dots, h_n)$	$ P_\tau $
Antichain	4	(1, 4, 6, 4, 1)	16
Chain	4	(1, 10, 20, 10, 1)	42
Single equivalence class	4	(1, 16, 36, 16, 1)	70
One minimum, three maxima	4	(1, 7, 12, 7, 1)	28
Diamond	4	(1, 9, 18, 9, 1)	38
Example (8.1)	5	(1, 11, 34, 34, 11, 1)	92
Six-element crown	6	(1, 12, 48, 75, 48, 12, 1)	197

The crown has minimal elements a, b, c and maximal elements A, B, C , with $a, b < A, b, c < B$, and $c, a < C$. Every row of the table is recomputed by the supplied verifier; no floating-point root finder is involved. [Added 28 September 2026, batch 39: the crown is the case $r = 3$ of Part II's crown family. Its support polynomial has the closed form (20.2) and only simple negative zeros (Proposition 20.1), yet no lattice triangulation of its C_τ , pulling or not, is flag (Theorem 13.6).]

8.5 Why arbitrary reflexive relations are not enough

Transitivity cannot simply be dropped from the fiber theorem. Consider the neighborhoods

$$D_1 = \{1\}, \quad D_2 = \{1, 2\}, \quad D_3 = \{2, 3\}.$$

They define a reflexive but nontransitive relation. Its draconian inequalities reduce to

$$x_i \geq 0, \quad x_1 \leq 1, \quad x_1 + x_2 \leq 2, \quad x_3 \leq 2, \quad x_1 + x_2 + x_3 \leq 3.$$

The support counts are (1, 5, 6, 1): one-coordinate supports contribute $1 + 2 + 2 = 5$, two-coordinate supports contribute $1 + 2 + 3 = 6$, and the full support contributes just (1, 1, 1). This is the nonpalindromic example also recorded in [1, Remark 5.5].

The general support-enumerator/root-polytope theorem remains valid for this graph. What fails is the reduction of all weighted Hall inequalities to matching ideal differences $u(I) - v(I)$. In particular, subtracting a common coordinate from the two margins can damage an inequality involving different row and column subsets. Thus the argument does not inadvertently imply palindromicity for all graph support-enumerators. [Added 29 September 2026, batch 44: Part IV explains this example exactly. For every reflexive relation R the coefficients of degrees 1 and $n - 1$ are $|R|$ and the size of its transitive closure, and palindromicity holds if and only if R is transitive (Theorem 48.3); here $(1, 5, 6, 1) = (1, |R|, |\bar{R}|, 1)$ with one missing transitive pair (Example 53.3).]

9 Exact computational verification

The program code/verify.py uses only Python's standard library and integer arithmetic. It stores a preorder by bitmasks

$$D_i = \{j : j \leq_\tau i\}.$$

It enumerates every reflexive relation on $[n]$ and retains precisely those for which $j \in D_i$ implies $D_j \subseteq D_i$. This is a direct test of transitivity. All weak compositions with total at most n are then checked against the ideal inequalities to obtain h_τ .

Ground-set size	Labeled preorders checked	Result
0	1	Pass
1	1	Pass
2	4	Pass
3	29	Pass
4	355	Pass
5	6,942	Pass
Total	7,332	Pass

For each of these preorders, the code checks duality, palindromicity, unimodality, the endpoint coefficients, the relation-count formula for h_1 , and the special-simplex slack identities. These are finite consistency checks, not the reason the theorem holds for arbitrary n .

Stronger, independent comparisons are performed for all 35 preorders on at most three elements. The program enumerates spanning trees directly, deduplicates their left degree-minus-one vectors, and compares them with $\{(n - |x|, x) : x \in P_\tau\}$. It computes internal inactivity by testing all allowed transfers to smaller coordinates, rather than replacing inactivity by support size in the code. Across these instances it enumerates 18,009 spanning trees and checks 399 explicit matching-tree certificates.

For the same cases it counts sums of edge-root vertices directly at every dilation $0 \leq m \leq 2n$. Those counts are compared both with Hall-margin enumeration and with (7.3). Finite differences recover all $2n + 1$ coefficients of the root-polytope Ehrhart numerator, including the predicted zero coefficients above degree n . Independently, integer points of C_τ are counted using the ternary polar description, compared with the gauge formula, and checked through $m = n$ against (7.2). Full-margin fiber identities and the interior translation are also checked exactly.

Five selected four-element preorders undergo full root Ehrhart checks through $m = 8$, quotient checks through $m = 4$, and direct edge-sum checks through $m = 3$. Further examples include the weighted five-element preorder and the six-element crown. The JSON file records all individual results, and the CSV file contains every preorder and support vector through size five.

The original conjecture paper reports tests through larger sizes for some coefficient properties [1]. The present computations are not offered as an improved size record. Their purpose is to compare independent realizations of the newly proposed construction and detect normalization, orientation, or lattice-index errors.

10 What remains unresolved and how to audit the claim

The proof gives an ordinary simplicial polytope, not a flag one. A flag complex is determined by its edges: every collection of pairwise adjacent vertices must form a face. Neither a pulling triangulation nor a general simplicial polytope automatically has that property. Therefore Conjecture 5.1(d) is not settled by the argument. [Added 28 September 2026, batch 39: Part II shows more. If τ contains an induced crown on $2r \geq 6$ elements, every lattice triangulation of C_τ or of ∂C_τ has a minimal nonface of size r (Theorem 13.6); so the polytope S_τ of Theorem 1.1 is then never flag. Conjecture 5.1(d), which asks for *some* flag polytope, remains open; see Section 11.3.]

Likewise, palindromic unimodality and the g -theorem do not establish γ -positivity or

real-rootedness. The manuscript proves neither the nonnegativity of all coefficients in

$$h_\tau(t) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_j t^j (1+t)^{n-2j}$$

nor the assertion that every zero of h_τ is real. Accordingly it does not claim to settle Conjectures 5.2 or 5.3 of [1]. [Added 28 September 2026, batch 39: the two preceding sentences describe Part I at its state of 20 September 2026. Conjecture 5.3 is false: the first counterexample known here is an eight-element height-two poset posted by Shivam Patel [11], which Part II checks and extends to the family of Theorem 13.3. Conjecture 5.2 holds for posets of height at most two, with γ_k the number of perfectly matchable vertex sets of size $2k$ in the comparability graph (Theorem 13.2), and is open in general. See Section 11.3.] [Added 29 September 2026, batch 44: Conjecture 5.2 is no longer open. Part IV proves it for every finite preorder, with γ_k the number of ordered pairs of disjoint k -element sets joined by a τ -increasing bijection (Theorem 48.2); see Section 46.3. Its proof does not use Part I's geometric argument. Conjecture 5.3 stays false.]

For independent mathematical review, the shortest audit is to verify four links: the identification of the draconian vectors with P_τ ; Proposition 3.2; the unique integer fibers in Theorem 4.1; and the determinant argument for cones over a polytopal pulling boundary. Once these are accepted, the cited support-enumerator theorem and the pulling lemma give the asserted simplicial-polytopal realization. Section 6 is a separate geometric cross-check, not a circular use of the desired symmetry.

The result should be understood as a proposed research proof with reproducible support, not as a claim of community acceptance. The distinction between the known August 2026 result and the additional construction developed here is part of the mathematical claim, not a bibliographic afterthought.

Part II. Height-two preorders: matching supports, nonreal roots, crown obstructions

Sections 1–10 and the appendices form the original report (Part I) of 20 September 2026, unchanged except for dated pointers to this addition. Part II was added on 28 September 2026 in batch 39 of ProveIt’s incoming-report intake, from one manuscript written against Part I as printed above. It addresses the three questions Part I left open in Section 10: Conjectures 5.3, 5.2 and 5.1(d) of Athanasiadis and Chapoton [1]. Its headline counterexample to Conjecture 5.3 is **not new**: it was posted earlier by Shivam Patel on MathDB [11], and the manuscript says so. Section 11 records the provenance, fixes the notation and states what is answered.

11 Provenance, conventions and what Part II answers

11.1 One manuscript, one addition

Part II is batch-39 manuscript 02, *Height-Two Preorder Polynomials: Matching supports, nonreal-root families, and crown obstructions*. With a spectral treatment of bipartite cactus graphs, a research manuscript developed with ChatGPT for Vladimir Reshetnikov and dated 28 September 2026 (26 pages as delivered). It arrived as the archive `ProveIt_Height_Two_Preorder_Research.zip` (inner directory `Preorder_Matching_Research/`) in ProveIt commit `74f7f5bdb` and was placed in `e2b1f016a`. It names no ProveIt commit; it records the Git blob `4c350f5652096b7db91d4a10075eb7905910f647` of Part I’s `article.tex` as the file it consulted. That blob is Part I exactly as printed above (it is unchanged at the placement commit), so every statement the manuscript makes about “the ProveIt report” refers to Part I. The manuscript’s code and data are shipped beside Part I’s files with the prefix `02-height-two-` (Part I’s unprefix files count as 01), and its status and source notes as `02-height-two-STATUS.md` and `02-height-two-sources.md`. Its manuscript, delivery README and PDF are not shipped and survive in the arrival commit; its checksum list was verified at placement and retired.

The manuscript is printed here as Sections 12 to 25: its Section k is Section $k + 11$ here, and its theorem and equation numbers shift accordingly (its Theorem 2.2 is Theorem 13.2); its two figures and ten research questions keep their numbers. Every result, proof, example, remark, limitation, question and non-claim of the manuscript is printed. For the reader’s convenience, its abstract is reproduced here as delivered.

Abstract of manuscript 02 (as delivered). For a finite poset P , let $h_P(t)$ enumerate the integer points of its preorder polytope by support size. If P has height at most two and G is its bipartite comparability graph, we prove

$$h_P(t) = (1+t)^{|P|} p_G\left(\frac{t}{(1+t)^2}\right), \quad p_G(u) = \sum_{\substack{W \subseteq V(G) \\ G[W] \text{ has a perfect matching}}} u^{|W|/2}.$$

Thus the γ -coefficients count distinct matchable vertex sets, not matchings, proving the height-two case of Athanasiadis–Chapoton Conjecture 5.2. For the graph Θ_m formed by m internally disjoint three-edge paths with common endpoints, we obtain an explicit polynomial and a complete real-root count. For $m \geq 3$, the associated height-two support polynomial has exactly two real roots when m is even and four when m is odd. Consequently its proportion of nonreal roots tends to one, even though the graph has treewidth two. The eight-element case was previously posted by Shivam Patel; we verify it and extend it rather than claiming its discovery. A separate 17-element construction transfers the classical Stembridge–Ohsugi–Tsuchiya example.

An orthant argument proves the height-two instance of the repository’s root-polytope identity. For bipartite cactus graphs, a sixth-root-of-unity matrix gives a determinantal formula and a multivariate stability refinement, with univariate real-rootedness credited to prior work. Induced crowns force arbitrarily large minimal nonfaces in every lattice triangulation of the canonical root polytope and indispensable toric relations of arbitrarily large degree. Exact Python and C++ certificates accompany the proofs.

Status and scope (manuscript 02). The proposed contributions are the general height-two connection, the complete root-distribution theorem for Θ_m , and the geometric and spectral refinements. The first counterexample to Conjecture 5.3 is *not* claimed as new: a smaller posted example was found during the source audit. Two classical root-polytope results are imported explicitly. The manuscript has not been independently refereed or checked in Lean. General flag realizability and γ -positivity beyond height two remain unsettled here; absolute publication priority is not certified. [Added 29 September 2026, batch 44: γ -positivity beyond height two is proved in Part IV (Theorem 48.2).]

Priority of the counterexample. The first known counterexample to Conjecture 5.3 of [1] printed in this report is the eight-element height-two poset P_{Θ_3} of Section 16. It is due to Shivam Patel, whose posted solution on MathDB [11] specifies the graph Θ_3 and both polynomials. The manuscript found that posting during its own source audit, after its 17-element example had been developed, and revised its claims accordingly; it reproduces and independently checks Patel’s example and claims only the all- m extension. The exact calendar date of the posting was not established: the page showed a relative date (about one month before 28 September 2026) and labelled the claim unverified. A withdrawn SSRN listing (abstract 7385158) for an eight-element counterexample was also found; its full text was unavailable, and it is not used as evidence for any mathematical or priority claim. The 17-element counterexample of Section 17 is a transfer of the classical Stembridge–Ohsugi–Tsuchiya graph polynomial [10, 8], neither a first disproof nor a minimum-size example. Univariate real-rootedness for bipartite cacti is Ohsugi–Tsuchiya’s [9, Theorem 7.2], and the even-cycle (crown) formula is Davis–Kohl’s [7]. The source audit behind these statements does not prove the absence of other publications, preprints or postings containing any of Part II’s proposed extensions.

Shortest audit (manuscript 02). Read the counting lemma in Section 14, the Boolean summation in Section 15, and the explicit graph and exact certificate in Section 17. Together with Section 16, these establish the root-distribution results without assuming any unreferenced theorem in ProveIt (in particular, without Part I's general theorem). Sections 18–21 develop its geometric and spectral context. The companion programs provide independent finite checks rather than substitutes for the infinite-family proofs.

11.2 Notation of Part II

Part II works with *posets* of height at most two, which are preorders; for $\tau = P$ its $\mathcal{J}(P)$, $Q(P)$, h_P and C_P are exactly Part I's $\mathcal{J}(\tau)$, Q_τ , h_τ and C_τ (compare (13.1) and (13.2) with the two displays opening Section 1, and Theorem 13.6 with (1.2)). Its γ -coefficients are taken in the basis $t^k(1+t)^{n-2k}$ of Section 10, and $\text{Vol}_\mathbb{Z}$ is the normalized volume of (7.4). Several letters mean different things in the two parts; Table 1 fixes them. No symbol of the manuscript was renamed and no normalization was changed.

Remark 11.1 (Merge note: Part I's input and Part II's input agree). This remark is the merge's own and is not in manuscript 02. Part I's augmented polytope B_τ of (2.1) is the edge polytope $B_{\widehat{H}}$ of Section 14 for $H = G_\tau$, with the same lattice. Proposition 2.2 and the imported identity (14.4) therefore give, for every finite preorder,

$$h_\tau(t) = h^*(B_\tau; t) = p_{G_\tau}(t) :$$

the support polynomial is itself the perfectly-matchable-set polynomial of the doubled graph G_τ . This is a consequence of two cited theorems, not a new result, and it is a different statement from Theorem 13.2, in which p_G of the n -vertex comparability graph is the γ -polynomial of h_{P_G} . As a normalization check, both sides were compared by direct enumeration (ideal inequalities against induced perfect matchings of G_τ) for all 390 labeled preorders with $n \leq 4$; they agree. That check is a scratch computation of the write phase and is not shipped.

11.3 What Part II answers in Section 10

Section 10 of Part I leaves three statements of [1] unsettled. Their status after Part II, as of 28 September 2026:

- **Conjecture 5.3 (real-rootedness of h_τ) is false.** The first counterexample known to this report is Patel's eight-element poset P_{Θ_3} [11], *not* a discovery of Part II. Part II proves that $h_{P_{\Theta_m}}$ has exactly $2m$ (m even) or $2m - 2$ (m odd) nonreal zeros for every $m \geq 3$ (Theorem 13.3, Corollary 16.2), and transfers the classical 17-element example (Theorem 13.4). Part I never asserted real-rootedness, so nothing in it is retracted.
- **Conjecture 5.2 (γ -positivity) holds for posets of height at most two** (Theorem 13.2: $\gamma_k = |\mathcal{M}_k(G)|$). It remains open for general preorders, including height-two posets whose elements are blown up into nontrivial equivalence classes (Research question 1). [Added 29 September 2026, batch 44: it holds in height two and, by Part IV, in general: Theorem 48.2 proves it for every finite preorder, blowups included (Example 54.2), and answers Research question 1; see Section 46.3.]
- **Conjecture 5.1(d) (a flag simplicial polytope) remains open.** Part II shows that no lattice triangulation of C_τ or of its boundary is flag when τ contains an induced crown on $2r \geq 6$ elements (Theorem 13.6). A pulling triangulation of ∂C_τ uses only vertices of C_τ , so for such τ the simplicial polytope S_τ of Theorem 1.1 is never flag; the smallest case is Part I's

Table 1: Symbols of Part II. The last column prints the tempting false reading, usually a Part I symbol with the same letter.

Symbol	Meaning in Part II	Not to be confused with
$P, P_G, P_0, P_{\Theta_m}, P_r$	finite posets: P_G has $a < b$ exactly for the edges ab of G ; $P_0 = P_{G_0}$; P_r is the crown	Part I's lattice-point set $P_\tau = Q_\tau \cap \mathbb{Z}^n$, which Part II writes $Q(P) \cap \mathbb{Z}^V$. P_r is not " P_τ at $\tau = r$ ".
$V, n = V $	ground set of P	Part I's ground set $E = [n]$.
$G = (A \sqcup B, E)$	bipartite comparability graph of a height-two poset on its n elements; E its edge set	Part I's doubled graph G_τ on $2n$ vertices i_L, j_R (edge iff $j \leq_\tau i$, loops $i_L i_R$ included) and its augmentation $\widehat{G}_\tau$; Part I's ground set E .
$p = A , q = B $	shore sizes (Section 13)	Part I's functions $p(y), q(y)$ of (4.3); the polynomial $q_G(x)$ of (19.5).
$p_G(u), \mathcal{M}_k(G)$	perfectly matchable <i>vertex sets</i> of G ; p_G is the γ -polynomial of h_{P_G}	the matching polynomial $M_G(u) = \sum_k m_k(G)u^k$; also p_{G_τ} of Part I's doubled graph, which is h_τ itself (Remark 11.1).
$H = (X \sqcup Y, E), \widehat{H}, B_{\widehat{H}}, \mathcal{R}_H$	any bipartite graph, its augmentation by x_0, y_0 , its edge polytope, its demand vectors	Part I's B_τ , which is $B_{\widehat{H}}$ for $H = G_\tau$ (same augmentation, same lattice); Part I's hyperplanes H_1, H_m . Hypertrees here are <i>right</i> hypertrees; Appendix A uses left ones.
$\Theta_m, p_m, R_m(y), k = m - 2$	m parallel three-edge paths, its matchable-set polynomial	Part I's dilation m . The vertex d of Θ_m is not a diagonal vertex d_i of (6.2).
$G_0, p_0, F_{i,j}(u)$	the 17-element example, its γ -polynomial, the board recurrence	—
D_G, T_G, g_T, g_D	one-sided root polytope lattice-equivalent to C_{P_G} , its positive part, their gauges	Part I's gauge g_τ on C_τ ; g_D is g_τ after negating the A -coordinates. Part I's neighbourhoods D_i .
$I(G), I$ in Section 18	independent vertex sets	order ideals $I \in \mathcal{J}(\tau)$ (also written I in Section 15); the toric ideal $I_{\mathcal{A}}$. Part I's triangulation K in (5.2).
$K, \zeta, H_G, \Phi_G, q_G, \lambda_j$	cactus matrix, sixth root of unity, Hermitian block matrix, stable polynomial, its diagonal specialization, eigenvalues of KK^*	
S	a subset of B or Y ; in (19.7) the support size of a uniformly random lattice point	Part I's simplicial polytope S_τ .
$F, U, W, v_i, w_i, \mathcal{A}$	the exposed circuit face of a crown and its Radon halves	W also denotes a perfectly matchable set elsewhere in Part II; $F_{i,j}$ is the board recurrence.
$\nu(G)$	maximum matching size	—

six-element crown (Section 8). This does not disprove Conjecture 5.1(d), which asks for *some* flag polytope with h -polynomial h_τ .

Part II also reproves, for height two, three statements of Part I by different routes: the identity $h^*(C_{P_G}) = h_{P_G}$ and reflexivity by an orthant decomposition (Theorem 18.2, compare Corollary 4.2 and Proposition 5.2), invariance under reversal (Corollary 15.2, compare Section 7), and $h_1 = n + |E|$ (compare (1.4)). Its hypertree identification in Section 14 is the argument of Lemma A.1 for a general bipartite graph and right hypertrees. These are printed as the manuscript gives them, as second routes.

11.4 Where the merge had to choose

- *Placement.* Part II follows Section 10 and precedes Part I's appendices, whose letters and numbers are unchanged.
- *Self-citation.* The manuscript cites Part I as an external repository report (its reference "RepoRoot", with the blob above). Here those citations are replaced by references to Part I's sections, and the phrases "the ProvelIt report" and "the repository's construction" by "Part I".
- *File names.* The manuscript's delivered names (`code/verify.py`, `data/board_certificate.csv`, ...) are replaced by the shipped names in Section 22, because `code/verify.py` in this directory is Part I's different verifier. Its build commands are replaced by commands that run on a copy.
- *Bibliography.* The manuscript's entries for Athanasiadis–Chapoton and Kálmán–Postnikov are Part I's references [1] and [3] (the manuscript adds the DOI 10.1112/plms.12015 for the latter); its other five entries are appended.
- *Labels and front matter.* Labels carry the prefix `htwo:`. Two unlabelled corollaries, the conclusion and the build subsection received labels so that this section can cite them. The manuscript's title page, abstract, status box and audit box are printed in this section.

12 The research target and the distinction from prior work

Part I, the ProvelIt report *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, constructs a canonical root polytope for every finite preorder and gives a proposed proof of simplicial-polytopal realization of its support polynomial (Theorem 1.1). Its final section, Section 10, explicitly leaves flag realizability, γ -positivity, and real-rootedness unresolved. These are different properties, and separating them is essential here.

The underlying questions are Conjectures 5.1(d), 5.2, and 5.3 of Athanasiadis and Chapoton [1]. The first asks for *some* flag simplicial polytope with the prescribed h -vector. The second asks for a nonnegative expansion in the basis $t^k(1+t)^{n-2k}$. The third asserts that all zeros of the support polynomial are real. We focus on posets of height at most two: they are preorders, but unlike arbitrary preorders they admit a direct bipartite-graph description.

The graph polynomial needed for this description is classical. The perfectly matchable set polynomial is studied by Davis and Kohl [7]; its root-polytope realization originates in Ohsugi and Tsuchiya [8]. The nonreal-rooted example used below comes from Stembridge's naturally labeled, width-two counterexample to the Neggers conjecture [10], as translated into a graph polynomial by Ohsugi and Tsuchiya [8, Example 3.7]. It is important not to confuse that older *width-two* poset with the new *height-two* poset obtained by orienting its bipartite incomparability graph.

12.1 What the article proves

The height-two matching-support identity is the central theorem. It proves the relevant γ -positivity conjecture for an entire class and converts real-root questions into graph enumeration. The principal negative-family result concerns Θ_m : a complete real-root count shows that almost all zeros become nonreal while graph treewidth stays two.

The source audit found an earlier eight-element counterexample posted by Shivam Patel on MathDB [11]. It is exactly P_{Θ_3} . The source labels the posting unverified; our independent finite calculation agrees with it, and the general proof below includes it. Consequently a claim to have first disproved Conjecture 5.3 would be inappropriate. The site's exact posting date was not available, and its broad historical metadata is not used as evidence. The 17-element construction is retained as an independent bridge to classical literature, not as the smallest or first counterexample.

A second theme concerns a limitation of the canonical root-polytope construction. Every induced crown of size $2r$, $r \geq 3$, forces a minimal nonface of size r in any lattice triangulation. This rules out obtaining a flag complex by choosing a more clever lattice triangulation of that particular polytope. It does *not* rule out another flag polytope with the same h -vector.

12.2 A provenance ledger

Statement	Provenance and status
$h^*(T_G; u) = p_G(u)$ for bipartite G	Established input; Ohsugi–Tsuchiya, reformulated by Davis–Kohl.
Hypertrees count normalized root-polytope volume	Established input from Kálmán–Postnikov; used at $u = 1$.
Height-two formula $h_P = (1 + t)^n p_G(t/(1+t)^2)$	Proved here by a support partition and a counting lemma.
Nonreal-rooted degree-eight graph polynomial	Classical Stembridge–Ohsugi–Tsuchiya example, independently recomputed here.
Eight-element counterexample to AC Conjecture 5.3	Previously posted by Patel; independently checked here, not claimed as new.
Complete root count for P_{Θ_m}	Proved here for every m , extending the eight-element case to a connected treewidth-two family.
17-element counterexample	Independent transfer of the classical graph example; no claim of first disproof or minimality.
Cactus p_G is real-rooted	Already follows from Ohsugi–Tsuchiya [9, Theorem 7.2]; not claimed anew.
Sixth-root matrix and multivariate refinement	Explicit proof supplied here; no absolute novelty claim for the matrix technique.
Induced-crown obstruction in the Provel root model of Part I	Proved here, including a nonregular-triangulation obstruction and a toric degree lower bound.

13 Definitions and main statements

For a finite poset P on a ground set V of size n , let $\mathcal{J}(P)$ be its order ideals, including $\emptyset$ and V . Write $x(S) = \sum_{v \in S} x_v$. Its preorder polytope and support polynomial are

$$\mathcal{Q}(P) = \{x \in \mathbb{R}_{\geq 0}^V : x(I) \leq |I| \text{ for all } I \in \mathcal{J}(P)\}, \quad (13.1)$$

$$h_P(t) = \sum_{x \in \mathcal{Q}(P) \cap \mathbb{Z}^V} t^{|\text{supp}(x)|}. \quad (13.2)$$

The full-set inequality makes the sum finite. Its constant and leading coefficients are both one: the all-zero and all-one vectors are the unique points with support sizes zero and n , respectively.

Here height at most two means that every chain has at most two elements; we consider ordinary posets, not arbitrary equivalence-class blowups of them. Such a poset can be written $P = P_G$ for a bipartite graph $G = (A \sqcup B, E)$: the only strict comparisons are $a < b$ for $ab \in E$. Isolated elements may be assigned to either shore. Put $p = |A|$, $q = |B|$, $n = p + q$, and denote by $N_G(S) \subseteq A$ the neighborhood of $S \subseteq B$.

Definition 13.1. A subset $W \subseteq V(G)$ is *perfectly matchable* if the induced graph $G[W]$ has a perfect matching. The empty set is perfectly matchable. Let

$$\mathcal{M}_k(G) = \{W \subseteq V(G) : |W| = 2k, G[W] \text{ has a perfect matching}\}, \quad p_G(u) = \sum_{k \geq 0} |\mathcal{M}_k(G)| u^k.$$

For comparison, $M_G(u) = \sum_k m_k(G) u^k$ counts individual k -edge matchings. These two polynomials are generally different.

Theorem 13.2 (Height-two matching-support formula). *For every finite bipartite graph G on n vertices,*

$$h_{P_G}(t) = \sum_{k \geq 0} |\mathcal{M}_k(G)| t^k (1+t)^{n-2k} = (1+t)^n p_G\left(\frac{t}{(1+t)^2}\right). \quad (13.3)$$

In particular, every height-at-most-two poset has a γ -positive support polynomial, with $\gamma_k(P_G) = |\mathcal{M}_k(G)|$.

The rational substitution in (13.3) denotes its polynomial expansion; it is not an instruction to evaluate an undefined expression at $t = -1$.

Theorem 13.3 (A complete root-distribution family). *Let Θ_m have lower elements $d, a_1, \dots, a_m$, upper elements $b_0, b_1, \dots, b_m$, and edges $a_i b_0, a_i b_i, d b_i$. For $m \geq 2$,*

$$p_{\Theta_m}(u) = (1+u)^{m-2} (1 + (2m+3)u + (m^2+m+3)u^2 + u^3) - u. \quad (13.4)$$

The support polynomial of P_{Θ_m} is real-rooted exactly when $m \leq 2$ (for $m \geq 1$). For $m \geq 3$, it has precisely two real roots if m is even, and four if m is odd; these real roots are simple and negative. Its graph is connected and has treewidth at most two.

Theorem 13.4 (An explicit negative answer to real-rootedness). *Let G_0 have shores $A = \{a_1, \dots, a_9\}$ and $B = \{b_1, \dots, b_8\}$, with neighborhoods*

$$\begin{array}{c|cccccccc} j & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \hline N(b_j) & [1, 1] & [1, 2] & [1, 3] & [1, 4] & [1, 5] & [2, 6] & [3, 8] & [4, 9] \end{array} \quad (13.5)$$

where $[l, r]$ means $\{a_l, a_{l+1}, \dots, a_r\}$. Then the height-two poset P_{G_0} is a counterexample to Conjecture 5.3 of [1]. Its γ -polynomial is

$$\begin{aligned} p_0(u) = & 1 + 32u + 336u^2 + 1420u^3 + 2534u^4 \\ & + 1946u^5 + 658u^6 + 86u^7 + 3u^8. \end{aligned} \quad (13.6)$$

Its support polynomial is γ -positive despite having nonreal zeros.

Theorem 13.5 (Cactus matrix representation). *For every bipartite cactus graph $G = (A \sqcup B, E)$ there is an explicitly constructible complex matrix K , supported exactly on E , such that*

$$p_G(u) = \det(I_{|A|} + uKK^*). \quad (13.7)$$

The nonzero entries of K are sixth roots of unity, and each square minor has squared absolute value either zero or one, according as its support has no perfect matching or has one. Consequently h_{P_G} is real-rooted. A multivariate stable determinant and a vertex-deletion interlacing theorem are given in Section 19.

Theorem 13.6 (Induced-crown obstruction). *Let P be a finite preorder containing an induced height-two crown on $2r$ distinct elements, $r \geq 3$. The canonical root polytope*

$$C_P = \text{conv}(\{\pm \mathbf{e}_v : v \in P\} \cup \{\mathbf{e}_b - \mathbf{e}_a : a \neq b, a \leq_P b\})$$

uses all nonzero roots for distinct comparable elements; for equivalent elements both directions are included. Every lattice triangulation of C_P or of its boundary has a minimal nonface of cardinality r . The toric ideal on $C_P \cap \mathbb{Z}^n$ has an indispensable binomial of degree r , and every initial ideal has a minimal generator of degree r .

For preorders, all distinct pairs $a \leq_P b$ are used, including both directions within an equivalence class.

14 A matching–polymatroid counting lemma

For a bipartite graph $H = (X \sqcup Y, E)$, define the finite set

$$\mathcal{R}_H = \{c \in \mathbb{Z}_{\geq 0}^Y : c(S) \leq |N_H(S)| \text{ for every } S \subseteq Y\}. \quad (14.1)$$

This set describes nonnegative integer demands that can be supplied by unit-capacity vertices in X , with adjacency specifying allowable supply.

Lemma 14.1 (Counting demands by matching supports). *For every finite bipartite graph H ,*

$$|\mathcal{R}_H| = p_H(1) = \#\{W \subseteq V(H) : H[W] \text{ has a perfect matching}\}. \quad (14.2)$$

We give the reduction to two established root-polytope statements in full. This avoids treating (14.2) as an experimental identity or as an unexplained application of a lattice-point package.

14.1 The two classical inputs

Adjoin x_0 and y_0 to the shores of H , joining x_0 to every old right vertex, y_0 to every old left vertex, and x_0 to y_0 . Write $\widehat{H}$ for this connected augmented bipartite graph. Its edge polytope is

$$B_{\widehat{H}} = \text{conv}\{\mathbf{e}_x + \mathbf{f}_y : xy \in E(\widehat{H})\}.$$

A *right hypertree* is a vector $(\deg_T(y) - 1)_{y \in Y \cup \{y_0\}}$ realized by a spanning tree T of $\widehat{H}$. Multiple spanning trees with the same vector count only once.

We use the following published facts, with the full affine integer lattice on the edge polytope:

$$\text{Vol}_{\mathbb{Z}}(B_{\widehat{H}}) = \#\{\text{right hypertrees of } \widehat{H}\}, \quad (14.3)$$

$$h^*(B_{\widehat{H}}; u) = p_H(u). \quad (14.4)$$

The first is the evaluation at $u = 1$ of the interior-polynomial/root-polytope theorem of Kálmán and Postnikov [3]. The second is Ohsugi–Tsuchiya’s matching-support interpretation, stated explicitly in Davis–Kohl [7, Theorem 3.10]. Recall that $h^*(B; 1) = \text{Vol}_{\mathbb{Z}}(B)$. No result about preorder real-rootedness is an input to either statement.

14.2 Explicit identification of the hypertrees

Let $s = |X|$, $r = |Y|$. A spanning tree of $\widehat{H}$ has $s + r + 1$ edges. Its right degree-minus-one vector has total s . Write it as $(c_0, (c_y)_{y \in Y})$. For nonempty $S \subseteq Y$, the tree edges incident with S form a forest on $S \cup N_{\widehat{H}}(S)$. Their number is $|S| + c(S)$, so

$$|S| + c(S) \leq |S| + |N_{\widehat{H}}(S)| - 1 = |S| + |N_H(S)|.$$

Thus $c \in \mathcal{R}_H$, and $c_0 = s - c(Y)$.

Conversely, take $c \in \mathcal{R}_H$ and replace each $y \in Y$ by c_y clones, each with neighborhood $N_H(y) \subseteq X$. A set of clones with labels in S has at most $c(S) \leq |N_H(S)|$ members. Hall's criterion therefore matches all clones to distinct vertices of X . One can prove this form of Hall's criterion directly by augmenting paths: failure to match a clone produces an alternating-reachable set of clones with fewer neighbors than members, contradicting the stated inequality.

Start a spanning tree with the star joining x_0 to every vertex of $Y \cup \{y_0\}$. Attach each old $x \in X$ as a leaf to the label of its matched clone, or to y_0 if no clone was assigned to it. The resulting graph is a tree and has right degree-minus-one vector

$$(s - c(Y), (c_y)_{y \in Y}).$$

Hence hypertree vectors correspond bijectively to $\mathcal{R}_H$. Combining this bijection with (14.3) and (14.4) proves Lemma 14.1. This argument also handles an empty shore: $\mathcal{R}_H$ then contains only the zero demand and $p_H = 1$.

Remark 14.2. The lemma equates two cardinalities; it does not provide a canonical bijection between demand vectors and perfectly matchable vertex sets. The tree construction is a bijection with *hypertree vectors*, not with their realizing trees. Counting matchings or spanning trees with multiplicity at this step would invalidate the main formula.

15 The height-two formula and its consequences

15.1 Partitioning lattice points by zeros and support

Write a lattice point of $\mathcal{Q}(P_G)$ as (a, b) on the two shores. Each singleton in A is an ideal, so $a_i \in \{0, 1\}$. Let

$$Z = \{i \in A : a_i = 0\}, \quad S = \{j \in B : b_j > 0\}.$$

An ideal is $U \sqcup T$ with $T \subseteq B$ and $N_G(T) \subseteq U \subseteq A$. Its inequality is

$$b(T) \leq |T| + |U \cap Z|.$$

For fixed T , the strongest choice is $U = N_G(T)$. Thus all ideal inequalities are equivalent to

$$b(T) \leq |T| + |N_G(T) \cap Z| \quad (T \subseteq B). \quad (15.1)$$

On S put $b_j = 1 + c_j$, with $c_j \geq 0$, and put $b_j = 0$ elsewhere. For $T \subseteq S$, equation (15.1) becomes

$$c(T) \leq |N_G(T) \cap Z|.$$

These inequalities also imply (15.1) for arbitrary $T \subseteq B$, by applying them to $T \cap S$. Therefore the possible c are exactly $\mathcal{R}_{G[Z \sqcup S]}$.

Every point in this part has support size $p - |Z| + |S|$. Lemma 14.1 gives

$$h_{P_G}(t) = \sum_{Z \subseteq A} \sum_{S \subseteq B} t^{p - |Z| + |S|} \sum_{W \in \mathcal{M}(G[Z \sqcup S])} 1, \quad (15.2)$$

where $\mathcal{M}(H)$ denotes all perfectly matchable vertex sets of H .

15.2 The Boolean summation

Fix a perfectly matchable set $W = X \sqcup Y$ of G , with $X \subseteq A$, $Y \subseteq B$, and $|X| = |Y| = k$. It is counted in (15.2) exactly when $X \subseteq Z$ and $Y \subseteq S$. Its total contribution is

$$\begin{aligned} \left(\sum_{Z \supseteq X} t^{p-|Z|} \right) \left(\sum_{S \supseteq Y} t^{|S|} \right) &= (1+t)^{p-k} t^k (1+t)^{q-k} \\ &= t^k (1+t)^{n-2k}. \end{aligned}$$

Summing over W proves Theorem 13.2. In particular, the γ -positivity conclusion is not deduced from an assumed flag complex; it follows from an explicit nonnegative counting interpretation.

15.3 Roots, matching number, and small coefficients

Let $\nu(G)$ be the maximum matching size. Then $\deg p_G = \nu(G)$, and all its coefficients from degree zero through $\nu(G)$ are positive: a maximum matching has submatchings of every smaller size.

Proposition 15.1 (Exact transfer of real-rootedness). *For a bipartite graph G , the polynomial h_{P_G} is real-rooted if and only if p_G is real-rooted. If p_G has only real roots, they are strictly negative. In all cases, the root -1 of h_{P_G} has multiplicity $n - 2\nu(G)$. In the real-rooted case, its other roots occur in reciprocal negative pairs.*

Proof. Write $p_G(u) = \prod_{j=1}^{\nu} (1 + \lambda_j u)$ over $\mathbb{C}$, with $\lambda_j \neq 0$. Formula (13.3) gives

$$h_{P_G}(t) = (1+t)^{n-2\nu} \prod_{j=1}^{\nu} ((1+t)^2 + \lambda_j t). \quad (15.3)$$

If p_G is real-rooted, its positive coefficients and constant term one exclude nonnegative roots. Hence every $\lambda_j > 0$, and each quadratic $t^2 + (2 + \lambda_j)t + 1$ has two distinct negative reciprocal roots. Its value at -1 is $-\lambda_j \neq 0$, proving the stated multiplicity.

Conversely, if u_0 is a nonreal zero of p_G , then $u_0(1+t)^2 - t = 0$ has two roots different from -1 . Neither is real, since $t/(1+t)^2$ is real for real $t \neq -1$. Both are zeros of h_{P_G} . Thus real-rootedness of h_{P_G} forces that of p_G . $\square$

Corollary 15.2. *The height-two polynomial is unchanged by reversing the poset, and it is multiplicative under disjoint unions. Its first coefficients satisfy*

$$\gamma_0 = 1, \quad \gamma_1 = |E|, \quad \gamma_2 = \binom{|E|}{2} - \sum_{v \in V} \binom{\deg(v)}{2} - c_4(G), \quad h_1 = n + |E|,$$

where $c_4(G)$ is the number of four-cycles, each counted once.

Proof. Reversing the order exchanges the shores of G , leaving p_G unchanged. Perfectly matchable subsets split independently across components. For γ_2 , subtract adjacent pairs of edges from all edge pairs to obtain $m_2(G)$. A four-vertex bipartite subgraph has two perfect matchings exactly when it is a four-cycle; otherwise it has at most one. Thus $\gamma_2 = m_2 - c_4$. Finally, the coefficient of t in (13.3) is $n + \gamma_1$. $\square$

For a forest, every perfectly matchable induced subgraph has a unique perfect matching: the symmetric difference of two perfect matchings would contain a cycle. Hence $p_G = M_G$. More generally, $|\mathcal{M}_k(G)| \leq m_k(G)$, and equality for all k holds for a bipartite graph exactly when it is a forest. An even cycle supplies two distinct matchings with the same covered set, proving the converse.

16 A complete nonreal-root family at treewidth two

This section strengthens a single counterexample to a uniform theorem with an exact count of real roots. The mechanism is distinct from taking disjoint unions: every graph in the family is connected, and its treewidth is bounded.

16.1 Parallel paths and the matching-support polynomial

Let Θ_m be the union of the m paths

$$d - b_i - a_i - b_0 \quad (1 \leq i \leq m),$$

whose only common vertices are the endpoints d, b_0 . Its bipartition is $A = \{d, a_1, \dots, a_m\}$ and $B = \{b_0, b_1, \dots, b_m\}$. For $m = 1$ this is a four-vertex path; for $m = 2$ it is a six-cycle. For $m = 3$ the corresponding height-two poset is the previously posted counterexample of Patel [11].

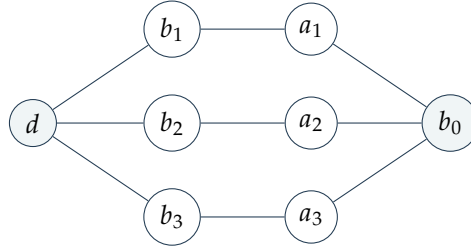


Figure 1: The graph Θ_3 . To obtain its height-two poset, orient every edge from the shore $\{d, a_1, a_2, a_3\}$ to the other shore. The horizontal layout shows paths, not increasing order in the poset.

Proposition 16.1. *For $m \geq 2$, the perfectly matchable set polynomial of Θ_m is (13.4). For $m = 1$, it is $1 + 3u + u^2$.*

Proof. Partition matchable vertex sets according to the presence of d, b_0 . If neither endpoint is present, each branch contributes either its two internal vertices or neither; the contribution is $(1 + u)^m$.

If d alone is present, it must be matched to a unique included vertex b_i whose partner a_i is absent. Every other included branch is a complete internal pair. There are m choices of i , giving $mu(1 + u)^{m-1}$. The case with b_0 alone has the same contribution.

Suppose both endpoints are present. If some branch contributes just b_i , then d must match it. There must be exactly one different branch contributing just a_j , to be matched to b_0 ; all other branches are empty or complete internal pairs. This gives $m(m - 1)u^2(1 + u)^{m-2}$. In the remaining case all selected branches are complete pairs. At least one such branch is needed to match the endpoints, and the selected vertex set is counted once, regardless of how many of its branches could be used for this purpose. This gives $u((1 + u)^m - 1)$.

Thus the sum is

$$\begin{aligned} & (1 + u)^m + 2mu(1 + u)^{m-1} + m(m - 1)u^2(1 + u)^{m-2} + u((1 + u)^m - 1) \\ &= (1 + u)^{m-2}(1 + (2m + 3)u + (m^2 + m + 3)u^2 + u^3) - u. \end{aligned}$$

The four-vertex path is counted directly. Notice that replacing matchable sets by matchings would overcount the case where both endpoints and several complete branches are present; that distinction is essential to the formula. $\square$

16.2 A positive interval with no roots

Write $p_m = p_{\ominus_m}$. For $m \geq 3$, put $k = m - 2 \geq 1$. A useful rearrangement, valid for $0 < x < 1$, is

$$p_m(-x) = (1-x)^k (1-(m+1)x)^2 + x [1 - (1-x)^k (1+kx+x^2)]. \quad (16.1)$$

The second line is strictly positive. Indeed, the binomial series gives

$$(1-x)^{-k} = 1 + kx + \frac{k(k+1)}{2}x^2 + \sum_{j \geq 3} \binom{k+j-1}{j} x^j > 1 + kx + x^2.$$

All terms are real and positive on this interval; the inequality holds also for $k = 1$ because the tail is then nonzero. Moreover, $p_m(0) = p_m(-1) = 1$. We have proved the particularly strong exclusion

$$p_m(u) > 0 \quad (-1 \leq u \leq 0, \quad m \geq 3). \quad (16.2)$$

The coefficients of p_m are nonnegative by their interpretation as counts, so there are no nonnegative real roots either.

Already this disproves real-rootedness: p_m is monic with constant term one. If all its roots were real, (16.2) would put them strictly below -1 , making the absolute value of their product larger than one. We next obtain the exact count, rather than just this contradiction.

16.3 Exact root counts by two sign changes

Set $u = -1 - y$ with $y > 0$ and define

$$R_m(y) = m(m-1) + 2m^2y + m(m+1)y^2 - y^3.$$

An exact change of variables gives

$$p_m(-1-y) = 1 + y + (-1)^{m-2} y^{m-2} R_m(y). \quad (16.3)$$

We use Descartes' elementary rule of signs: the number of positive roots, counted with multiplicity, is at most the number of changes in the sequence of nonzero coefficients. One way to prove the needed bound is to factor out positive linear factors successively; multiplying a real polynomial by $y - a$, $a > 0$, increases the number of sign changes by a positive odd integer. Factoring over $\mathbb{R}$ then yields the bound.

If m is even, $m \geq 4$, the polynomial on the right of (16.3) has positive coefficients except for its leading coefficient, which is negative. It has one sign change, is positive at $y = 0$, and tends to $-\infty$. It consequently has exactly one positive root, and that root is simple.

If m is odd, $m \geq 3$, its coefficients have signs

$$+, \quad \underbrace{- , \dots , -}_{\text{nonzero middle block}}, \quad +$$

when zero coefficients are omitted; for $m > 3$ there is an additional initial positive coefficient, which does not change the count of two sign changes. For $m = 3$, the merged coefficient of y is $1 - m(m-1) < 0$. At $y = 0$ the value is one, at $y = 1$ it is

$$2 - (4m^2 - 1) = 3 - 4m^2 < 0,$$

and it tends to $+\infty$ as $y \rightarrow \infty$. There is a root in $(0, 1)$ and another in $(1, \infty)$. The sign bound makes these the only positive roots, both simple.

By (16.2), these exhaust the real roots of p_m . It has degree $m + 1$, so it has m nonreal roots when m is even and $m - 1$ when m is odd, counted with multiplicity. The substitution in Proposition 15.1 sends each negative root to two distinct negative reciprocal roots of $h_{P_{\Theta_m}}$ and each nonreal root to two nonreal roots. There is no root at -1 , because $n = 2m + 2 = 2 \deg p_m$. This proves the root count in Theorem 13.3. The cases $m = 1, 2$ are a path and a cycle, respectively, and are real-rooted by the independent cactus proof in Section 19.

For completeness, a width-two tree decomposition is explicit. Use bags $\{d, b_0, a_i\}$ and $\{d, a_i, b_i\}$ for each i , connect the first bags in a path, and attach each second bag to its corresponding first bag. Every edge is covered; bags containing any fixed vertex form a connected subtree. All bags have size three. Thus the asserted treewidth bound requires no general graph-minor theorem.

Corollary 16.2 (Most roots nonreal in a bounded-treewidth family). *The connected height-two posets P_{Θ_m} satisfy*

$$\#\{\text{nonreal zeros of } h_{P_{\Theta_m}}\} = \begin{cases} 2m, & m \text{ even}, \\ 2m - 2, & m \text{ odd}, \end{cases} \quad (m \geq 3),$$

with multiplicities. Their proportion among all $2m + 2$ zeros tends to one. Thus bounded treewidth, even together with height two and connectedness, does not force real-rootedness or even a positive limiting proportion of real roots.

16.4 The eight-element specialization and its status

The specialization $m = 3$ gives

$$p_3(u) = 1 + 9u + 24u^2 + 16u^3 + u^4,$$

$$h_{P_{\Theta_3}}(t) = 1 + 17t + 106t^2 + 303t^3 + 427t^4 + 303t^5 + 106t^6 + 17t^7 + t^8.$$

These are the polynomials in Patel's posting [11], not newly claimed examples. Our direct enumeration from the ideal inequalities finds 1281 lattice points and reproduces every coefficient. Exact arithmetic also verifies the posted Laguerre obstruction

$$(p'_3(-1/4))^2 - p_3(-1/4)p''_3(-1/4) = -95/1024 < 0.$$

The additional result here is the all- m formula and complete root count, not a priority claim for this specialization.

m	Poset size	Real roots of p_m	Real roots of h	Nonreal roots of h
3	8	2	4	4
4	10	1	2	8
5	12	2	4	8
6	14	1	2	12
7	16	2	4	12
8	18	1	2	16

17 The 17-element counterexample in exact arithmetic

17.1 A complete graph specification

Use the graph G_0 of (13.5); there are no other edges. Orient each edge from a_i to b_j to define $P_0 = P_{G_0}$. Since all strict comparisons go from A to B , the relation is automatically transitive and antisymmetric, and has height two. There are $1 + 2 + 3 + 4 + 5 + 5 + 6 + 6 = 32$ strict comparisons.

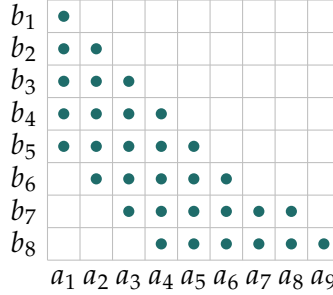


Figure 2: The complete 8×9 incidence board of the counterexample. A dot means $a_i < b_j$. Increasing pairs of dots give the canonical matching of any matchable support. The diagram is drawn from the exact interval list, not inferred from a numerical polynomial.

17.2 Why an increasing matching suffices

Lemma 17.1 (Uncrossing on a monotone interval board). *Suppose $N(b_j) = [\ell_j, r_j]$ with both (ℓ_j) and (r_j) nondecreasing. For sorted subsets $X = \{a_{i_1}, \dots, a_{i_k}\}$ and $Y = \{b_{j_1}, \dots, b_{j_k}\}$, the induced graph on $X \sqcup Y$ has a perfect matching if and only if every $a_{i_s} b_{j_s}$ is an edge.*

Proof. The backward implication is immediate. In a matching with an inversion, suppose $j < j'$ and $i < i'$ but the matching uses $a_i b_{j'}$ and $a_{i'} b_j$. Then

$$\ell_j \leq \ell_{j'} \leq i < i' \leq r_j \leq r_{j'}.$$

Therefore $a_i b_j$ and $a_{i'} b_{j'}$ are also edges. Swap the inverted pair. Choosing an adjacent inversion in the permutation of matched row indices reduces the inversion number; iteration gives the unique sorted matching. Thus existence is equivalent to the sorted-edge test. $\square$

Let $F_{i,j}(u)$ count increasing matched pairs in the first i vertices of A and first j vertices of B , by their number of pairs. With $\varepsilon_{i,j} = 1$ for an edge and zero otherwise,

$$\begin{aligned} F_{i,0}(u) &= F_{0,j}(u) = 1, \\ F_{i,j}(u) &= F_{i-1,j}(u) + F_{i,j-1}(u) + (u\varepsilon_{i,j} - 1)F_{i-1,j-1}(u). \end{aligned} \tag{17.1}$$

Indeed, those objects omitting the last row or the last column are counted by inclusion–exclusion. An object using both must pair them with each other and leave an arbitrary increasing matching on the smaller board. Lemma 17.1 implies $F_{9,8} = p_{G_0}$.

For a convenient hand-checkable summary, the last row of this polynomial recurrence is as follows. The complete 90-cell table, including boundary cells, is in `data/02-height-two-board_certificate.csv`.

j	$F_{9,j}(u)$
0	1
1	$1 + u$
2	$1 + 3u + u^2$
3	$1 + 6u + 6u^2 + u^3$
4	$1 + 10u + 20u^2 + 10u^3 + u^4$
5	$1 + 15u + 50u^2 + 50u^3 + 15u^4 + u^5$
6	$1 + 20u + 105u^2 + 175u^3 + 105u^4 + 21u^5 + u^6$
7	$1 + 26u + 205u^2 + 595u^3 + 665u^4 + 301u^5 + 49u^6 + 2u^7$
8	$1 + 32u + 336u^2 + 1420u^3 + 2534u^4 + 1946u^5 + 658u^6 + 86u^7 + 3u^8$

Thus the coefficient formula (13.6) is an exact finite recurrence calculation, not a fit to sampled values. There are $p_0(1) = 7016$ perfectly matchable sets.

17.3 An elementary obstruction to real-rootedness

Lemma 17.2 (Laguerre inequality). *If a nonzero real polynomial f has only real zeros, then*

$$(f'(x))^2 - f(x)f''(x) \geq 0 \quad (x \in \mathbb{R}). \quad (17.2)$$

Proof. Write $f(x) = c \prod_i (x - r_i)$ with $r_i \in \mathbb{R}$, repeated according to multiplicity. Away from the roots,

$$\frac{(f')^2 - ff''}{f^2} = -\left(\frac{f'}{f}\right)' = \sum_i \frac{1}{(x - r_i)^2} \geq 0.$$

Continuity extends the inequality to the roots. □

Direct substitution into (13.6) gives

$$p_0(-2) = 65, \quad p_0'(-2) = -560, \quad p_0''(-2) = 4912. \quad (17.3)$$

Consequently

$$(p_0'(-2))^2 - p_0(-2)p_0''(-2) = 313600 - 319280 = -5680 < 0. \quad (17.4)$$

By Lemma 17.2, p_0 has nonreal roots. Proposition 15.1 proves Theorem 13.4. This proof requires no isolation of a complex root and no trust in a floating-point solver.

Expanding the height-two formula gives

$$\begin{aligned} h_{P_0}(t) = & 1 + 49t + 952t^2 + 9828t^3 + 61302t^4 + 248816t^5 \\ & + 688516t^6 + 1337738t^7 + 1857081t^8 + 1857081t^9 \\ & + 1337738t^{10} + 688516t^{11} + 248816t^{12} + 61302t^{13} \\ & + 9828t^{14} + 952t^{15} + 49t^{16} + t^{17}. \end{aligned} \quad (17.5)$$

Its coefficient sum is 8 408 566. An independent C++ enumeration of *all* these lattice points, based only on integer capacity allocations, reproduces (17.5); see Section 22.

For orientation only, a nonreal pair of p_0 is approximately $-1.8588440 \pm 0.1497675 i$. Its images yield two nonreal reciprocal pairs of h_{P_0} , approximately

$$-2.0461493 \pm 0.0565649 i, \quad -0.4883497 \pm 0.0135002 i.$$

The exact negative Laguerre value, rather than these approximations, is the certificate used in the proof. The accompanying exact computation does not need numerical roots.

17.4 What is old, what is transferred, and what is not settled

Polynomial (13.6) is the normalized descent polynomial in the older Stembridge example, and its interpretation as a bipartite matching-support polynomial is recorded in [8, Example 3.7]. We independently recompute it from the interval board so that the counterexample is specified without relying on a figure or a prior decimal root calculation.

For an explicit comparison, the older poset has chains $1 < 3 < \dots < 17$ and $2 < 4 < \dots < 16$, together with the relations

$$2 < 3, 4 < 5, 6 < 7, 8 < 9, 10 < 11, 12 < 13, 14 < 17, \quad 1 < 12, 3 < 14, 5 < 16.$$

Its cross-chain incomparable pairs have precisely the neighborhoods in (13.5), under $a_i = 2i - 1, b_j = 2j$. In the present counterexample these incomparable pairs are made into *new* comparabilities, all oriented from A to B ; the old chain comparisons are discarded. The two posets are therefore not the same object.

This transfer supplies an independent negative answer to the assertion about the support polynomial of *every preorder*; the existence of such counterexamples was already claimed in the smaller example [11]. It does not contradict γ -positivity, palindromic unimodality, or simplicial-polytopal realizability. The 17-element construction is not smallest: the preceding family already gives an eight-element example. Adding isolated elements multiplies the support polynomial by $(1 + t)$, so counterexamples exist in every size $n \geq 17$; this operation alone says nothing about connected counterexamples of every size.

18 The canonical root polytope by orthant decomposition

Part I's construction (1.2) has a particularly concrete form in height two. Negate the coordinates in the lower shore A . This unimodular map sends the roots $\mathbf{e}_b - \mathbf{e}_a$ to $\mathbf{e}_b + \mathbf{e}_a$ and preserves the collection of signed coordinate vectors. Thus C_{P_G} is lattice equivalent to

$$D_G = \text{conv}(\{\pm \mathbf{e}_v : v \in V\} \cup \{\mathbf{e}_a + \mathbf{e}_b : ab \in E\}). \quad (18.1)$$

Only the positive sums are added; the polytope is not generally centrally symmetric. Also put

$$T_G = \text{conv}(\{0\} \cup \{\mathbf{e}_v : v \in V\} \cup \{\mathbf{e}_a + \mathbf{e}_b : ab \in E\}). \quad (18.2)$$

The full coordinate simplex is present, so both polytopes have dimension $n = |V|$, with the zero-dimensional empty case understood separately.

18.1 An integral gauge from a capacitated matching problem

For $y \in \mathbb{R}_{\geq 0}^V$, let $g_T(y)$ be the least $s \geq 0$ for which $y \in sT_G$. Write $I(G)$ for the independent vertex sets of G .

Lemma 18.1. *For every bipartite graph G and nonnegative y ,*

$$g_T(y) = \max_{I \in I(G)} y(I). \quad (18.3)$$

Consequently g_T is coordinatewise monotone and takes integer values on nonnegative integer vectors.

Proof. Represent y as a nonnegative combination of coordinate vectors and edge-sum vectors. If an edge receives weight w_{ab} , it supplies that amount at each endpoint, reducing the total

coefficient mass by w_{ab} compared with using only coordinate vectors. The minimum total mass is therefore

$$|y| - \max \left\{ \sum_{ab \in E} w_{ab} : w_{ab} \geq 0, \sum_{e \ni v} w_e \leq y_v \right\}.$$

The maximum is the value of a flow from a source through A , then B , to a sink, with vertex capacities y_a, y_b and across-edge capacities larger than $|y|$. A minimum cut never cuts an across-edge arc. Such cuts correspond exactly to vertex covers C , with capacity $y(C)$: the cover consists of lower-shore vertices on the sink side and upper-shore vertices on the source side. Max-flow/min-cut gives the maximum as $\min_C y(C)$. Complements of vertex covers are independent sets, proving (18.3). This argument is valid for real capacities; compactness and the usual residual-path proof give max-flow/min-cut in this finite network. The two final conclusions follow directly from the displayed maximum over independent sets. $\square$

For $x \in \mathbb{R}^V$, let $x_v^+ = \max(x_v, 0)$ and $x_v^- = \max(-x_v, 0)$. Since $D_G = \text{conv}(T_G, \{-\mathbf{e}_v\})$, its gauge is

$$g_D(x) = g_T(x^+) + |x^-|. \quad (18.4)$$

Indeed, any representation $x = y - z$ with $y, z \geq 0$ has coefficient mass at least $g_T(y) + |z|$. Necessarily $y \geq x^+$ and $z \geq x^-$, so the lower bound in (18.4) follows from monotonicity. Taking $y = x^+$ and $z = x^-$ achieves it.

This also proves reflexivity without a triangulation. The gauge can be written as the finite maximum of integer linear forms

$$g_D(x) = \max_{I \in I(G), R \subseteq V \setminus I} \left(\sum_{v \in I} x_v - \sum_{v \in R} x_v \right). \quad (18.5)$$

Every form is bounded by the right side of (18.4); equality is obtained by taking a maximizing independent subset of the positive coordinates and all negative coordinates in R . Thus D_G is cut out by integer inequalities with right side one. The origin is interior because D_G contains the crosspolytope. Its polar is the convex hull of the integer normals in (18.5), hence a lattice polytope.

18.2 Counting the lattice points in every orthant

The augmented edge polytope $B_{\widehat{G}}$ from Section 14 is lattice equivalent to T_G . Project away the two added coordinates. The inverse recovers them as

$$x_0 = 1 - \sum_{a \in A} x_a, \quad y_0 = 1 - \sum_{b \in B} y_b$$

on the affine span of $B_{\widehat{G}}$. These formulas preserve the full affine integer lattice. The projected vertices are exactly 0, the coordinate vectors, and the edge sums. Therefore the established identity (14.4) gives

$$h^*(T_G; t) = p_G(t). \quad (18.6)$$

For an integral gauge g of a bounded polytope containing the origin,

$$\text{Ehr}(t) = \frac{1}{1-t} \sum_{x \in \mathbb{Z}^n} t^{g(x)};$$

a lattice point of least dilation $g(x)$ occurs in every subsequent integer dilation. For T_G , the sum is over nonnegative lattice points. All series here are formal series with finitely many contributions to each coefficient.

Partition the lattice points x of $\mathbb{R}^V$ by the set R of strictly negative coordinates, and put $S = V \setminus R$. The negative coordinates contribute $(t/(1-t))^{|R|}$. The nonnegative coordinates contribute $(1-t) \text{Ehr}_{T_{G[S]}}(t)$, since independent sets restricted to S give precisely the gauge of the induced graph. Using (18.4),

$$\begin{aligned} \text{Ehr}_{D_G}(t) &= \frac{1}{1-t} \sum_{R \subseteq V} \left(\frac{t}{1-t} \right)^{|R|} (1-t) \text{Ehr}_{T_{G[V \setminus R]}}(t) \\ &= \frac{\sum_{R \subseteq V} t^{|R|} p_{G[V \setminus R]}(t)}{(1-t)^{n+1}}. \end{aligned} \quad (18.7)$$

In the numerator, a fixed matchable set W of size $2k$ occurs once for each $R \subseteq V \setminus W$. Its contribution is $t^k(1+t)^{n-2k}$. Hence:

Theorem 18.2 (Height-two root identity). *For every bipartite graph G , D_G is reflexive and*

$$h^*(C_{P_G}; t) = h^*(D_G; t) = \sum_{k \geq 0} |\mathcal{M}_k(G)| t^k (1+t)^{n-2k} = h_{P_G}(t). \quad (18.8)$$

This is a direct height-two proof of the corresponding identity in Part I (Corollary 4.2), independent of its general preorder fiber argument.

Remark 18.3 (A normalization trap). The symmetric edge polytope of type B in Ohsugi–Tsuchiya includes all signed edge sums. Its γ -polynomial is $p_G(4u)$ [8, 9]. Our D_G has only the positive edge sums, and its γ -polynomial is $p_G(u)$. These are different polytopes. The quick check is $h_1(P_G) = n + |E|$, not $n + 4|E|$.

19 Cactus graphs: matrices, stability, and interlacing

A cactus graph is one in which every edge belongs to at most one simple cycle. Distinct cycles may meet in a vertex, but not in an edge; isolated vertices and disconnected components are allowed. In the bipartite case every cycle has even length. Univariate real-rootedness of p_G for these graphs is already contained in Ohsugi–Tsuchiya [9, Theorem 7.2]. The purpose here is a concrete matrix proof that retains a multivariate and weighted structure.

19.1 A unit-modulus certificate for every matchable minor

Let $\zeta = e^{\pi i/3}$, so $\zeta^2 - \zeta + 1 = 0$ and $|1 - \zeta| = |1 - \bar{\zeta}| = 1$. Form an A -by- B matrix K supported on the edges of G . Initially every edge entry is one. For each cycle of length $2r$, choose one of its edges and replace that entry by $(-1)^r \zeta$. These choices do not conflict because cycles are edge-disjoint. All nonzero entries are sixth roots of unity.

Lemma 19.1 (The minor identity). *For $X \subseteq A$ and $Y \subseteq B$ with $|X| = |Y|$,*

$$|\det K_{X,Y}|^2 = \begin{cases} 1, & G[X \cup Y] \text{ has a perfect matching,} \\ 0, & \text{otherwise.} \end{cases} \quad (19.1)$$

Proof. If there is no perfect matching, every term in the determinant is zero. Otherwise fix a perfect matching M_0 of the induced subgraph. The symmetric difference of M_0 with another perfect matching is a disjoint union of M_0 -alternating cycles. In a cactus, all M_0 -alternating cycles are vertex-disjoint: if two shared a vertex, the unique matching edge there would

have to belong to both cycles, contrary to edge-disjointness. Conversely, flipping any chosen subcollection of these alternating cycles gives a perfect matching. Thus all determinant terms are indexed by independent choices of cycle flips.

For an alternating cycle of length $2r$, the ratio of the products of edge entries in its two matching halves is $\Gamma = (-1)^r \zeta$ or $(-1)^r \bar{\zeta}$. There is exactly one modified entry on that cycle; modifications on other cycles are not on its edges. The permutation sign changes by $(-1)^{r-1}$ when the cycle is flipped. Factoring out the unit-modulus term for M_0 , the determinant therefore factors as

$$\text{a complex unit} \prod_{\text{alternating cycles}} (1 + (-1)^{r-1} \Gamma).$$

Each factor is $1 - \zeta$ or $1 - \bar{\zeta}$, hence has absolute value one. The determinant has absolute value one, proving the claim. $\square$

Theorem 19.2 (Determinant and spectral factorization). *Let ν be the maximum matching size of a bipartite cactus graph, and let $\lambda_1, \dots, \lambda_\nu$ be the positive eigenvalues of KK^* , with multiplicities. Then*

$$p_G(u) = \det(I + uKK^*) = \prod_{j=1}^{\nu} (1 + \lambda_j u), \quad (19.2)$$

$$h_{P_G}(t) = (1+t)^{n-2\nu} \prod_{j=1}^{\nu} ((1+t)^2 + \lambda_j t). \quad (19.3)$$

In particular, all roots of p_G and h_{P_G} are real and negative.

Proof. The principal-minor expansion and Cauchy–Binet give

$$\det(I + uKK^*) = \sum_{k \geq 0} u^k \sum_{\substack{X \subseteq A, Y \subseteq B \\ |X|=|Y|=k}} |\det K_{X,Y}|^2.$$

Lemma 19.1 turns its coefficients into $|\mathcal{M}_k(G)|$. The largest nonzero minor has size ν , so $\text{rank } K = \nu$. The Hermitian positive-semidefinite matrix KK^* has exactly ν positive eigenvalues, proving (19.2). Substitute in Theorem 13.2 to get (19.3). Each quadratic factor has discriminant $\lambda_j(\lambda_j + 4) > 0$ and two negative roots. $\square$

The distinction between a cactus and a general series-parallel graph is now sharp in a useful sense: the connected graphs Θ_m still have treewidth two, but for $m \geq 3$ their cycles overlap along paths and their support polynomials fail real-rootedness dramatically.

19.2 A stable multivariate refinement

Let

$$H_G = \begin{pmatrix} 0 & K \\ K^* & 0 \end{pmatrix}.$$

This is an n -by- n Hermitian matrix indexed by all vertices. For a vertex set W , its principal minor is zero if the two shores have unequal sizes. When $|W \cap A| = |W \cap B| = k$, it equals $(-1)^k |\det K_{W \cap A, W \cap B}|^2$. Expanding a determinant along its diagonal entries therefore proves

$$\Phi_G(\mathbf{z}) := \det(\text{diag}(z_v) - H_G) = \sum_{W \text{ perfectly matchable}} (-1)^{|W|/2} \prod_{v \notin W} z_v. \quad (19.4)$$

In particular, the right side has real coefficients and does not depend on the particular chosen edges or phases in the construction.

Theorem 19.3 (Stability and deletion interlacing). *For a bipartite cactus graph, Φ_G is real stable: it is nonzero whenever $\operatorname{Im} z_v > 0$ for every v . Its diagonal specialization*

$$q_G(x) := x^n p_G(-x^{-2}) = \det(xI - H_G) \quad (19.5)$$

is real-rooted, and q_{G-v} interlaces q_G for every vertex v .

Proof. If $(\operatorname{diag}(\mathbf{z}) - H_G)w = 0$ for $w \neq 0$, taking the imaginary part of its scalar product with w would give

$$0 = \sum_v \operatorname{Im}(z_v) |w_v|^2 > 0,$$

because $w^* H_G w$ is real. This proves stability. The diagonal identity follows from (19.4); it denotes a polynomial even at $x = 0$. Deleting v gives the principal submatrix of H_G . Its characteristic polynomial is q_{G-v} by the same minor identity, regardless of whether a previously modified edge survives as a bridge. The eigenvalues of a Hermitian principal submatrix interlace those of the whole matrix: the variational characterization of the j th eigenvalue, restricted to the coordinate hyperplane omitting v , gives the two inequalities between consecutive eigenvalues. This proves the stated polynomial interlacing. $\square$

The signed complementary-support polynomial in (19.4) is the stable object. We are not claiming that the unsigned polynomial $\sum_W \prod_{v \in W} z_v$ is stable; already $1 + z_a z_b$ for a single edge vanishes at $z_a = z_b = i$.

[Added 29 September 2026, batch 42: Part III shows that this construction is exhaustive. Only cacti carry a matrix satisfying (19.1), and on a cactus every such matrix is gauge-equivalent to one of $2^{\beta(G)}$ sixth-root matrices, the choice $(-1)^r \zeta$ being allowed on each cycle as well (Theorems 29.1 and 30.3). Real stability of the signed polynomial extends beyond cacti (Theorem 37.1).]

19.3 Positive weights and a probabilistic consequence

Given positive vertex weights w_v , replace K by $K_w = \operatorname{diag}(\sqrt{w_a}) K \operatorname{diag}(\sqrt{w_b})$. The minor identity gives

$$\sum_{W \text{ perfectly matchable}} u^{|W|/2} \prod_{v \in W} w_v = \det(I + u K_w K_w^*). \quad (19.6)$$

Thus the weighted polynomial is real-rooted as well, and its γ -transform has the same negative-root factorization. This is a weighted graph refinement; no weighted preorder-polytope interpretation is presumed.

For the original unweighted preorder polytope, choose a lattice point uniformly and let S be its support size. Its probability generating function is $h_{P_G}(t)/h_{P_G}(1)$. Equation (19.3) identifies its distribution with a sum of $n - 2v$ independent Bernoulli variables of parameter $1/2$ and v independent variables $Y_j \in \{0, 1, 2\}$ with

$$\Pr(Y_j = 0) = \Pr(Y_j = 2) = \frac{1}{4 + \lambda_j}, \quad \Pr(Y_j = 1) = \frac{2 + \lambda_j}{4 + \lambda_j}.$$

Hence

$$\mathbb{E}S = \frac{n}{2}, \quad \operatorname{Var}(S) = \frac{n - 2v}{4} + \sum_{j=1}^v \frac{2}{4 + \lambda_j}. \quad (19.7)$$

This is an equality in distribution, not a claimed bijection assigning independent coordinates to individual lattice points.

20 Crowns: explicit formulas with simple negative roots

Let the crown P_r have lower elements $a_1, \dots, a_r$ and upper elements $b_1, \dots, b_r$, with comparisons $a_i < b_i$ and $a_{i+1} < b_i$, where indices are cyclic. For $r \geq 2$, its comparability graph is C_{2r} . The graph polynomial formula below is known [7, Corollaries 3.4 and 3.14]; we derive it to identify precisely the polynomials in the geometric obstruction that follows.

A proper induced subgraph of a cycle is a forest, so each of its matchable vertex sets has a unique perfect matching. The whole even cycle has two perfect matchings, but contributes only one vertex set. Therefore

$$p_{C_{2r}}(u) = M_{C_{2r}}(u) - u^r = \sum_{k=0}^r \frac{2r}{2r-k} \binom{2r-k}{k} u^k - u^r. \quad (20.1)$$

To check the matching coefficient, distinguish one of the k selected edges. There are $2r$ choices of its position and $\binom{2r-k-1}{k-1}$ choices of positive intervening gap lengths; divide by k to forget the distinction. The $k=0$ case contributes one. The top matching coefficient is two and becomes one after the correction.

Put

$$L_{\pm}(t) = \frac{1+t \pm \sqrt{1+6t+t^2}}{2}.$$

The cycle matching recurrence, or the two characteristic roots of the path recurrence, gives

$$h_{P_r}(t) = L_+(t)^{2r} + L_-(t)^{2r} - t^r. \quad (20.2)$$

The radical expression is a polynomial: its conjugate terms cancel all odd powers of the square root. In particular,

$$|Q(P_r) \cap \mathbb{Z}^{2r}| = (1 + \sqrt{2})^{2r} + (1 - \sqrt{2})^{2r} - 1. \quad (20.3)$$

For $r=3$, this gives the six-element crown vector $(1, 12, 48, 75, 48, 12, 1)$ and total 197 of Part I's table in Section 8.

Proposition 20.1 (Explicit location of every crown root). *For $r \geq 2$, all r roots of $p_{C_{2r}}$ are simple and less than $-1/4$. They are*

$$u = -\frac{1}{4 \cos^2 \theta}, \quad 0 < \theta < \frac{\pi}{2}, \quad \cos(2r\theta) = \frac{(-1)^r}{2}. \quad (20.4)$$

All $2r$ roots of h_{P_r} are simple and belong to

$$(-3 - 2\sqrt{2}, -1) \cup (-1, -3 + 2\sqrt{2}).$$

Proof. The roots of $z^2 - z - u = 0$ are $z_{\pm} = (1 \pm \sqrt{1+4u})/2$, and $M_{C_{2r}}(u) = z_+^{2r} + z_-^{2r}$. Under the substitution in (20.4), these become $z_{\pm} = e^{\pm i\theta}/(2 \cos \theta)$. Hence

$$M_{C_{2r}}(u)/u^r = 2(-1)^r \cos(2r\theta).$$

The correction in (20.1) gives exactly the angle equation. As $2r\theta$ traverses $(0, r\pi)$, each interval $(j\pi, (j+1)\pi)$ contains one solution. Thus there are exactly r distinct roots, all strictly below $-1/4$. Each gives two negative reciprocal roots through

$$t = \frac{1 - 2u \pm \sqrt{1 - 4u}}{2u}.$$

This map has no double root for $u < 0$. Its endpoint behavior as $u \rightarrow -1/4$ and $u \rightarrow -\infty$ gives the two asserted open intervals. No roots from different u can coincide, because $u = t/(1+t)^2$ recovers the original value. $\square$

21 Induced crowns obstruct every lattice triangulation

A simplicial complex is *flag* if every set of pairwise adjacent vertices is a face. A *minimal nonface* is a set that is not a face although all its proper subsets are faces. Flagness is equivalent to all minimal nonfaces having size two. We show that the canonical root model has forced minimal nonfaces of unbounded size, even when its h^* -polynomial has only simple negative roots.

21.1 An exposed circuit face

Let a finite preorder P contain an induced crown on distinct elements a_i, b_i ($1 \leq i \leq r$), $r \geq 3$. Thus the only distinct-element comparisons among these $2r$ points are the crown comparisons. Consider C_P as in Theorem 13.6. Give a linear functional the coordinate values $-1/2$ at the a_i , $1/2$ at the b_i , and zero on all other elements. Every coordinate root has value at most $1/2$, and every difference root has value at most one. Equality occurs precisely at the crown roots

$$v_i = \mathbf{e}_{b_i} - \mathbf{e}_{a_i}, \quad w_i = \mathbf{e}_{b_i} - \mathbf{e}_{a_{i+1}}. \quad (21.1)$$

They therefore span an exposed face F of C_P .

The coordinate equations show that the unique affine dependence, up to scale, is

$$\sum_{i=1}^r v_i = \sum_{i=1}^r w_i. \quad (21.2)$$

Indeed, a dependence among the cycle edge vectors forces consecutive coefficients alternately equal and opposite. There is one free scalar, and the coefficients sum to zero. Hence F has dimension $2r - 2$, its $2r$ vertices form an affine circuit, and its Radon partition consists of $U = \{v_1, \dots, v_r\}$ and $W = \{w_1, \dots, w_r\}$.

After negating the a_i coordinates, the vertices of F are 0–1 edge-incidence vectors. A convex combination of 0–1 vectors can be integral only if all vectors with positive coefficient agree in every coordinate: an integer coordinate in $[0, 1]$ is an endpoint, forcing agreement. Thus

$$F \cap \mathbb{Z}^n = \{v_1, \dots, v_r, w_1, \dots, w_r\}. \quad (21.3)$$

There are no additional lattice points with which to subdivide this face.

21.2 The nonflag obstruction does not assume regularity

Lemma 21.1. *Every triangulation of F using its vertices has a minimal nonface of size r .*

Proof. Every proper subset T of U spans a simplex face of F . To see this, assign height zero to T , height $-r$ to $U \setminus T$, and height $-(r - |T|)$ to W . The sums of the assigned heights on the two halves are equal, so the unique affine relation (21.2) is respected. There is consequently an affine functional with exactly these values on the vertices. Its maximum face is precisely T . The vertices in T are affinely independent because a proper subset of an affine circuit is independent. The same argument works for proper subsets of W .

On the other hand, $\text{conv}(U)$ and $\text{conv}(W)$ meet in their relative interiors at the common average in (21.2). They cannot both be simplices of one triangulation: simplices with disjoint vertex sets may not intersect in a point which is not a common face. At least one of U, W is therefore missing. All of its proper subsets are forced simplex faces, so it is a minimal nonface of size r . $\square$

A lattice triangulation of C_P , or of its boundary, restricts to a triangulation of its exposed face F . By (21.3) the restriction has no extra vertices. Every proper subset used in the lemma is already a simplex face of C_P and cannot disappear under such a triangulation. A missing half cannot occur elsewhere as a simplex, since its geometric simplex lies in F . This proves the triangulation part of Theorem 13.6 for *all* lattice triangulations, not merely coherent, pulling, or regular ones. Since $r \geq 3$, the missing half is a clique in the one-skeleton, so the triangulation is not flag.

For a height-two graph, the hypothesis is exactly an induced cycle of length at least six in the bipartite comparability graph. Thus chordal bipartiteness (absence of such induced cycles) is a necessary condition for a flag lattice triangulation of D_G . Sufficiency for this particular root model is not proved here. Results about a fully signed symmetric edge polytope cannot be substituted without checking the different configuration.

21.3 An indispensable degree- r toric relation

Fix any field $\mathbb{k}$. Let $\mathcal{A} = C_P \cap \mathbb{Z}^n$, and consider the toric homomorphism

$$\mathbb{k}[X_a : a \in \mathcal{A}] \longrightarrow \mathbb{k}[s, z_1^{\pm 1}, \dots, z_n^{\pm 1}], \quad X_a \longmapsto sz^a.$$

Its kernel $I_{\mathcal{A}}$ is homogeneous in total degree. Define

$$M_U = \prod_{i=1}^r X_{v_i}, \quad M_W = \prod_{i=1}^r X_{w_i}.$$

Equation (21.2) gives $M_U - M_W \in I_{\mathcal{A}}$.

Proposition 21.2. *The binomial $M_U - M_W$ occurs, up to a nonzero scalar, in every binomial generating set of $I_{\mathcal{A}}$. For every monomial order, its leading monomial is a minimal generator of the corresponding initial ideal. Consequently neither a quadratic binomial generating set nor a quadratic Gröbner basis is possible when $r \geq 3$.*

Proof. Consider the monomial fiber with total degree r and vector sum $\sum_i v_i$. The exposing functional of F has value r on this sum, and every lattice point of C_P has value at most one. Every factor of any monomial in the fiber must therefore be a point of F . By (21.3), only its cycle vertices may occur. After the coordinate sign change, the required sum has a one at every cycle vertex. Its decompositions into edges are exactly the two perfect matchings of the cycle. Thus the fiber contains only M_U and M_W .

Moreover, every proper divisor of either monomial has a singleton fiber. The same face-functional argument restricts any competing factorization to cycle edges, while its endpoint set is a proper subset of the cycle. The induced subgraph is a forest and has at most one perfect matching. No proper divisor can therefore be a term of a nonzero toric binomial.

A binomial generating set connects each monomial fiber by multiplying its generators by monomials. In this two-element fiber, a connection cannot come from a generator of smaller degree, because that would require a proper divisor with another monomial in its fiber. The binomial itself is therefore indispensable. For an initial ideal, the leading monomial of this binomial has no proper divisor in the initial ideal by the same singleton-fiber argument. It is a minimal generator of degree r . $\square$

Corollary 21.3 (A spectral–triangulation separation). *For every $r \geq 3$ there is a height-two poset on $2r$ elements whose canonical root polytope has a real-rooted h^* -polynomial with simple zeros, but every lattice triangulation has a minimal nonface of size r and every toric initial ideal has a minimal generator of degree r .*

Proof. Take the crown P_r and combine Theorem 18.2, Proposition 20.1, and the preceding results. $\square$

The existential flag question is different. These obstructions apply to triangulations and the toric configuration of the canonical C_P . Athanasiadis–Chapoton Conjecture 5.1(d) asks whether *some* flag simplicial polytope has h -polynomial h_P . A different polytope might do so even when every lattice triangulation of C_P fails. The present result does not disprove that conjecture.

22 Exact verification and reproducibility

Part II ships three source programs and their recorded outputs, under the names given in Section 22.4; below they are called by their delivered names `verify.py`, `verify_theta.py` and `direct_counterexample.cpp`. (Part I’s own verifier, `code/verify.py` in this directory, is a different program; see Section 9.) No numerical root finder is used by the certificate programs. Python computations use integers, rational numbers, or pairs of integers in $\mathbb{Z}[\zeta]$; the C++ program uses exact finite state sets and integer counts. The proofs establish the infinite-family results. The programs check finite instances and offer independent routes to the explicit counterexamples.

22.1 Independent representations, not repeated substitutions

The program `verify.py` represents a bipartite graph by the bitmask neighborhood of each upper vertex. It computes p_G by adding upper vertices one at a time and keeping a set of pairs of covered-shore masks. Different matchings with the same covered vertices are merged. For the 17-element graph it independently runs the ordered-board recurrence from Section 17; the entire 10-by-9 polynomial table is saved in `data/02-height-two-board_certificate.csv`.

For every one of the 512 labeled bipartite graphs with three vertices on each shore, the program directly enumerates lattice points from the ideal inequalities and compares their support polynomial with the matching-support transform. It independently compares the cardinality of $\mathcal{R}_G$ with $p_G(1)$. A further 24 graphs with shores of sizes four and three use the fixed seed 20260928.

The cactus tests do not infer determinants from matching counts. They compute determinants in $\mathbb{Z}[\zeta]$ by subset-memoized Laplace expansion, using $\zeta^2 = \zeta - 1$. If $z = a + b\zeta$, its exact squared modulus is $a^2 + ab + b^2$. The code compares that value with an independently enumerated matching-support indicator for every square minor. The tested graphs are cycles of lengths 4, 6, 8, 10, a square and a hexagon sharing a vertex, and that cactus with a two-edge bridge extension. There are 936 square minors in total, including empty minors.

22.2 The theta-family audit

The separate program `verify_theta.py` computes (13.4) by binomial expansion and compares it with the covered-shore enumeration for $1 \leq m \leq 9$. For $1 \leq m \leq 4$, it also enumerates the original lattice points directly from the ideal inequalities. In particular, the eight-element specialization has exactly 1281 lattice points with the coefficients displayed in Section 16.

A rational-arithmetic Sturm implementation independently checks the number of real roots of p_m for $1 \leq m \leq 20$. Its polynomial remainders use exact fractions, normalized only by positive factors. The variations at $-\infty$ and $+\infty$ are stored alongside every polynomial. These checks agree with the theorem, but the theorem does not extrapolate from them: (16.1) and

the two sign-change arguments prove the count for all m . The small example's exact Laguerre defect is also recomputed.

22.3 All 8,408,566 points of the larger example

The C++17 program `direct_counterexample.cpp` is a third route, not a translation of the polynomial transform. For each lower-zero set Z , each upper coordinate has its own optional private unit, and there is one optional unit for each element of Z , assignable only to an adjacent upper coordinate. The set of resulting upper-load vectors is deduplicated.

This is exact because the upper-demand inequalities

$$b(T) \leq |T| + |N(T) \cap Z|$$

are precisely Hall's conditions for assigning demand clones to the upper private resources and the lower resources. Integral demand vectors are therefore realizable by these allocations if and only if they satisfy the original ideal inequalities. Counting the positive coordinates and the fixed lower support gives h_p directly.

Each of the eight upper coordinates is encoded in three bits. Its maximum load is at most its degree plus one, at most seven in this graph, so the encoding is injective and never overflows a digit. A generation array deduplicates states without hash-dependent behavior. The resulting coefficient vector agrees exactly with (17.5), and its sum is 8 408 566. The source is included; a precompiled executable is not.

Certificate	Recorded verification
All labeled 3-by-3 graphs	512 direct support-polynomial and demand-count comparisons.
Additional 4-by-3 graphs	24 direct comparisons with a recorded seed.
Crown graphs	Formula checks for $2 \leq r \leq 8$; direct lattice checks for $r \leq 4$.
Cactus matrix certificate	936 exact square-minor norm comparisons.
Theta family	Support enumeration through $m = 9$, direct lattice enumeration through $m = 4$, exact Sturm counts through $m = 20$.
17-element example	Two graph-polynomial algorithms and a separate direct enumeration of all 8 408 566 lattice points.

All the listed checks passed in the supplied run. None is a Lean proof or a substitute for independent mathematical review.

22.4 Build commands

The manuscript's own build commands, run from its package directory, were `python3` on `code/verify.py` and then on `code/verify_theta.py`, the compiled C++ program, and `latexmk` for the PDF. In this report every delivered file name carries the prefix `02-height-two-`:

- programs: `code/02-height-two-verify.py`, `code/02-height-two-verify_theta.py`, `code/02-height-two-direct_counterexample.cpp` and `code/02-height-two-Makefile`;
- recorded outputs: `data/02-height-two-verification.json`, `data/02-height-two-theta_verification.json`, `data/02-height-two-direct_counterexample.json` and `data/02-height-two-board_certificate.csv`.

The programs still use the delivered names internally: `verify_theta.py` imports a module `verify`, which in this directory would be Part I's verifier, and the Makefile's `verify` target would run Part I's verifier and overwrite Part I's data. Therefore rerun them only in a scratch copy with the delivered names restored, from this report's directory:

```
W=/path/to/scratch; mkdir -p "$W/02/code" "$W/02/data"
for f in code/02-height-two-*; do cp "$f" "$W/02/code/${f#code/02-height-two-}";
done
for f in data/02-height-two-*; do cp "$f" "$W/02/data/${f#data/02-height-two-}";
done
mv "$W/02/code/Makefile" "$W/02/Makefile"; cd "$W/02"
python3 code/verify.py
python3 code/verify_theta.py
c++ -O3 -std=c++17 -Wall -Wextra -pedantic code/direct_counterexample.cpp \
-o direct_counterexample
./direct_counterexample > data/direct_counterexample.json
```

The Python programs require Python 3.10 or later and only its standard library. The C++ program needs a C++17 compiler. This article is rebuilt as described in the report's README; its preamble adds TikZ to Part I's packages. No external image or font files are required.

23 Proof dependencies and a formalization plan

The main route has a short dependency structure:

$$\begin{aligned} &\text{two published root-polytope inputs} \implies \text{counting lemma} \\ &\implies \text{height-two formula} \implies \text{theta root distribution.} \end{aligned}$$

The canonical root identity is proved separately by orthants. The cactus minor proof and the crown circuit proof are independent of one another. The crown toric result depends only on an explicit exposed lattice face, not on the general root-polytope theorem of Part I.

The following checks are particularly important in an independent audit. First, all matching counts concern *covered vertex sets*; confusing these with individual matchings destroys both the theta and crown formulas. Second, a height-two poset here has singleton elements, not nontrivial preorder-equivalence blocks. Third, h_P counts support sizes of lattice points; it is not initially the Ehrhart numerator of $Q(P)$. The identity with an Ehrhart numerator concerns the different polytope C_P . Fourth, a nonflag triangulation obstruction for C_P does not preclude an unrelated flag polytope having the same h -vector.

A staged formalization could begin with finite graphs on two finite types, covered-shore pairs, and the elementary theta-support classification. The polynomial identity (13.4), the positive-interval rearrangement, and the Descartes/interval root count then form a compact algebraic project once the required real-polynomial lemmas are available. This would verify the entire graph-family theorem without any polytope formalization.

Transferring that theorem to preorder support polynomials requires either formalizing Lemma 14.1 and its classical inputs, or finding a direct combinatorial bijection. A finite-only route to the eight-element counterexample is different: certify its bounded ideal-inequality enumeration, then verify its polynomial and negative Laguerre value. That route need not formalize the general graph-to-polytope bridge, but it also does not establish the all-height-two theorem.

For the cactus result, exact sixth-root arithmetic can be implemented as a quadratic extension with relation $z^2 - z + 1 = 0$. The combinatorial symmetric-difference lemma is a natural

intermediate target. The analytic stability proof can then be isolated as a general Hermitian determinant lemma. For the crown obstruction, the finite affine dependence and the two-monomial fiber give a clean algebraic certificate before tackling full triangulation definitions. No uncompiled Lean source is presented as if it had been checked.

24 Further research questions and topics

The results replace an undifferentiated real-rootedness question with several sharper problems. The following questions are proposed directions, not assertions that they have been settled by this article. Except for explicitly attributed conjectures, they are directions arising from these proofs, not claims that no answer exists anywhere in the literature.

Research question 1 (Extend the matching-support interpretation). Does the γ -vector of a larger class of posets admit a similarly concrete nonnegative interpretation? Height three and preorders with nontrivial equivalence classes are natural next cases. Determine exactly which step of the lower-zero-set argument survives weighted blocks.

A successful extension must account for the fact that a coordinate in a nontrivial equivalence block need not be individually bounded by one. Simply replacing the preorder by its quotient poset loses information. [Added 29 September 2026, batch 44: answered in Part IV. For every finite preorder, γ_k counts ordered pairs of disjoint k -element sets joined by a τ -increasing bijection (Theorem 48.2); in height two these are Part II’s matchable vertex sets (Section 55.3). The step that survives is the zero-set partition, taken over all zero coordinates rather than a minimal shore (Lemma 50.1), and it allows coordinates larger than one. Positive block formulas are Theorems 54.1 and 54.3. See Section 46.3.]

Research question 2 (A purely combinatorial bridge). Construct an explicit bijection between $\mathcal{R}_G$ and perfectly matchable vertex subsets, or an equally elementary proof of their equal cardinality. Can the choice be made compatible with induced subgraphs, vertex deletion, or positive weights?

Such a construction would remove the two root-polytope inputs from the height-two theorem and substantially reduce its formalization burden. The hypertree-vector bijection in Section 14 does not itself solve this problem. [Added 29 September 2026, batch 44: still open. Part IV’s general theorem uses the same counting lemma and constructs no such bijection; its Research question 20 asks for one that also preserves the receiver support.]

Research question 3 (Which graph structures preserve real-rootedness?). Characterize useful hereditary classes of bipartite graphs for which p_G is real-rooted. Cacti are a positive class, while the treewidth-two family Θ_m shows that series-parallel structure alone is inadequate. What happens under vertex sums, edge subdivisions, or controlled cycle intersections?

The precise boundary between one shared vertex and a shared path is worth studying. The present examples do not imply that every noncactus graph fails real-rootedness. [Added 29 September 2026, batch 42: indeed not every noncactus fails. Part III shows that $K_{2,3}$ and every graph obtained from it by attaching trees have a real stable Φ_G , hence a real-rooted p_G (Corollary 37.4, Section 39). The question itself remains open; see Section 26.3.]

Research question 4 (Beyond the sixth-root cactus matrices). Which bipartite graphs admit a complex matrix supported on their edges whose square-minor norms are exactly the matchability indicators? Is this condition equivalent to a useful multivariate stability property, or strictly stronger?

A positive answer for a class immediately gives positive-weight real-rootedness through Cauchy–Binet. The theta counterexamples show that such representations cannot exist for all bipartite graphs. [Added 29 September 2026, batch 42: answered in Part III. Exactly the cacti admit such a matrix (Theorem 29.1), and the condition is strictly stronger than real stability of Φ_G : $K_{2,3}$ and all its tree attachments are real stable without it (Theorem 37.1, Corollary 37.4). See Section 26.3.]

Research question 5 (Asymptotic zero geometry for Θ_m). The real-root count is exact, but how are the nonreal roots distributed as $m \rightarrow \infty$? Determine their limiting curves and scaling regimes, including the roots near zero and the large negative real root. What is the corresponding distribution after $u = t/(1+t)^2$?

Formula (13.4) gives a concrete exponential-versus-polynomial equation from which such asymptotics may be developed; the present paper does not assert a limiting measure.

Research question 6 (Flag triangulations of the canonical model). For height-two posets, does chordal bipartiteness suffice for D_G to admit a flag unimodular triangulation? More generally, characterize which induced configurations obstruct flagness beyond the crown circuits.

The necessity of chordal bipartiteness follows here. Sufficiency is a separate problem for D_G , even if related statements hold for other signed root configurations.

Research question 7 (Existential flag realizations). Can the crown polynomial h_{P_r} be realized as the h -polynomial of a flag simplicial polytope by a construction unrelated to lattice triangulations of C_{P_r} ? Can one do this for all height-two posets?

This is the distinction needed to make progress on AC Conjecture 5.1(d), rather than only testing one canonical construction.

Research question 8 (Exact toric degree bounds). How much of the largest indispensable degree of $I_{C_P \cap \mathbb{Z}^n}$ is controlled by induced crowns? Is there an upper bound in terms of the largest such crown, or do other faces produce independent unbounded obstructions?

The theorem supplies a lower bound, not a characterization. Additional coordinate roots may change global toric relations while leaving the exposed-face obstruction intact.

Research question 9 (Minimum counterexamples with checkable completeness). Give a fully reproducible exhaustive certificate for the smallest counterexample to preorder support real-rootedness, including nontrivial preorder blocks. The eight-element example is explicit, while Athanasiadis–Chapoton report successful tests through size seven [1]; a complete independent certificate for that lower bound is not supplied here.

A smaller search space restricted to height-two posets may be practical for identifying every minimal forbidden example and testing structural conjectures about their bipartite graphs.

Research question 10 (Efficient exact enumeration and formal certificates). Develop a tree-decomposition algorithm for matchable *vertex-set* counts that avoids counting matching multiplicities. Can it output a small certificate that a proof assistant can check independently of the search procedure?

The theta family gives a useful stress test: treewidth is small, yet root behavior is not controlled by real-rootedness. Enumeration and spectral regularity should therefore be treated as distinct algorithmic issues.

25 Conclusion

The height-two identity identifies the γ -coefficients of preorder support polynomials with a classical graph invariant. Its value is not merely that it reproduces an isolated counterexample: it exposes a complete root-distribution family, a determinantal positive class, and a sharp distinction between root behavior and canonical triangulations.

The negative examples do not destroy γ -positivity in height two; that property is guaranteed by the matching-support formula. Likewise, real-rooted crown polynomials do not rescue flag lattice triangulations of the canonical root polytope. The remaining questions therefore have different mathematical mechanisms and require different constructions. The accompanying proofs and exact certificates are intended to make those mechanisms, and the limits of the novelty claims, available for direct independent review.

Part III. Cactus rigidity: exact square-minor models and stability

Part III was added on 29 September 2026 in batch 42 of ProveIt's incoming-report intake, from one manuscript written against Parts I and II as printed above. It answers Part II's Research question 4, "Beyond the sixth-root cactus matrices": the bipartite graphs that admit a complex matrix on their edges whose square-minor norms are exactly the matchability indicators are precisely the cacti, and this property is strictly stronger than real stability of the signed matching-support polynomial. Parts I and II are unchanged except for dated pointers to this addition. Section 26 records the provenance, fixes the notation and states what is answered.

26 Provenance, conventions and what Part III answers

26.1 One manuscript, one addition

Part III is batch-42 manuscript 05, *Cactus Rigidity for Matching-Support Determinants: Exact classification, a sharp approximation gap, and stability beyond vertex determinants*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated 29 September 2026 (24 US-Letter pages as delivered). It arrived as the archive `ProveIt_Cactus_Rigidity.zip` (inner directory `ProveIt_Cactus_Rigidity/`) in ProveIt commit 8315d24e3 and was placed in 3609d0473. It is pinned to ProveIt commit 9754e8360 (the full hash is printed in Section 27) and records the Git blob 8739a224fa8956adab5d8df9f2d5012a6a220959 of this report's `article.tex`. That blob is Parts I and II exactly as printed above: the file was last changed by the batch-39 write 6a354f52f and is unchanged at the pin and at the placement commit. The manuscript therefore cites Part II in its printed numbering; its "Section 19", "Section 24" and "Theorem 13.2" are Section 19, Section 24 and Theorem 13.2, and are replaced here by cross-references. The manuscript's program, recorded outputs, build file and notes are shipped beside Part I's files with the prefix 03-cactus- (Part I's unprefixed files count as 01, Part II's are 02-); Section 40.3 lists them. Its manuscript, delivery README and PDF are not shipped and survive in the arrival commit; its checksum list `SHA256SUMS.txt` was verified (10/10) at placement and retired.

The manuscript is printed here as Sections 27 to 45: its Section k is Section $k + 26$ here, its Appendices A and B are Sections 44 and 45, and its theorem numbers shift accordingly (its Theorem 3.1 is Theorem 29.1); its equations, numbered (1)–(26) consecutively there, are numbered by section here, as in the rest of this report. Its nine research questions continue Part II's numbering as Research questions 11–19. Every result, proof, remark, limitation, question and non-claim of the manuscript is printed; three displays that repeat Part II's formulas are printed once, in Part II (Section 26.4). Its title-page abstract and status note are reproduced as delivered.

Abstract of manuscript 05 (as delivered). For a finite bipartite graph, can one assign unit complex numbers to its edges so that every square minor has squared absolute value equal to the indicator that its vertex support is perfectly matchable? The ProveIt preorder-polytope report constructs such assignments for cactus graphs and explicitly asks for the exact boundary of the construction. We prove that the boundary is precisely the cactus class. Equivalently, the matching-support polynomial has a Hermitian vertex-diagonal determinant representation exactly for cacti; equivalently, the graph is the support pattern of a matrix over $\mathbb{F}_3$ with no vanishing structurally nonzero minor. A cactus of cycle rank β has exactly 2^β phase-gauge classes of complex certificates.

The obstruction is quantitative. Every unit-phase matrix on every noncactus has a matchable minor whose squared absolute value differs from one by at least

$$\varepsilon_* = \frac{\sqrt{17} - 3}{2},$$

and this constant is attained on $K_{2,3}$. We obtain the classification and the uniform gap by reducing induced theta graphs to three explicit matrices. In contrast, the five-variable matching-support polynomial of $K_{2,3}$ is real stable, as certified by three exact sums-of-squares identities. Attaching arbitrary trees produces infinitely many connected stable examples without a Hermitian vertex determinant. We also prove that bounded-order minor tests cannot recognize exact representability, even on subcubic graphs of treewidth two.

Status (manuscript 05, as delivered). Complete ordinary mathematical proofs are supplied for the stated results. This AI-assisted manuscript is unrefereed and has not been verified in Lean or another proof assistant. The cactus construction and its established real-rootedness consequences are prior work, not claimed here. The classification answers a specified repository question; worldwide priority for its formulations and refinements is not asserted.

Priority and prior work (from the manuscript’s status and source notes). The cactus sixth-root construction, its determinant factorization and cactus stability are Part II’s (Section 19); univariate cactus real-rootedness is Ohsugi–Tsuchiya’s [9, Theorem 7.2]; the height-two matching-support transform is Part II’s Theorem 13.2. The binary (forest) observation and the general phenomenon of stability without a determinant representation are not claimed as new; the latter is established elsewhere, for example for Vámos-type polynomials [15]. A related earlier characterization of binary fundamental transversal matroids [16] was located (publisher abstract only; its full text was not available); possible connections to older fundamental-transversal-matroid characterizations deserve further literature review. The source audit was targeted, not exhaustive, and does not establish the absence of an older equivalent theorem; no worldwide publication priority is claimed for the classification, the finite-field reformulations, the gauge count, the quantitative refinements or the stability formulations. The only specialized external theorem used is Brändén’s multiaffine Rayleigh criterion [12, Theorem 5.6], for the stability of $K_{2,3}$.

26.2 Notation of Part III

Part III uses Part II’s graph conventions: $G = (A \sqcup B, E)$ is a finite simple bipartite graph with $p = |A|$, $q = |B|$, $n = p + q$; a cactus is a graph in which every edge lies on at most one simple cycle; p_G and $\mathcal{M}_k(G)$ count perfectly matchable *vertex sets*; $\zeta = e^{\pi i/3}$; ν is the matching number and λ_j are the positive eigenvalues of KK^* . Table 2 lists the symbols Part III adds or uses differently. One symbol of the manuscript was renamed: its theta graph $\Theta(a, b, c)$ with three paths of lengths a, b, c is written $\vartheta(a, b, c)$ here, because Part II’s Θ_m is a different family with m paths. No normalization was changed.

Table 2: Symbols of Part III. The last column prints the tempting false reading, usually a Part I or Part II symbol with the same letter.

Symbol	Meaning in Part III	Not to be confused with
$\vartheta(a, b, c)$	a theta: three internally disjoint paths of lengths a, b, c between two common endpoints; one length may be 1 (renamed from the manuscript's $\Theta(a, b, c)$)	Part II's Θ_m , which has m paths, all of length three. Only $\Theta_3 = \vartheta(3, 3, 3) = T$ is both; Θ_m for $m \neq 3$ is no $\vartheta(a, b, c)$. Part II's "theta family" means Θ_m .
E, D, T	the three cores $\vartheta(2, 2, 2) = K_{2,3}$, $\vartheta(1, 3, 3)$, $\vartheta(3, 3, 3)$	the edge set E ; Part II's polytope D_G and its positive part T_G .
$\Phi_G(\mathbf{z})$	the signed complementary-support polynomial (28.2), defined for <i>every</i> bipartite G	Part II defines $\Phi_G := \det(\text{diag}(\mathbf{z}) - H_G)$ for cacti; the two agree there (19.4). For a noncactus, Φ_G is <i>no</i> such determinant (Theorem 29.1).
$\Phi_G(s, \dots, s)$	$s^n p_G(-s^{-2})$, for every G	Part II's $q_G(s)$ of (19.5), stated there for cacti only.
K , faithful, H , H_K	a phase matrix; faithful if (28.3) holds; a Hermitian vertex-indexed matrix; the bipartite Hermitian matrix of K	Part II's K is <i>one</i> faithful certificate (choice $(-1)^r \zeta$ on every cycle) and its H_G is H_K for it; Part II's bipartite graph $H = (X \sqcup Y, E)$; Part I's triangulation K . Part II's D_G ; Part I's neighbourhoods D_i .
D_A, D_B , gauge	diagonal unitary row and column scalings	
T	a spanning forest	Part II's T_G ; Part I's τ .
$\beta(G), c(G)$	cycle rank $ E - A - B + c(G)$; number of components	—
$\delta_G(K), d(G), \varepsilon_*, d_0$	minor defect of K , its minimum over phases, the constant $(\sqrt{17} - 3)/2$, the constant $(\sqrt{5} - 1)/2$	the vertex d of Θ_m ; Part I's diagonal vertices d_i .
p, q, r in Lemma 36.6; q in Proposition 35.3	negated real inner products of unit vectors; a field size ($\mathbb{F}_q$)	the shore sizes $p = A $, $q = B $.
I, J	a row set and a column set of a square minor	Part I's order ideals $I \in \mathcal{J}(\tau)$; Part II's independent sets.
U, V, W in (37.2)	the monomials xy, xz, yz	W as a perfectly matchable set elsewhere.
L in Theorem 38.1	an odd path length	Part I's shore index i_L .

26.3 What Part III answers in Part II

Status after Part III, as of 29 September 2026:

- **Research question 4 (“Beyond the sixth-root cactus matrices”) is answered.** A bipartite graph has a complex matrix on its edges whose square-minor norms are exactly the matchability indicators if and only if it is a cactus; equivalently, Φ_G is a Hermitian vertex-diagonal determinant, or G carries a support-exact matrix over $\mathbb{F}_3$ (Theorem 29.1). Part II’s sixth-root matrices are therefore all there is: every such matrix is gauge-equivalent to one of $2^{\beta(G)}$ sixth-root matrices (Theorem 30.3). The condition is **strictly stronger** than real stability of Φ_G , the multivariate property Part II proves for cacti (Theorem 19.3): $\Phi_{K_{2,3}}$ and Φ_G for every tree attachment to $K_{2,3}$ are real stable with no such matrix (Theorem 37.1, Corollary 37.4). The obstruction is quantitative, with the sharp constant ε_* (Theorem 36.3). Graphs with real stable Φ_G are not classified (Research question 11 below).
- **Part II’s remark after that question** (“the theta counterexamples show that such representations cannot exist for all bipartite graphs”) is consistent with Part III and gives a second, non-quantitative route for one core. *Merge note, not in manuscript 05:* a faithful matrix gives $p_G(u) = \det(I + uKK^*)$ (Section 39), which is real-rooted, whereas $h_{p_{\Theta_3}}$, and hence p_{Θ_3} (Proposition 15.1), is not real-rooted (Theorem 13.3); so $T = \Theta_3$ has no faithful matrix. Every Θ_m with $m \geq 3$ contains Θ_3 as an induced subgraph. Part III’s Lemma 36.6 gives the quantitative form, and Lemma 31.1 explains why some core occurs in every noncactus.
- **Research question 3 (which structures preserve real-rootedness) is re-scoped, not answered.** Its remark notes that the examples of Part II do not imply that every noncactus fails real-rootedness. Part III supplies such noncacti: $K_{2,3}$, with $p_{K_{2,3}}(u) = 1 + 6u + 3u^2$, and every connected graph obtained from it by attaching trees has a real stable Φ_G and hence a real-rooted p_G (Section 39). The question of useful hereditary classes stays open.
- Research question 1 (a γ -interpretation beyond height two) remains open; Research question 18 below proposes stable signed-support methods as one route. [Added 29 September 2026, batch 44: Research question 1 is answered in Part IV, by transport pairs rather than stability (Theorem 48.2); see Section 46.3.]

Part III refutes no statement of Parts I and II, and nothing in them is retracted.

26.4 Where the merge had to choose

- *Placement.* Part III follows Part II and precedes Part I’s appendices, whose letters and numbers are unchanged. The manuscript’s own Appendices A and B are printed as the ordinary Sections 44 and 45 of Part III.
- *Self-citation.* The manuscript cites this report as an external repository report (its reference “Repo”, with the pinned commit, Sections 19 and 24 and Theorem 13.2). Here those citations are replaced by references to Part II, and “the repository(’s)” by “Part II(’s)” where it means this report in running text; two headings, in Sections 27 and 39, keep the manuscript’s wording. The pin itself is kept in Section 27.
- *Duplicates.* The manuscript’s cactus factorization (Lemma 30.1) is Part II’s Lemma 19.1, by the same route; it is printed as the manuscript states it because it displays both admissible phases on each cycle, which Theorem 30.3 needs. The manuscript’s height-two formula, Cauchy–Binet identity and two spectral factorizations (its displays (23)–(26), in its Section 13) are Theorem 13.2 and Theorem 19.2; they are printed once, in Part II, and cited in Section 39. The

manuscript’s recall of the cactus stability argument (Section 37) is the proof of Theorem 19.3 and is kept as a short recall.

- *Notation.* $\Theta(a, b, c)$ is renamed $\vartheta(a, b, c)$ (Table 2); nothing else is renamed.
- *File names.* The delivered names (`code/verify.py`, `data/verification.json`, the `Makefile`) are replaced by the shipped names in Section 40, because `code/verify.py` in this directory is Part I’s verifier; the build commands are replaced by commands that run on a copy.
- *Bibliography.* The manuscript’s entries for Ohsugi–Tsuchiya’s matching-generating-polynomial paper and for Davis–Kohl are Part II’s references [9] and [7]; its five other external entries are appended.
- *Labels and front matter.* Labels carry the prefix `cac:`. The conclusion and the dependency ledger received labels, and Part II’s Research question 4 received the label `cac:q:beyond` so that Part III can cite it. The manuscript’s title page, abstract and status note are printed in this section; its table of contents is dropped.

27 The question, its provenance, and the answer

27.1 The repository question being resolved

The starting point is Part II of this report, *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, whose height-two extension is entitled *Height-Two Preorder Polynomials: Matching supports, nonreal-root families, and crown obstructions*. Section 19 of Part II constructs a sixth-root-of-unity matrix for each bipartite cactus. Its Section 24 asks, under the heading “Beyond the sixth-root cactus matrices” (Research question 4), which bipartite graphs admit a matrix whose square-minor norms reproduce the matchability indicators, and whether this requirement is equivalent to multivariate stability.

This article answers both parts, with the precise meaning of the representation fixed below:

$$\boxed{\text{Exact minor representation} \iff \text{cactus graph} \implies \text{real stability},}$$

where the last implication is strict. The necessity direction is the main extension of Part II’s result. In particular, adding arbitrary complex phases does not extend the exact model to a larger graph class than the existing sixth-root construction.

The source is pinned to commit

9754e83603e223811b7515eb892f22c847144593.

The relevant `article.tex` has Git blob

8739a224fa8956adab5d8df9f2d5012a6a220959.

The main branch may subsequently change. All references to the source’s sections concern this pinned version, not an unspecified current checkout. [Merge note: that blob is this report’s `article.tex` before Part III was added, so the cited sections are those of Part II above; see Section 26.1.]

27.2 Why exact support matters

For an ordinary matrix, a determinant is a signed sum over perfect matchings. The invariant studied here counts a vertex set once whenever at least one perfect matching exists. It must not count the number of such matchings. These two tasks agree on forests but diverge on cycles. Phases can correct the overcount on isolated cycles; the question is whether they can do so consistently when cycles overlap.

This is a concrete obstruction problem with implications for determinant proofs of stability, weighted real-rootedness, and exact certificate design. It does not settle general γ -positivity or flag realization questions for preorders. Nor does it characterize all graphs with real-rooted univariate support polynomials.

27.3 A map of the contributions

The results are organized around four statements. Theorem 29.1 gives the exact graph classification, including its Hermitian and ternary versions. Theorem 30.3 classifies all complex certificates up to diagonal phase changes. Theorem 36.3 gives the sharp universal separation from noncacti. Theorem 37.1 and Corollary 37.4 separate real stability from the exact vertex determinant property. Theorem 38.1 shows why testing a fixed collection of minor orders cannot substitute for the structural theorem.

The proofs of the classification and approximation gap use only finite graph arguments, determinants, and elementary geometry of unit complex numbers. The stability example uses Brändén's established multiaffine Rayleigh criterion [12, Theorem 5.6]. This is the only specialized external theorem needed for the new stability conclusion.

28 Definitions and distinctions

Throughout, $G = (A \sqcup B, E)$ is a finite simple bipartite graph with a specified bipartition. Write $p = |A|$, $q = |B|$, $n = p + q$, and let $c(G)$ count connected components, including isolated vertices. Its cycle rank is

$$\beta(G) = |E| - |A| - |B| + c(G).$$

Empty shores and isolated vertices cause no difficulty. A cactus is a graph in which every edge belongs to at most one simple cycle. Equivalently, every nontrivial block is an edge or a cycle. We allow disconnected cacti.

Definition 28.1 (Matching supports). A vertex set $W \subseteq A \sqcup B$ is *perfectly matchable* when the induced graph $G[W]$ has a perfect matching. The empty set is perfectly matchable. Let $\mathcal{M}_k(G)$ be the set of such W with $|W| = 2k$. Define

$$p_G(u) = \sum_{k \geq 0} |\mathcal{M}_k(G)| u^k, \quad (28.1)$$

$$\Phi_G(\mathbf{z}) = \sum_{k \geq 0} \sum_{W \in \mathcal{M}_k(G)} (-1)^k \prod_{v \notin W} z_v. \quad (28.2)$$

The polynomial p_G is a classical perfectly matchable set polynomial; relevant prior work includes Ohsugi–Tsuchiya and Davis–Kohl [9, 7]. Our symbol Φ_G agrees with the signed multivariate refinement used in Section 19 of Part II, equation (19.4), on the cacti treated there; here it is defined for every bipartite graph. Every monomial in (28.2) has coefficient 1 or -1 , regardless of the number of matchings on its omitted vertex set.

Definition 28.2 (Faithful phase certificate). A matrix $K \in \mathbb{C}^{A \times B}$ is a *phase matrix* on G if

$$K_{ab} = 0 \quad (ab \notin E), \quad |K_{ab}| = 1 \quad (ab \in E).$$

It is *faithful* if, for every $I \subseteq A, J \subseteq B$ with $|I| = |J|$,

$$|\det K_{I,J}|^2 = \mathbf{1}\{G[I \sqcup J] \text{ has a perfect matching}\}. \quad (28.3)$$

The empty determinant is one.

There is no extra restriction in calling the edge entries phases: the one-by-one instances of (28.3) force precisely these moduli and this zero pattern. A nonmatchable square submatrix has determinant zero for every choice of entries on its pattern, since every permutation product contains a zero.

Definition 28.3 (Support-exact matrices over a field). For a field F , a matrix $L \in F^{A \times B}$ is *support-exact* for G if its nonzero entries occur precisely on E and every square submatrix whose support has a perfect matching is nonsingular.

This is the pattern-sensitive notion often called superregularity: structurally zero minors are permitted, but no other minor may vanish [13]. We use “support-exact” to avoid confusion with the different convention that a superregular matrix has *every* square minor nonzero.

Remark 28.4 (Two weaker conditions are not the same problem). Requiring merely that every nonzero complex minor have modulus one permits matchable minors to vanish. That weaker requirement belongs to the theory of complex unimodular and sixth-root-of-unity matrices [14]; it is not (28.3). Likewise, an univariate identity $p_G(u) = \det(I + uM)$ alone forgets the individual vertex supports and does not imply faithfulness.

29 The classification theorem

Theorem 29.1 (Exact boundary of the determinant model). *For a finite simple bipartite graph G , the following are equivalent:*

- (i) G is a cactus.
- (ii) G has a faithful complex phase certificate.
- (iii) G has a faithful certificate all of whose nonzero entries are sixth roots of unity.
- (iv) There is an $n \times n$ Hermitian matrix H , indexed by $V(G)$, for which

$$\Phi_G(\mathbf{z}) = \det(\text{diag}(z_v : v \in V(G)) - H). \quad (29.1)$$

- (v) G has a support-exact matrix over $\mathbb{F}_3$.

No zero pattern for H is assumed in (iv); the polynomial identity forces the required bipartite pattern.

The implication (i) $\Rightarrow$ (iii) is the construction already present in Part II (Theorem 13.5 and Lemma 19.1). We reproduce its proof in Section 30 to keep the argument self-contained and to prepare the classification of all certificates. Sections 31–33 prove the converse. Section 34 proves the precise Hermitian equivalence, and Section 35 proves the finite-field characterization.

Remark 29.2 (The strength and the limit of the answer). The theorem rules out every Hermitian matrix of the form (29.1), not only the matrix chosen by Part II’s construction. It does *not* rule out arbitrary larger determinantal pencils, substitutions involving auxiliary variables, sums of determinants, or representations of powers of a polynomial. Those are different questions.

30 Cactus certificates and their complete phase freedom

Set $\zeta = e^{\pi i/3}$. Thus $\zeta^2 - \zeta + 1 = 0$ and

$$|1 - \zeta| = |1 - \bar{\zeta}| = 1.$$

Choose a spanning forest T of a cactus. Each cycle has exactly one edge outside T , and no edge belongs to two cycles. Put $K_{ab} = 1$ on the forest edges. On the remaining edge of each cycle of length $2r$, put

$$K_{ab} = (-1)^r \zeta \quad \text{or} \quad (-1)^r \bar{\zeta}, \quad (30.1)$$

independently from cycle to cycle. All nonedges have entry zero.

Lemma 30.1 (The cactus factorization; prior construction). *Every matrix constructed by (30.1) is faithful.*

Proof. Fix a square induced support admitting a perfect matching M . The symmetric difference of M with any other perfect matching consists of vertex-disjoint M -alternating cycles. In a cactus, any two M -alternating cycles are themselves vertex-disjoint: if they met at a vertex, its unique edge of M would have to belong to both cycles, contradicting the cactus condition.

Consequently all perfect matchings on the support are obtained by independently flipping a subset of its M -alternating cycles. A flip on a cycle of length $2r$ changes the permutation sign by $(-1)^{r-1}$. The ratio of the products of edge entries in the two alternating halves is $(-1)^r \zeta$ or $(-1)^r \bar{\zeta}$, depending on the orientation of the comparison. Therefore, after factoring out the unit contribution of M , the determinant is a product of factors $1 - \zeta$ and $1 - \bar{\zeta}$. Its modulus is one. If the support has no perfect matching, every determinant term is zero. $\square$

This is the content of Part II's cactus minor lemma (Lemma 19.1, which uses the choice $(-1)^r \zeta$ on every cycle), now with both allowed choices on each cycle displayed. The proof does not invoke any general matrix representability theorem.

Definition 30.2 (Phase gauge). Two phase matrices on the same labeled bipartite graph are gauge-equivalent if

$$K' = D_A K D_B$$

for diagonal matrices D_A, D_B whose diagonal entries have modulus one. No row permutation, column permutation, pivot, or complex conjugation is included in this equivalence.

Such changes preserve all square-minor moduli. They are also precisely the relevant diagonal unitary changes of the corresponding bipartite Hermitian matrix, after conjugating the column-side factor as appropriate.

Theorem 30.3 (Rigidity and moduli of faithful certificates). *For a cactus G , every faithful certificate is gauge-equivalent to exactly one of the matrices (30.1), relative to a fixed spanning forest. There are exactly $2^{\beta(G)}$ gauge classes. In particular, arbitrary unit complex phases produce no additional classes beyond sixth-root phases.*

Proof. On each tree component, select a root. Choose row and column phases successively away from the root so that every tree edge becomes one. A residual gauge preserving all forest entries uses a common phase on the row vertices of a component and its reciprocal on the column vertices; it changes no edge entry anywhere in that component. Thus normalization gives a unique representative of each gauge class.

A simple cycle in a cactus is induced. In its normalized square submatrix, all entries are one except for the single non-tree entry t , and there are exactly two perfect matchings. Up to a unit sign, its determinant is

$$1 + (-1)^{r-1}t.$$

Faithfulness and $|t| = 1$ imply

$$\operatorname{Re}((-1)^{r-1}t) = -\frac{1}{2}.$$

The only possibilities are $t = (-1)^r \zeta$ and $t = (-1)^r \bar{\zeta}$. This proves necessity of every listed cycle choice. Lemma 30.1 shows that the choices are independent and sufficient. A cactus has $\beta(G)$ cycles, proving the count. $\square$

Corollary 30.4 (When real phases suffice). *A bipartite graph has a faithful real matrix if and only if it is a forest. For a forest, every assignment of signs to the edges is faithful and all assignments are gauge-equivalent.*

Proof. A forest has at most one perfect matching on each vertex support, because the symmetric difference of two would contain a cycle. Hence every nonzero determinant is a signed product of edge entries. Conversely, a faithful real matrix would be a faithful complex matrix and hence have cactus support. On any cycle the required normalized entry (30.1) is nonreal. One can also argue directly on a shortest induced cycle: its two determinant terms are ± 1 , so the determinant is 0 or ± 2 , never of modulus one. $\square$

31 Why every noncactus contains the right obstruction

A *theta* consists of three internally vertex-disjoint paths with common distinct endpoints. Our convention permits one of the paths to have length one; at most one can do so in a simple graph. An *induced theta* has no edges besides those in the three paths. This convention matters: excluding the length-one case would miss two cycles sharing an edge.

Lemma 31.1 (Induced-theta lemma for bipartite graphs). *Every finite noncactus bipartite graph contains an induced theta. The three path lengths have the same parity.*

Proof. A noncactus has a two-connected block B that is not a cycle or an edge. To see the relevant block fact, two cycles sharing an edge lie in a common two-connected subgraph, and a maximal such block is not a single cycle. Choose a shortest cycle C of B . It is induced, and B has vertices outside C ; otherwise an extra edge would be a chord contradicting shortestness.

First suppose that a vertex $v \notin C$ has at least two neighbors on C . If it has exactly two, $C \cup \{v\}$ is an induced theta. If it has at least three, take three consecutive neighbors x, y, z in their cyclic order. Retain v and the arc of C from x through y to z having no further neighbors of v in its interior. The three paths from v to y are the edge vy and the two paths through x and z . The arc has no chord, and its endpoints x, z are in the same bipartition class, so there is no extra edge xz . This is again an induced theta.

We may therefore assume that each vertex outside C has at most one neighbor on C . Some component of $B - V(C)$ has at least two distinct attachment vertices on C : if it had only one, that vertex would disconnect B . Choose a shortest path with distinct endpoints on C and all internal vertices outside C . It has no chord. It also has no additional edge from an internal vertex to C . Such an edge would produce a shorter path of the same kind to one of the two endpoints; at the first or last internal vertex, the assumed uniqueness of its neighbor on C excludes the only exceptional adjacency. The union of this path with C is therefore an induced theta.

In a bipartite graph, every path between two fixed vertices has parity determined by the shores of its endpoints. Hence all three lengths have the same parity. $\square$

The lemma is tailored to induced subgraphs because deleting arbitrary edges from a phase matrix need not preserve any minor identity. Our proof never makes that invalid inference. The only graph reductions used below are restriction to an induced subgraph and a precisely controlled path replacement.

32 A path-compression identity that preserves all supports

Consider a path

$$a - v - u - b, \quad a, u \in A, \quad v, b \in B,$$

whose internal vertices v, u have degree two. Assume that a and b are not adjacent. Delete u, v and replace the three-edge path by the edge ab . For a matrix supported on the original graph, define the new entry by

$$K'_{ab} = -\frac{K_{av}K_{ub}}{K_{uv}}, \quad (32.1)$$

and leave the other surviving entries unchanged. This operation shortens one path by two while preserving bipartiteness.

Lemma 32.1 (Compression preserves support and determinant magnitude). *Suppose $K_{uv} \neq 0$. For every equally sized pair I, J of surviving row and column sets,*

$$\det K_{I \cup \{u\}, J \cup \{v\}} = \pm K_{uv} \det K'_{I, J}. \quad (32.2)$$

Moreover, the two induced supports in (32.2) are perfectly matchable simultaneously. If all edge entries of K have modulus one, the same is true for K' , and the squared determinant magnitudes in (32.2) are equal.

Proof. Put row u and column v last. The block determinant identity gives the factor K_{uv} times the determinant of the Schur complement. The column at v has only one other possible nonzero entry, in row a , and the row at u has only one other possible nonzero entry, in column b . Thus the only altered surviving entry is (32.1). Restoring the original order contributes only a sign.

For matchings there is an explicit bijection. A matching on the compressed support not using ab lifts by adding uv . A matching using ab lifts by replacing it with av and ub . Conversely, a matching of the augmented original support either uses uv , or must match v to a and u to b . These alternatives also cover the cases in which I or J omits one of the endpoints. Finally, (32.1) is a quotient of unit complex numbers with a minus sign, hence has modulus one. $\square$

Corollary 32.2 (Preservation of exactness). *Faithful complex certificates and support-exact matrices over any field descend through this compression. Both properties also descend to induced subgraphs.*

Repeated compression of an induced theta produces one of the following three graphs:

Parity and short path	Final path lengths	Name used here
Even	(2, 2, 2)	$E = K_{2,3}$
Odd, a length-one path present	(1, 3, 3)	D
Odd, no length-one path	(3, 3, 3)	T

In the even case shorten only paths of length at least four. In the odd case shorten only paths of length at least five. Consequently no parallel edge is created. At each step the internal vertices have degree two, and the replacement edge does not already exist. These details are why the Schur-complement identity applies throughout.

33 Three finite phase obstructions

We now exclude a faithful matrix for each core. Row and column phase changes normalize a spanning tree to ones. Every remaining parameter below has modulus one.

33.1 The even core $E = K_{2,3}$

A normalized matrix is

$$K_E = \begin{pmatrix} 1 & 1 & 1 \\ 1 & z & w \end{pmatrix}. \quad (33.1)$$

Its three two-by-two supports are all matchable, so faithfulness requires

$$|1 - z| = |1 - w| = |z - w| = 1.$$

The first two equalities imply $z, w \in \{\zeta, \bar{\zeta}\}$. Their distance is then either zero or $\sqrt{3}$, contradicting the last equality. Geometrically, three points on the unit circle cannot have all three chord lengths equal to one.

33.2 Two squares sharing an edge: D

A normalized matrix is

$$K_D = \begin{pmatrix} 1 & 1 & 1 \\ x & 1 & 0 \\ y & 0 & 1 \end{pmatrix}. \quad (33.2)$$

The two cycle minors have determinants $1 - x$ and $1 - y$. They force $x, y \in \{\zeta, \bar{\zeta}\}$. The full determinant is $1 - x - y$, with squared modulus zero if the phases are conjugate and three if they are equal. The full support is matchable, so both possibilities are forbidden.

33.3 Three odd paths: T

A normalized matrix is

$$K_T = \begin{pmatrix} 0 & 1 & 1 & 1 \\ x_1 & 1 & 0 & 0 \\ x_2 & 0 & 1 & 0 \\ x_3 & 0 & 0 & 1 \end{pmatrix}. \quad (33.3)$$

Every pair of arms gives a three-by-three cycle minor with determinant $-(x_i + x_j)$. Faithfulness would imply

$$\operatorname{Re}(x_i \bar{x}_j) = -\frac{1}{2} \quad (i \neq j).$$

But the full determinant is $-(x_1 + x_2 + x_3)$, and these equalities give

$$|x_1 + x_2 + x_3|^2 = 3 + 2 \sum_{i < j} \operatorname{Re}(x_i \bar{x}_j) = 0.$$

Its support has a perfect matching, a contradiction.

Proof of (ii) $\Rightarrow$ (i) in Theorem 29.1. If G were a noncactus with a faithful matrix, Lemma 31.1 would give an induced theta. Restrict the matrix to it, and apply Lemma 32.1 until one of the three cores remains. The resulting faithful matrix contradicts the corresponding calculation above. $\square$

34 The Hermitian representation has no hidden extra freedom

For a phase matrix K , put

$$H_K = \begin{pmatrix} 0 & K \\ K^* & 0 \end{pmatrix}.$$

On a balanced vertex set $W = I \sqcup J$, with $|I| = |J| = k$, its principal determinant is

$$\det(H_K)_W = (-1)^k |\det K_{I,J}|^2. \quad (34.1)$$

On an unbalanced vertex set the principal determinant vanishes: one of the two off-diagonal rectangular blocks has more rows than columns and hence produces a kernel vector.

Proposition 34.1 (Coefficient recovery). *Identity (29.1) holds for a Hermitian H if and only if $H = H_K$ for a faithful phase matrix K , with vertices ordered by the two shores.*

Proof. The diagonal expansion is

$$\det(\text{diag } \mathbf{z} - H) = \sum_{W \subseteq V} (-1)^{|W|} \det H_W \prod_{v \notin W} z_v. \quad (34.2)$$

If K is faithful, (34.1) and the vanishing of unbalanced minors make this exactly (28.2).

Conversely, compare coefficients for omitted singleton sets. They give $H_{vv} = 0$. For an omitted pair $\{v, w\}$, the coefficient is

$$\det H_{\{v,w\}} = -|H_{vw}|^2.$$

The coefficient in Φ_G is -1 for an edge and zero otherwise. Therefore $H_{vw} = 0$ on nonedges and $|H_{vw}| = 1$ on edges. In particular its same-shore blocks vanish, so $H = H_K$. Comparing all remaining balanced coefficients using (34.1) yields faithfulness. $\square$

This proves (ii) $\Leftrightarrow$ (iv). It also explains exactly which information disappears under diagonal specialization. For example, $K_{2,3}$ has

$$p_{K_{2,3}}(u) = 1 + 6u + 3u^2 = (1 + (3 + \sqrt{6})u)(1 + (3 - \sqrt{6})u).$$

Thus its univariate polynomial has a positive-semidefinite two-by-two determinant representation using a diagonal matrix. Nevertheless, there is no representation (29.1) for its five-variable polynomial. A one-variable factorization is not a counterexample to the classification.

35 The same boundary over the smallest fields

Theorem 35.1 (Binary and ternary support patterns). *A finite bipartite graph admits a support-exact matrix over $\mathbb{F}_2$ exactly when it is a forest, and over $\mathbb{F}_3$ exactly when it is a cactus. A cactus has a unique support-exact $\mathbb{F}_3$ gauge class under independent nonzero row and column scalings.*

Proof. Over $\mathbb{F}_2$, all edge entries are one. A shortest cycle is induced, and its two perfect-matching terms sum to zero. Conversely, a forest has at most one perfect matching on any support, so every structurally nonzero determinant is one.

For $\mathbb{F}_3$, first consider a cactus. Normalize a spanning forest to ones. On the remaining edge of a cycle of length $2r$, choose

$$t = (-1)^{r+1} \in \mathbb{F}_3.$$

The flip ratio including its determinant sign is $(-1)^{r-1}t = 1$. The same independent-cycle factorization as in Lemma 30.1 expresses every structurally nonzero determinant as a nonzero matching term times a product of factors $1 + 1 = 2$. It cannot vanish in $\mathbb{F}_3$.

For necessity, induced restriction and path compression are valid over every field. Normalize each of the three cores as above, now with all parameters in $\mathbb{F}_3^* = \{1, -1\}$. In (33.1), three column ratios cannot all be distinct, so some two-by-two minor vanishes. In (33.2), nonvanishing of $1 - x$ and $1 - y$ forces $x = y = -1$, but $1 - x - y = 3 = 0$. In (33.3), nonvanishing of every $x_i + x_j$ forces the three parameters to be equal, but their sum is again zero. No core is support-exact, so the graph must be a cactus.

Finally, on a normalized cactus cycle, $1 + (-1)^{r-1}t$ is nonzero for precisely the choice $t = (-1)^{r+1}$. The spanning-forest gauge is unique as in Theorem 30.3, proving uniqueness. $\square$

Together with the preceding sections, this completes every equivalence in Theorem 29.1.

Remark 35.2 (Reduction of the complex certificate). The ring homomorphism $\mathbb{Z}[\zeta] \rightarrow \mathbb{F}_3$ sending ζ to -1 is well-defined because $(-1)^2 - (-1) + 1 = 3$. It sends both complex cycle choices to the unique ternary choice. This provides another explanation of why the 2^β complex classes collapse to one ternary class. The determinant factorization, rather than just entrywise reduction, ensures that no required minor is lost.

Proposition 35.3 (A first larger-field contrast). *For $m \geq 1$ and a finite field $\mathbb{F}_q$, the complete bipartite graph $K_{2,m}$ admits a support-exact matrix over $\mathbb{F}_q$ if and only if $m \leq q - 1$.*

Proof. Scale the columns so that the first row is all ones. The second row consists of nonzero elements $t_1, \dots, t_m$. The two-by-two determinants are $t_j - t_i$, so support-exactness is equivalent to the t_i being pairwise distinct. There are exactly $q - 1$ available elements. $\square$

This elementary observation shows that the cactus classification is genuinely tied to the ternary field in its finite-field formulation; it is not a field-independent prohibition on overlapping cycles. Broader finite-field classifications are not asserted.

36 A sharp universal approximation gap

Exact equality can be fragile in numerical experiments. Here the obstruction is not merely that some equations cannot hold simultaneously: they cannot all be approximately correct beyond an explicit dimension-independent threshold.

Definition 36.1 (Minor defect). For a phase matrix K on G , let

$$\delta_G(K) = \max_{\substack{I \subseteq A, J \subseteq B, |I|=|J| \\ G[I \sqcup J] \text{ perfectly matchable}}} \left| |\det K_{I,J}|^2 - 1 \right|, \quad d(G) = \min_K \delta_G(K). \quad (36.1)$$

The minimum is over unit phases on all edges. It exists because the phase torus is compact and the maximum is over finitely many continuous functions. The empty minor guarantees a nonempty indexing set, even for an edgeless graph.

Nonmatchable minors always have their exact target value zero and need not be included. The normalization $|K_{ab}| = 1$ is essential: allowing unrestricted edge magnitudes would define a different optimization problem.

Lemma 36.2 (Defect is monotone under the reductions). *Induced restriction and the path compression of Lemma 32.1 cannot increase the defect of a phase matrix.*

Proof. Restriction only removes minors from the maximum. Every minor of a compressed matrix is, by (32.2), equal in squared modulus to an augmented minor of the original matrix, and the two supports are matchable simultaneously. Thus every quantity tested after compression already occurs before it. $\square$

Theorem 36.3 (Sharp universal noncactus gap). *Put*

$$\varepsilon_* = \frac{\sqrt{17} - 3}{2} = 0.5615528128 \dots$$

For every finite bipartite noncactus G and every phase matrix K on it,

$$\delta_G(K) \geq \varepsilon_*. \quad (36.2)$$

The constant is optimal over all noncactus graphs: $d(K_{2,3}) = \varepsilon_$. In particular,*

$$G \text{ cactus} \iff d(G) = 0, \quad \inf_{G \text{ noncactus}} d(G) = \varepsilon_*.$$

The proof occupies the next two subsections. It yields more information than the exact classification: a phase fit with all matchable minor squared moduli in $(1 - \varepsilon_*, 1 + \varepsilon_*)$ certifies that the graph is a cactus. This statement concerns all minors, not a sampled subset.

36.1 The exact optimization on $K_{2,3}$

Proposition 36.4 (Three-chord minimax problem). *For the even core, $d(E) = \varepsilon_*$. A minimizing normalized matrix is*

$$K_* = \begin{pmatrix} 1 & 1 & 1 \\ e^{-i\varphi} & 1 & e^{i\varphi} \end{pmatrix}, \quad \cos \varphi = \frac{\sqrt{17} - 1}{4}. \quad (36.3)$$

Its three nontrivial minor squared moduli are $1 - \varepsilon_$, $1 - \varepsilon_*$, and $1 + \varepsilon_*$.*

Proof. After column normalization, the second-row entries are three unit complex numbers. Every nontrivial determinant is their pairwise difference, so the problem is to approximate one by all three squared chord lengths.

Suppose first that the defect δ is less than one; otherwise the desired lower bound is immediate. All squared chord lengths are then less than two, so all pairwise real inner products are positive. The three points lie in an arc of angular length $\theta < \pi/2$. To justify this, rotate one point to angle zero and represent the other two angles in $(-\pi/2, \pi/2)$. Their difference is in $(-\pi, \pi)$, and positivity of their mutual inner product forces its absolute value below $\pi/2$. Thus the interval between the extreme angles has the asserted length.

Order the points on that arc. At least one of the two successive gaps is at most $\theta/2$. Write $c = \cos(\theta/2)$, so $\sqrt{2}/2 < c \leq 1$. The smallest squared chord length is at most $2 - 2c$, whereas the largest is exactly $2 - 2\cos \theta = 4(1 - c^2)$. It follows that

$$\delta \geq \max\{2c - 1, 3 - 4c^2\}. \quad (36.4)$$

The first expression increases with c and the second decreases. Their unique intersection in the indicated interval occurs at

$$c_* = (\sqrt{17} - 1)/4, \quad 2c_* - 1 = 3 - 4c_*^2 = \varepsilon_*.$$

Therefore (36.4) gives $\delta \geq \varepsilon_*$.

For (36.3), the consecutive gaps are equal to φ . The corresponding squared chord lengths are $2 - 2c_* = 1 - \varepsilon_*$ and $4(1 - c_*^2) = 1 + \varepsilon_*$. Thus the lower bound is attained. The argument also shows that any minimizing triple has equal consecutive gaps, up to rotation, reflection, and permutation of its points. $\square$

36.2 The other two cores are separated by at least $3/5$

Lemma 36.5 (A stronger-than-needed bound for D). *Every phase matrix on D has defect at least $3/5$.*

Proof. Use (33.2) and suppose the defect is less than $3/5$. The two cycle minors imply

$$a = \operatorname{Re} x, \quad b = \operatorname{Re} y \in (1/5, 4/5).$$

Depending on whether the imaginary parts have the same or opposite signs, the squared modulus of the full determinant is

$$g_{\pm}(a, b) = 3 - 2a - 2b + 2ab \pm 2\sqrt{1 - a^2}\sqrt{1 - b^2}.$$

On the closed rectangle $[1/5, 4/5]^2$, g_- is convex in each variable separately and g_+ is concave in each variable separately. Indeed the second derivative with respect to a is respectively

$$\frac{2\sqrt{1 - b^2}}{(1 - a^2)^{3/2}} \quad \text{or its negative,}$$

and the same holds with the variables exchanged. Hence the maximum of g_- and the minimum of g_+ occur at corners. The corner values give

$$g_- \leq \frac{9}{25}, \quad g_+ \geq \frac{9}{5}.$$

For the off-diagonal corners the respective values are $(33 - 12\sqrt{6})/25$ and $(33 + 12\sqrt{6})/25$, so these bounds follow by direct comparison. Both alternatives are outside the interval $(2/5, 8/5)$ required for a full minor of defect less than $3/5$. This is a contradiction. $\square$

Lemma 36.6 (A planar Gram obstruction for T). *Every phase matrix on T has defect at least $3/5$.*

Proof. Use (33.3). Suppose its defect is less than $3/5$ and put

$$p = -\operatorname{Re}(x_1\bar{x}_2), \quad q = -\operatorname{Re}(x_1\bar{x}_3), \quad r = -\operatorname{Re}(x_2\bar{x}_3).$$

The three pair-cycle minors imply $p, q, r \in (1/5, 4/5)$. As three unit vectors in the real plane, the x_i have a singular real Gram matrix. Expanding its determinant yields

$$p^2 + q^2 + r^2 + 2pqr = 1. \tag{36.5}$$

Write $s = p + q + r$. The full minor has squared modulus $3 - 2s$, so its lower bound $3 - 2s > 2/5$ implies $s < 13/10$.

For each $t \in [1/5, 4/5]$, the inequality $(t - 1/5)(t - 4/5) \leq 0$ gives $t^2 \leq t - 4/25$. Also $pqr \leq (s/3)^3$ by the arithmetic–geometric mean inequality. Therefore

$$1 = p^2 + q^2 + r^2 + 2pqr \leq s - \frac{12}{25} + 2(s/3)^3 < \frac{13}{10} - \frac{12}{25} + 2(13/30)^3 = \frac{13267}{13500} < 1,$$

a contradiction. $\square$

Proof of Theorem 36.3. Take an induced theta in G and compress it to a core. The defect cannot increase by Lemma 36.2. On E it is at least ε_* by Proposition 36.4; on the other cores it is at least $3/5 > \varepsilon_*$ by Lemmas 36.5 and 36.6. Formula (36.3) attains ε_* on the noncactus $K_{2,3}$, proving optimality. Cacti have defect zero by Lemma 30.1. $\square$

Remark 36.7 (Why a weaker four-minor bound is not sharp). On T, let $a_{ij} = |x_i + x_j|^2 - 1$ and $b = |x_1 + x_2 + x_3|^2 - 1$. Then $b = a_{12} + a_{13} + a_{23} - 1$, so the triangle inequality alone gives $\delta \geq 1/4$. It loses the fact that the three complex phases live in a real plane. The Gram identity (36.5) restores that constraint and makes the sharp universal theorem possible.

37 Real stability is strictly more general

A nonzero polynomial with real coefficients is *real stable* if it does not vanish when every variable has strictly positive imaginary part. For a cactus certificate, (29.1) immediately gives stability: if $(\text{diag } \mathbf{z} - H)u = 0$ for a nonzero vector u , then

$$0 = \text{Im}(u^*(\text{diag } \mathbf{z} - H)u) = \sum_v \text{Im}(z_v)|u_v|^2 > 0,$$

a contradiction. This determinant argument is already used in Part II, as the proof of Theorem 19.3.

The converse fails on only five vertices. Denote the variables on the two shores of $K_{2,3}$ by a, b and x, y, z . Direct support enumeration gives

$$f = \Phi_{K_{2,3}} = abxyz - (a+b)(xy+xz+yz) + (x+y+z). \quad (37.1)$$

The size-four supports are the three choices of two vertices on the larger shore together with both vertices on the smaller shore, explaining the last three terms.

Theorem 37.1 (A stable polynomial without a Hermitian vertex determinant). *Polynomial (37.1) is real stable, but there is no Hermitian 5×5 matrix H with $f = \det(\text{diag}(a, b, x, y, z) - H)$.*

Proof. Nonrepresentability follows from Theorem 29.1, or directly from the three-chord obstruction (33.1). For stability, use Brändén's criterion: a nonzero real multiaffine polynomial f is real stable if and only if

$$\Delta_{ij}f = (\partial_i f)(\partial_j f) - f \partial_i \partial_j f \geq 0$$

for all real choices of all variables and all distinct pairs [12, Theorem 5.6]. Symmetry leaves only three types of pairs.

For the pair on the smaller shore, put $U = xy$, $V = xz$, $W = yz$. Direct differentiation gives

$$\begin{aligned} \Delta_{a,b}f &= U^2 + V^2 + W^2 + UV + UW + VW \\ &= \frac{1}{2}((U+V+W)^2 + U^2 + V^2 + W^2) \geq 0. \end{aligned} \quad (37.2)$$

For two variables on the larger shore,

$$\begin{aligned} \Delta_{x,y}f &= (a^2 + ab + b^2)z^2 - (a+b)z + 1 \\ &= \left(1 - \frac{a+b}{2}z\right)^2 + \frac{3a^2 + 2ab + 3b^2}{4}z^2 \geq 0. \end{aligned} \quad (37.3)$$

For a cross-shore pair,

$$\begin{aligned} \Delta_{a,x}f &= (byz)^2 - byz(y+z) + y^2 + yz + z^2 \\ &= \left(byz - \frac{y+z}{2}\right)^2 + \frac{3y^2 + 2yz + 3z^2}{4} \geq 0. \end{aligned} \quad (37.4)$$

The remaining quadratic forms are nonnegative because $3u^2 + 2uv + 3v^2 = 2(u^2 + v^2) + (u+v)^2$. The three identities cover all ten pairs by shore symmetries. The criterion proves stability. $\square$

This is not a claim that stability without determinant representations is new as a general phenomenon. Such separations are established in other settings, including work on Vámos-type polynomials [15]. The result here answers Part II's specific question (Research question 4) for its particular matching-support refinement, with a small explicit example and elementary certificates.

37.1 Leaf extension and an infinite connected family

Lemma 37.2 (Exact leaf recurrence). *Let $G + \ell$ be obtained by adding a new leaf ℓ adjacent only to a vertex v . Then*

$$\Phi_{G+\ell} = z_\ell \Phi_G - \Phi_{G-v} = z_\ell \Phi_G - \partial_{z_v} \Phi_G. \quad (37.5)$$

Proof. A matchable support omitting ℓ gives the first term. A support containing ℓ must also contain v , and its perfect matching must use ℓv . Removing these two vertices gives a unique perfectly matchable support in $G - v$, with the opposite sign in (28.2). Finally, differentiation with respect to z_v retains exactly the supports omitting v and removes that variable from their monomials. $\square$

Lemma 37.3 (Stability of the leaf operation). *If a real multiaffine polynomial f is stable, then $wf - \partial_{z_v} f$, where w is a new variable, is stable.*

Proof. Fix all variables other than z_v in the upper half-plane. Write $f = Az_v + B$ at these fixed values. It is not the zero polynomial, since stability would otherwise fail. If $A = 0$, then $f = B \neq 0$ and the claimed expression is $wB \neq 0$. If $A \neq 0$, its root $\rho = -B/A$ has imaginary part at most zero. For z_v in the upper half-plane,

$$\frac{\partial_{z_v} f}{f} = \frac{1}{z_v - \rho}$$

has strictly negative imaginary part. It cannot equal w in the upper half-plane. Multiplying by the nonzero f proves the claim. $\square$

Corollary 37.4 (Infinitely many connected examples). *Every graph obtained from $K_{2,3}$ by attaching finite trees at its vertices has a real stable polynomial Φ_G and has no Hermitian vertex determinant representation (29.1). These examples can be arbitrarily large and connected.*

Proof. Construct the attached trees by successive leaf additions and apply Theorem 37.1 and Lemmas 37.2–37.3. The original $K_{2,3}$ remains an induced subgraph, so the result is still a noncactus. The classification excludes the representation. $\square$

38 No fixed minor order can test exact representability

The classification has a finite structural obstruction mechanism, but not a bounded matrix-size obstruction. Compression may require adjoining many eliminated rows and columns before a core minor becomes a minor of the original matrix.

Theorem 38.1 (Unbounded order of minor witnesses). *For every positive integer r , there is a connected bipartite noncactus G of maximum degree three and treewidth at most two such that every phase matrix on G satisfies (28.3) for every square minor of order at most r . Nevertheless no phase matrix on G satisfies all orders.*

Proof. Choose an odd integer $L > r$, $L \geq 3$, and take three internally disjoint paths of length L between two endpoints. This theta is bipartite, subcubic, connected, and not a cactus. Every simple cycle consists of two of the three paths, so its girth is $2L > 2r$.

A square minor of order at most r uses at most $2r$ vertices. Its induced graph therefore has no cycle. Such a forest has at most one perfect matching; its determinant is zero when there is none and a unit product when there is one, for every phase assignment. Thus all the small-minor tests pass. Theorem 29.1 rules out a faithful matrix globally.

For completeness, a width-two tree decomposition is explicit. Let the common endpoints be s, t and write the internal vertices of path i as $v_{i,1}, \dots, v_{i,L-1}$. Use a central bag $\{s, t\}$. Along path i use the chain of bags

$$\{s, v_{i,1}, v_{i,2}\}, \dots, \{s, v_{i,L-2}, v_{i,L-1}\}, \{s, v_{i,L-1}, t\},$$

attaching its last bag to the central bag. Every edge is covered and the bags containing any given vertex are connected. Each bag has at most three vertices. $\square$

Corollary 38.2 (Failure of uniform low-order numerical testing). *For every r , unit-phase matrices can pass all minor tests through order r exactly while some higher-order matchable minor has defect at least ε_* .*

Proof. Combine Theorem 38.1 with Theorem 36.3. $\square$

Thus the approximation gap does not justify replacing the complete minor maximum by a fixed-order sample. Even zero observed error on all two-by-two, three-by-three, or any other bounded range of minors can be misleading.

39 Consequences for the repository's preorder program

For a height-two poset P_G with comparability graph G on n vertices, Part II establishes the height-two formula $h_{P_G}(t) = (1+t)^n p_G(t/(1+t)^2)$ with a precise lattice-support interpretation (Theorem 13.2). We use this as a prior bridge rather than reprove it or claim its γ -positivity conclusion.

For a faithful matrix, the principal-minor expansion and Cauchy–Binet give $p_G(u) = \det(I + uKK^*)$. Indeed, the coefficient of u^k is the sum of $|\det K_{I,J}|^2$ over all k -element row and column sets, hence counts $\mathcal{M}_k(G)$. If $\lambda_1, \dots, \lambda_v$ are the positive eigenvalues of KK^* , where v is the matching number, the factorizations (19.2) and (19.3) of Part II's Theorem 19.2 follow; that proof uses only faithfulness, so it applies to every certificate of Theorem 30.3. [Merge note: manuscript 05 printed these four formulas again as displays; they are printed once, in Part II.] These cactus formulas and their real-rootedness consequences are old relative to this article; univariate cactus real-rootedness is also contained in Ohsugi–Tsuchiya [9, Theorem 7.2]. Our contribution identifies the exact boundary of this *support-by-support* determinant route.

For the stable noncactus $K_{2,3}$, the diagonal specialization of (37.1) is

$$\Phi_{K_{2,3}}(s, s, s, s, s) = s^5 - 6s^3 + 3s = s(s^2 - (3 + \sqrt{6}))(s^2 - (3 - \sqrt{6})),$$

and Theorem 13.2 gives

$$h_{P_{K_{2,3}}}(t) = 1 + 11t + 31t^2 + 31t^3 + 11t^4 + t^5.$$

The graph is not excluded from real-rootedness; only its exact vertex determinant is excluded. More generally, stable Φ_G implies real-rooted p_G : for $s \neq 0$,

$$\Phi_G(s, \dots, s) = s^n p_G(-s^{-2}).$$

The left-hand side has only real zeros. Any nonzero zero of p_G would otherwise give a nonreal zero on the left after solving $u = -s^{-2}$. Since p_G has positive coefficients and constant term one, its zeros must consequently be negative. This applies to all the tree-attached examples of Corollary 37.4.

There are three logically different classes: faithful support determinants, stable signed support polynomials, and real-rooted univariate support polynomials. We prove the first is properly contained in the second. The second is contained in the third by specialization. We do not assert a classification of the latter two classes or strictness of that second containment.

40 Exact verification and reproducibility

The included verifier, shipped as `code/03-cactus-verify.py` (delivered as `code/verify.py`), uses only the Python standard library and exact arithmetic. Its checks were executed with Python 3.13.5. The full output is saved in `data/03-cactus-verification.json` (delivered as `data/verification.json`). The tests supplement the proofs; in particular, checking sixth-root assignments does not by itself exclude arbitrary complex phases.

40.1 Independent combinatorial and algebraic representations

Graphs with fixed labeled shores are encoded by binary edge masks. Cactus recognition enumerates simple cycles and checks that no edge belongs to two of them. This recognition routine does not use determinants. A separate spanning-forest procedure identifies the free normalized edge entries.

For each support pattern, the verifier enumerates every normalized assignment over $\mathbb{F}_3^*$, or over the six complex sixth roots, as appropriate. Structurally nonzero determinant terms are obtained by permutation expansion. Minors with one matching term automatically have unit magnitude; ambiguous minors are evaluated exactly. Complex values are pairs of integers $a + b\zeta$, with

$$(a + b\zeta)(c + d\zeta) = (ac - bd) + (ad + bc + bd)\zeta, \quad |a + b\zeta|^2 = a^2 + ab + b^2.$$

The canonical cactus certificates are checked again by evaluating *all* square minors, including empty minors. Their expected zero-or-one values come from a separate covered-shore dynamic program that merges different matchings with the same covered vertices. Thus the determinant checker is compared with a support-set enumeration, not with its own permutation count.

Check	Recorded count
All labeled 3-by-3 bipartite graphs	512
Cacti among those graphs	460
Normalized sixth-root assignments tested	5,872
Ambiguous minors evaluated in that search	7,792
All labeled 3-by-4 bipartite graphs	4,096
Cacti among those graphs	3,122
Normalized ternary assignments tested	10,576
Ambiguous minors evaluated in that search	14,717
Canonical cactus minor checks, both graph families	118,470
Compression: augmented-minor norm/support comparisons	100
Rayleigh identity classes checked symbolically	3
Independent leaf-attachment support enumerations	5

All checks passed. In a separate check, each of the three small cores has zero valid assignments among its 36 normalized sixth-root assignments and four normalized ternary assignments. The 3-by-3 cacti consist of 328 forests and 132 one-cycle graphs. The 3-by-4 cacti consist of 1,856 forests, 1,248 one-cycle graphs, and 18 two-cycle graphs. The exhaustive normalized counts agree with 2^β complex classes and one ternary class on every tested cactus, and zero on every tested noncactus.

40.2 Symbolic and compression checks

A small rational multivariate polynomial implementation expands and compares the three Rayleigh identities (37.2)–(37.4); by symmetry they cover all ten pairs. The polynomial (37.1)

is independently reconstructed from matchable supports. Each of the five possible leaf attachments is enumerated directly and compared with (37.5).

The compression checker uses theta path lengths (2, 2, 4), (1, 3, 5), and (3, 3, 5) with deterministic, nonconstant sixth-root phases. For all compressed square minors it compares the augmented original determinant norm and matching-support indicator. Exact arithmetic in $\mathbb{Q}(\sqrt{17})$ also verifies the two algebraic equalities defining the extremal constant. None of these finite checks proves the compactness, geometric minimax, or all-graph statements by extrapolation; those are proved in the text.

40.3 Build and rerun

The manuscript's own commands, run from its package root, were `python3 code/verify.py` followed by two `pdflatex` runs on its `article.tex`; its Makefile offered the same. In this report every delivered file name carries the prefix `03-cactus-`:

- program and build file: `code/03-cactus-verify.py` and `code/03-cactus-Makefile`;
- recorded outputs: `data/03-cactus-verification.json` (the recorded run),
`data/03-cactus-verification_log.txt` (its console output, byte-identical to the JSON)
and `data/03-cactus-build_status.json` (the manuscript's record of its own PDF build);
- notes: `03-cactus-STATUS.md` and `03-cactus-SOURCES.md`.

The program still uses the delivered names. Run in this directory, it would write a new unprefixed `data/verification.json` beside Part I's files; the Makefile's `verify` target would run Part I's `code/verify.py`, which without an output option rewrites Part I's shipped data; and its default target would rebuild this whole article in place. Rerun the program only in a scratch copy with the delivered names restored, from this report's directory:

```
W=/path/to/scratch; mkdir -p "$W/03/code" "$W/03/data"
for f in code/03-cactus-*; do cp "$f" "$W/03/code/${f#code/03-cactus-}"; done
for f in data/03-cactus-*; do cp "$f" "$W/03/data/${f#data/03-cactus-}"; done
mv "$W/03/code/Makefile" "$W/03/Makefile"; cd "$W/03"
python3 code/verify.py # rewrites data/verification.json of the copy
```

The program needs Python 3.10 or later and only its standard library; no network access or repository checkout is needed, and its elapsed-time field differs from run to run. This article is rebuilt as described in the report's README; Part III needs no package beyond those of Parts I and II.

41 Proof audit, priority, and formalization boundary

41.1 The parts most worth independent scrutiny

The decisive combinatorial step is the induced-theta lemma, especially the length-one path case. The decisive algebraic step is (32.2), together with its exact matching bijection. Deleting arbitrary edges or suppressing arbitrary degree-two vertices would not be a valid substitute. In the Hermitian equivalence, the singleton and pair coefficients must be compared before concluding that the matrix has the graph's support. In the quantitative theorem, unit moduli of the pivot entries are essential for preserving the defect.

The sharp constant is not inferred from optimization software. Proposition 36.4 proves the lower bound and constructs an extremizer. The other cores are bounded away by $3/5$, a

strictly larger rational number. The final step uses induced restriction and compression, not a supposition that every noncactus literally contains an induced copy of one of the three *small* cores.

For stability, all ten Rayleigh inequalities are covered by symmetry, and the input theorem applies to nonhomogeneous multiaffine polynomials as well as homogeneous ones. The leaf-preservation proof is independent of a general classification of stability-preserving operators.

41.2 What is and is not claimed to be new

The following separation of status is part of the mathematical result, not a bibliographic afterthought.

Statement	Status in this report
Cactus sixth-root construction and its determinant factorization	Prior result of Part II; reproduced and credited.
Cactus univariate real-rootedness	Prior literature and Part II result; not claimed.
Height-two preorder matching-support transform	Prior result of Part II; used only for consequences.
Cactus necessity, full gauge classification, and ternary support characterization	Proved here as answers/extensions to Part II's Research question 4; absolute priority not established.
Sharp universal defect ε_* and the extremal $K_{2,3}$ phases	Proved here; no global publication-priority claim.
Explicit stable $K_{2,3}$ separation and its tree extensions	Proved here in Part II's signed-support setting; general stability/determinant separation is prior.
Failure of all fixed minor-order tests on subcubic width-two graphs	Proved here as a consequence of high-girth theta supports.

The targeted source audit inspected the pinned repository discussion (Part II), the perfectly matchable set literature [9, 7], Brändén's stability criterion [12], pattern-sensitive superregularity [13], and sixth-root-of-unity matrix terminology [14]. Related fundamental-transversal-matroid literature was also located, including an earlier binary characterization [16]; the present report does not claim novelty for the elementary binary corollary. This audit is not a complete survey of representable transversal matroids or every equivalent formulation of the support-pattern problem. A literature-priority determination would require that broader comparison. The proofs and the resolution of the identified repository question do not depend on a claim that no prior equivalent theorem exists.

41.3 A feasible formalization decomposition

A formal development should begin with finite bipartite graphs on two finite types and supports encoded by pairs of finite subsets. One layer proves that a forest support has at most one perfect matching and that symmetric differences decompose into alternating cycles. A second layer proves the exact compression bijection. A third proves the block determinant identity and the three core obstructions.

The graph-theoretic necessity can then be isolated from analysis. The ternary version needs only finite graph arguments and algebra over $\mathbb{F}_3$. The complex classification additionally needs elementary norm identities and the two allowed unit-circle solutions. The sharp gap adds real inequalities, the compact phase torus, and the planar Gram identity; the existence of the minimizer can be omitted if one first states the theorem for every phase assignment and gives the explicit extremizer.

Finally, the stability example can be represented as exact polynomial identities plus a separately imported or formalized Rayleigh criterion. This separation prevents the external analytic theorem from entering the trust boundary of the cactus classification. No uncompiled Lean files are supplied or described as checked. The Python verifier is executable evidence for finite cases, not a substitute for these formal layers.

42 Further research questions and topics

The original determinant-classification question is answered under the precise hypotheses stated here. The following directions remain outside the proved results. Except where attributed, they are questions generated by this investigation, not assertions that no answer is known anywhere in the literature.

Research question 11 (Classify real-stable signed support polynomials). Which bipartite graphs have real stable Φ_G ? The class contains all cacti and all tree attachments to $K_{2,3}$, but the determinant classification shows that no purely faithful-phase characterization can cover it. Determine a useful structural description, at least for graphs of treewidth two.

The three Rayleigh identities suggest searching for graph decompositions that preserve nonnegative Rayleigh differences without preserving a vertex determinant. A first target is a stable gluing theorem more general than leaf attachment.

Research question 12 (Determine the remaining core optimization exactly). Compute $d(T)$ exactly and classify its minimizing phase configurations. The present bound $d(T) \geq 3/5$ suffices for the sharp universal theorem but is not claimed optimal. Determine how the value changes when individual minors receive prescribed positive penalties.

The constraints reduce to three unit vectors and their rank-two real Gram matrix. This gives a small semialgebraic problem, but a numerical stationary point alone is not an exact certificate.

Research question 13 (Describe graphwise sharp defects). Which noncactus graphs satisfy $d(G) = \varepsilon_*$? Having an induced or compressed $K_{2,3}$ gives a lower bound, not an extension of its optimal phases to every other edge and minor. Characterize when the three-point extremizer can be extended without introducing a larger defect.

This asks for more than the universal infimum, which Theorem 36.3 already determines. Additional cycles can impose new norm constraints not visible in one chosen theta.

Research question 14 (Larger finite fields and prescribed support). For $q \geq 4$, characterize useful classes of bipartite support patterns admitting a support-exact $\mathbb{F}_q$ matrix. Proposition 35.3 gives a first field-size obstruction. Can the general question be expressed by a manageable family of induced-and-compressed obstructions?

Any proposed characterization must distinguish a fixed zero pattern from unrestricted representability of an associated abstract matroid. Changing the presentation can change the support question.

Research question 15 (Larger determinantal models). For the stable polynomial $\Phi_{K_{2,3}}$, determine the smallest useful Hermitian pencil representation allowing auxiliary dimensions or structured substitutions. How does that size grow under the tree attachments of Corollary 37.4?

Theorem 29.1 addresses only the vertex-indexed diagonal pencil. A no-go result for that form should not be used as an obstruction to every possible determinant construction.

Research question 16 (Quantitative obstructions with variable edge magnitudes). Replace the unit-phase constraint by $1 - \eta \leq |K_{ab}|^2 \leq 1 + \eta$ on edges. Find an explicit lower bound on the largest support-minor defect, in terms of η , for noncacti. Does a useful dimension-independent gap persist for all sufficiently small η ?

Path compression now multiplies and divides magnitudes, so the proof of Lemma 36.2 no longer gives an exact contraction. Long subdivisions are the natural stress test.

Research question 17 (Short checkable obstruction certificates). Design an algorithm that, given a noncactus graph and edge phases in an exact number field, outputs a matchable minor with defect at least ε_* , together with a compact certificate. Optimize the number of rows and columns of the output witness.

The constructive proof finds a theta and traces a failing core minor back through augmented minors. Theorem 38.1 proves that no constant bound on witness order is possible; it does not determine the best bound in terms of girth, cycle overlap, or input size.

Research question 18 (Transfer beyond height-two posets). Can stable signed-support methods, rather than faithful matrices, extend Part II's γ -interpretation (Theorem 13.2; compare Research question 1) to height-three posets or nontrivial preorder-equivalence blocks? Identify an explicit intermediate class where the combinatorial support model remains valid.

Neither the cactus theorem nor the $K_{2,3}$ example answers general γ -positivity or existential flag-polytope realization. Those questions require an additional bridge beyond Theorem 13.2. [Added 29 September 2026, batch 44: the aim of this question is reached in Part IV by a different route. The combinatorial support model is valid for every finite preorder, with ordered transport pairs in place of matchable sets (Theorem 48.2, Section 55.3); no stable signed-support method is used, and whether one gives a second route is not addressed. Existential flag realization remains open (Research question 22).]

Research question 19 (Kernel-checked certificate theory). Formalize the ternary support-pattern classification first, then extend it to the complex phase and sharp-gap theorems. Can the induced-theta reduction and the augmented-minor map be packaged as a reusable verified obstruction framework?

A successful implementation would convert a structural nonexistence proof into independently checkable finite witnesses while leaving the harder stability theory in a separate module.

43 Conclusion

The phase construction for cactus matching supports is not merely one successful family among many exact vertex determinant models: it exhausts them. Cycle overlaps produce induced theta obstructions, and a support-preserving Schur compression reduces every such obstruction to one of three small matrices. The same reduction determines the ternary support boundary and yields an optimal, dimension-independent approximation threshold.

At the same time, real stability survives beyond that boundary. The explicit $K_{2,3}$ sums of squares and their leaf extensions show why searching only for a Hermitian vertex determinant would miss valid stability phenomena. The next stage of this report's program should therefore distinguish structural determinant certificates from more general stability certificates, and should distinguish both from one-variable real-rootedness tests.

44 A sharper calculation for the shared-edge core

(Appendix A of manuscript 05.) Although the bound $3/5$ is enough for the universal theorem, the other small core D also has an exact elementary minimax value.

Proposition 44.1 (The exact defect of two squares sharing an edge). *For $D = \mathfrak{D}(1, 3, 3)$,*

$$d(D) = d_0 := \frac{\sqrt{5} - 1}{2}.$$

Proof. In (33.2), every ambiguous proper square minor is one of the two cycle minors $1 - x$, $1 - y$; the only remaining ambiguous minor is the full determinant $1 - x - y$. Thus these three norms determine the defect.

Assume a defect $\delta < d_0$. The real parts a, b lie strictly inside the rectangle $[l, h]^2$, where

$$l = (1 - d_0)/2, \quad h = (1 + d_0)/2, \quad l + h = 1.$$

Use the separately convex/concave functions g_- , g_+ from Lemma 36.5. The two equal-coordinate corners of g_- have value $d_0^2 = 1 - d_0$. At an off-diagonal corner,

$$g_-(l, h) = (\sqrt{1 - l^2} - \sqrt{1 - h^2})^2 = \frac{(h - l)^2}{(\sqrt{1 - l^2} + \sqrt{1 - h^2})^2} \leq d_0^2.$$

The denominator is at least one: $l + h = 1$ gives $(1 - l^2) + (1 - h^2) = 1 + 2lh \geq 1$. Therefore opposite imaginary signs force $|1 - x - y|^2 \leq 1 - d_0 < 1 - \delta$, impossible.

For the same-sign case, the equal-coordinate corners of g_+ have values $3 - 2d_0$ and $3 + 2d_0$. Both exceed $1 + d_0$, since $d_0 < 2/3$. For the off-diagonal corner, $lh = d_0/4$ and

$$g_+(l, h) = 1 + \frac{d_0}{2} + 2\sqrt{\frac{d_0}{2} + \frac{d_0^2}{16}} > 1 + d_0.$$

The last strict inequality follows from $\sqrt{d_0/2 + d_0^2/16} > d_0/4$. Separate concavity therefore forces $|1 - x - y|^2 > 1 + d_0 > 1 + \delta$, another contradiction. Hence $d(D) \geq d_0$.

For attainment, take $x = e^{i\alpha}$ and $y = e^{-i\alpha}$ with $\cos \alpha = (1 + d_0)/2$. The two cycle squared norms are $1 - d_0$, while the full squared norm is $d_0^2 = 1 - d_0$. Each has defect d_0 , and all other nonzero minors have norm one. This proves equality. $\square$

This additional calculation is not needed for the main classification or for the sharp universal constant. It leaves the three-odd-path core as the remaining small exact optimization target.

45 A compact dependency ledger

(Appendix B of manuscript 05.)

Result	Inputs
Cactus sufficiency	Symmetric differences of matchings; independent cycle flips; sixth-root norm identity.
Cactus necessity	Induced-theta lemma; support-preserving Schur compression; three core calculations.
Hermitian equivalence	Diagonal coefficient expansion; singleton and pair recovery; bipartite block determinant.
Ternary classification	Same graph reduction; three computations in $\mathbb{F}_3$; independent cycle factors.
Gauge count	Spanning-forest normalization; two possible normalized phases per cycle.
Sharp universal gap	Defect monotonicity; exact three-chord minimax; rational bounds on D, T .
Stable noncactus family	Three Rayleigh identities; Brändén's criterion; direct leaf-preservation argument.
No bounded minor-order test	High-girth odd theta; uniqueness of a forest perfect matching; classification.

Part IV. Gamma-positivity for every finite preorder

Part IV was added on 29 September 2026 in batch 44 of ProveIt's incoming-report intake, from two manuscripts that prove the same theorem: for every finite preorder τ , the k th γ -coefficient of h_τ is the number of ordered pairs of disjoint k -element sets joined by a τ -increasing bijection. This proves Conjecture 5.2 of Athanasiadis and Chapoton [1] in general and answers Part II's Research question 1, "Extend the matching-support interpretation", which Part III's Research question 18 had re-posed. Part II's height-two theorem (Theorem 13.2) is the special case of height at most two. Parts I–III are unchanged except for dated pointers to this addition. Section 46 records the provenance, fixes the notation and states what is answered.

46 Provenance, conventions and what Part IV answers

46.1 Two manuscripts, one addition

Part IV merges two batch-44 manuscripts, both AI-assisted research manuscripts prepared with ChatGPT for Vladimir Reshetnikov and dated 29 September 2026, both 22 A4 pages as delivered. They arrived in ProveIt commit ae28ea2db and were placed in 203016015. Both archives wrap an inner directory of the same name, ProveIt_Preorder_Gamma/, but they are different packages: no file of one coincides with a file of the other.

- **Manuscript 06 (the base):** *Gamma-Positivity for Every Finite Preorder: Disjoint transport supports, Boolean expansions, block formulas, and a transitivity criterion for palindromicity*, archive ProveIt_Preorder_Gamma_Research.zip. It is pinned to ProveIt commit 9b24a3a8d (full hash in Section 47), at which this report's article.tex was the Git blob 8739a224fa8956adab5d8df9f2d5012a6a220959: Parts I and II exactly as printed above, before Part III. Its statements about "the repository" therefore concern Parts I and II. Its program, recorded outputs, build script and two notes are shipped with the prefix 05-gamma-positivity-.
- **Manuscript 01:** *The Gamma-Vector of Every Finite Preorder: Disjoint matching supports, Boolean expansions, and sharp degree laws*, archive ProveIt_Preorder_Gamma.zip. It is pinned to ProveIt commit 077672c84 (full hash in Section 47), at which this report's article.tex was the blob cd80088f04daeabdf6e320e08bdd1b61a5c9cf4c, Parts I–III exactly as printed above; it cites Part III once. Its program, recorded outputs and build script are shipped with the prefix 04-gamma-vector-; it delivered no audit notes.

The two prefixes continue this report's local sequence (Part I's unprefix files count as 01, Part II's are 02-, Part III's 03-) in arrival order; the base took the later number, as in earlier merges of the collection. The manuscripts, delivery READMEs and PDFs of both are not shipped and survive in the arrival commit. Manuscript 01's checksum list SHA256SUMS was verified (7/7) at placement and retired; manuscript 06 shipped none. Section 59 lists the shipped files.

Why 06 is the base. The two manuscripts prove the same main theorem by the same route and share most corollaries. Manuscript 06 is the more general: its second theorem (Theorem 48.3) concerns every *reflexive* relation, shows that palindromicity, γ -positivity and transitivity are equivalent there, and has an elementary proof independent of the imported counting identity. Manuscript 01 treats preorders only. Proofs taken from manuscript 01 were re-read for hidden uses of antisymmetry or bounded height; there are none (its own words: its reduction "applies

unchanged to arbitrary height and to supports which split equivalence classes"). What only manuscript 01 contains is printed under its own headings: the Boolean orbits of commuting toggles (Section 52), multivariate complement-duality and the low coefficients (Section 55), the leading Taylor coefficient at -1 (Section 56), the induced-deck identity with odd-cardinality reconstruction (Section 57), an Ehrhart reinterpretation, three false variants, a dependency graph, and a compact proof certificate (Section 64).

Abstract of manuscript 06 (as delivered). Let $h_\tau(t)$ count the lattice points of the preorder polytope of a finite preorder τ by support size. We prove

$$h_\tau(t) = \sum_{k=0}^{\lfloor n/2 \rfloor} |\mathcal{T}_k(\tau)| t^k (1+t)^{n-2k}, \quad n = |\tau|,$$

where $\mathcal{T}_k(\tau)$ consists of ordered pairs of disjoint k -element sets (U, V) for which a bijection $f : U \rightarrow V$ with $u \leq_\tau f(u)$ exists. The objects counted are support pairs, not individual bijections. This establishes Conjecture 5.2 of Athanasiadis and Chapoton for all finite preorders, without a height or antisymmetry assumption. A support-resolved multivariate identity is proved at the same time. The essential new step is that, after fixing the zero coordinates, every preorder support fibre is a bipartite demand polymatroid. An established matching-support counting identity then turns the fibre sum into Boolean intervals.

We also prove that, for any reflexive relation R on n elements, the coefficients of degrees 1 and $n-1$ of its doubled matching-support polynomial are $|R|$ and $|R^*|$, respectively, where R^* is transitive closure. Consequently that polynomial is palindromic if and only if R is transitive, and in that case it is gamma-positive. This gives an exact classification for balanced bipartite graphs admitting a perfect matching. Further consequences include positive quotient-block formulas, coefficientwise monotonicity, sharp universal bounds, an exact multiplicity formula at -1 , and a centered-binomial mixture description of support size. Reproducible integer computations check all 7,332 labeled preorders through size five, 140 larger examples, and all reflexive relations through size four.

Abstract of manuscript 01 (as delivered). For a finite preorder τ on n elements, let $h_\tau(t)$ count the lattice points of its preorder polytope by the number of nonzero coordinates. We prove that its k th gamma coefficient counts ordered pairs of disjoint k -element sets U, V admitting a bijection $f : U \rightarrow V$ with $u \leq_\tau f(u)$. Thus

$$h_\tau(t) = \sum_{\substack{U \cap V = \emptyset \\ U \rightsquigarrow_\tau V}} t^{|V|} (1+t)^{n-2|V|}.$$

This establishes gamma-positivity for every finite preorder, with no restriction on height or on the sizes of its equivalence classes. It extends the height-two matching-support theorem in Provel and proves the full Athanasiadis–Chapoton conjecture in the cited version.

The proof partitions a lattice point by its entire zero set, converts its remaining excess coordinates into a bipartite demand vector, and applies a classical demand–matching-support counting identity. A second reduction uses transitivity to cancel the overlap of two matched sets; it realizes the gamma expansion as Boolean orbits in a support-pair model. We obtain a multivariate refinement, complement-duality, coefficientwise gamma-monotonicity under relation extension, and the exact degree law $\deg \Gamma_\tau = \nu(C_\tau)$, where C_τ is the undirected comparability graph. Consequently the multiplicity of -1 is exactly the matching deficiency $n - 2\nu(C_\tau)$. Sharp extremal bounds, an explicit equivalence-block formula, recoveries of known special cases, exhaustive finite checks, and further research directions are included.

The two abstracts are reproduced as delivered, in their own notation: manuscript 06's R^* (transitive closure) is written $\bar{R}$ below, and manuscript 01's C_τ (comparability graph) is written $\text{Comp}(\tau)$, because C_τ is Part I's polytope (Table 3).

Status and dependence (manuscript 06, as delivered). Full mathematical proofs are supplied. The general preorder theorem uses a classical bipartite counting identity, whose reduction to explicitly cited root-polytope theorems is included. The transitivity/palindromicity classification has a separate elementary proof. This manuscript is AI-assisted and unrefereed, and is not verified in Lean or Rocq. Computation supports the arguments but does not replace them. The source audit establishes the relation to the inspected sources; it does not certify absolute publication priority.

Status (manuscript 01, as delivered). The article supplies ordinary mathematical proofs with explicitly cited classical inputs. It has not been independently refereed or verified in Lean or another proof assistant. The literature and repository comparison is bounded; worldwide priority is not certified. Computations are independent finite checks, not substitutes for the proofs.

Priority and prior work (from both manuscripts and their notes). The matching-support interpretation of the augmented bipartite root polytope is Ohsugi–Tsuchiya’s [8, Propositions 3.3–3.4], restated by Davis–Kohl [7, Theorem 3.10]; the hypertree count of root-polytope volume is Kálmán–Postnikov’s [3]; the demand/support counting identity these give is Part II’s Lemma 14.1. None of these is claimed as new. The support-enumerator/root-polytope identity and preorder duality $h_\tau = h_{\tau^*}$ are Dai, Hou, Liu, Thawinrak and Wang’s [2, Theorems 1.2–1.3]; Part IV recovers duality and refines it, and claims no first proof. Menon [6] gives a bijective proof of γ -positivity for octopuses and lopsided octopuses; the octopus formula (attributed there to Athanasiadis, Xiao and Yan) is recovered in Section 55, not discovered. The height-two case is Part II’s Theorem 13.2. The nonreal-root counterexamples to Conjecture 5.3, with priority credited to Shivam Patel [11], are Part II’s and are not re-claimed. The classical chain (Narayana), binomial and universal-equivalence formulas are checks, not novelty claims; the matching-theoretic ingredients of the reflexive-relation criterion are standard, and its relation to older matching and Gorenstein characterizations deserves further historical review. Targeted searches on 29 September 2026 (manuscript 06: “preorders”, “gamma-positivity”, “Athanasiadis”, “Chapoton”, “Conjecture 5.2”, “perfectly matchable set polynomial”, “transitive”) located no earlier general proof of the main formula, but some search responses were poor or unrelated; this is a bounded search, not a proof of the absence of a preprint, an unpublished argument or parallel work. No worldwide publication priority is claimed for any result of Part IV.

46.2 Notation of Part IV

Part IV works with an arbitrary finite preorder τ on a ground set E of size n , as Part I does, and uses Part I’s $\mathcal{J}(\tau)$ (all order ideals), Q_τ , h_τ and $\downarrow$; its γ -coefficients are those of Section 10, in the basis $t^k(1+t)^{n-2k}$, and $\text{Vol}_{\mathbb{Z}}$ is the normalized volume of (7.4). For a poset of height at most two, $\tau = P_G$ and Part II’s $\mathcal{M}_k(G)$, p_G and $\nu(G)$ are recovered (Section 55.3). Table 3 lists what Part IV adds or uses differently. Four symbols of the manuscripts were renamed and one was retypeset; no normalization was changed:

- manuscript 06’s transitive closure R^* is written $\overline{R}$, because τ^* is the *opposite* preorder in Part I and in manuscript 01;
- manuscript 06’s opposite preorder τ^{op} is written τ^* , as in Part I;
- manuscript 01’s comparability graph C_τ is written $\text{Comp}(\tau)$, manuscript 06’s notation, because C_τ is Part I’s polytope (manuscript 01 itself warns of this);
- manuscript 01’s $g_k(\tau)$, $\mathcal{D}_k(\tau)$ and variables $\mathbf{t} = (t_v)$ are written $\gamma_k(\tau)$, $\mathcal{T}_k(\tau)$ and $\mathbf{z} = (z_v)$, manuscript 06’s notation for the same objects;
- the preorder polytope, Q_τ in both manuscripts, is printed $Q_\tau = Q_\tau$, as in Part I.

Table 3: Symbols of Part IV. The last column prints the tempting false reading, usually a symbol of Parts I–III with the same letter.

Symbol	Meaning in Part IV	Not to be confused with
$\mathcal{T}_k(\tau), \mathcal{T}(\tau)$	ordered pairs (U, V) of disjoint k -subsets of E joined by a bijection $f : U \rightarrow V$ with $u \leq_\tau f(u)$, counted once however many bijections exist (<i>transport pairs</i>); all k (manuscript 01's $\mathcal{D}_k(\tau)$)	Part II's $\mathcal{M}_k(G)$, a set of <i>unordered</i> vertex sets of size $2k$ (Section 55.3 relates them); the number of such bijections.
$\gamma_k(\tau), \Gamma_\tau(u)$	$ \mathcal{T}_k(\tau) $ (manuscript 01's $g_k(\tau)$); $\Gamma_\tau(u) = \sum_k \gamma_k(\tau) u^k$	$\Gamma(1/4)$ in Section 58 is Γ_τ at $1/4$, <i>not</i> Euler's Gamma function; Part III's cycle phase Γ .
$A \rightsquigarrow_\tau B$	$ A = B $ and a bijection $f : A \rightarrow B$ with $a \leq_\tau f(a)$ exists; A, B may overlap	a relation between elements.
$\mathbf{z} = (z_i), z_A, H_\tau(\mathbf{z})$	support variables, $z_A = \prod_{i \in A} z_i$, the multivariate support enumerator $\sum_{x \in Q_\tau \cap \mathbb{Z}^E} z_{\text{supp}(x)}$	Part III's $\mathbf{z}$ in $\Phi_G(\mathbf{z})$; Part II's graph H and matrix H_G .
G_τ, G_R	doubled graph on $E_L \sqcup E_R$ with edge $a_L b_R$ iff $a \leq_\tau b$ (iff $a R b$)	Part I's G_τ , which has $i_L j_R$ iff $j \leq_\tau i$: the same graph with its two shores exchanged. Matchable vertex sets correspond, so p_{G_τ} is the same polynomial; the receiver side B here is Part I's left side.
$H = (X \sqcup Y, F), \mathcal{R}_H, \mathcal{M}(H), H_{Z,S}$	a bipartite graph with edge set F ; its demand vectors (Part II's $\mathcal{R}_H$); its matchable pairs (U, V) , $U \subseteq X, V \subseteq Y$; the zero/support graph of Lemma 50.1 (manuscript 01's H_S)	$\mathcal{M} = \mathcal{M}$ in Part II indexes vertex sets $W = U \sqcup V$ by size; F is not a face or F_H .
$p_H, p_R, F_H(\mathbf{z})$	perfectly-matchable-set polynomial (Part II's); p_{G_R} ; the demand support enumerator	Part I's $p(y)$; $p = A $.
$R, \bar{R}, R $	a reflexive relation, its transitive closure (manuscript 06's R^*), number of related ordered pairs including the diagonal	τ^* , the <i>opposite</i> preorder (Part I, manuscript 01, Theorem 55.2); for a preorder $\bar{\tau} = \tau$.
τ^*	opposite preorder (manuscript 06's τ^{op})	transitive closure.
$\text{Comp}(\tau), \nu$	simple undirected comparability graph on E (manuscript 01's C_τ), its matching number	Part I's polytope C_τ ; a maximum matching of G_τ , which is perfect (size n).
$P, E_v, s_v, \mathcal{D}(P, \mathbf{s}), D(d), B_{s,r}, A_{s,r}$	quotient poset of τ , blocks, block sizes, feasible signed block vectors and $\sum_v \max(d_v, 0)$, local polynomials (manuscript 06)	Part I's P_τ and B_τ ; Part II's P_G, D_G .
$E_i, m_i, b, (\alpha, \beta), \mathcal{A}(P, \mathbf{m})$	the same blocks, sizes and number of blocks, source/receiver profiles, feasible profiles (manuscript 01)	Part II's m indexing Θ_m ; a dilation m in (52.3).
d	a signed vector in $\{-1, 0, 1\}^E$ or $\mathbb{Z}^P$; in Corollary 56.2 the deficiency $n - 2\nu$	Part I's diagonal vertices d_i ; Part III's defect $d(G)$.
$D_\tau(t), D_2(\tau), R_{2,2}(\tau)$	the one-element deletion deck polynomial; two counts in Proposition 55.5	Part I's neighbourhoods D_i ; $\mathcal{D}(P, \mathbf{s})$.
K, L	mixing variable, uniform support size (Section 58)	Part II's cactus matrix K and its support size S in (19.7); Part I's triangulation K .
$L_n, \mathbf{E}_n, C_k$	the n -chain, the universal equivalence preorder, the Catalan number	Part I's C_τ ; the ground set E .
T, Z, S, W	a subset; the zero set and the support of a lattice point; a common subset	Part I's S_τ ; Part II's T_G and matchable sets W .

46.3 What Part IV answers

Status after Part IV, as of 29 September 2026:

- **Conjecture 5.2 of [1] (γ -positivity) holds for every finite preorder** (Theorem 48.2), with no height or antisymmetry assumption and an explicit count: $\gamma_k(\tau) = |\mathcal{T}_k(\tau)|$. This settles what Part I's Section 10 left open for Conjecture 5.2 and what Part II's Section 11.3 left open beyond height two, including height-two posets blown up into nontrivial equivalence classes (Example 54.2). Both manuscripts prove it, by the same route. Its proof does not use Part I's unrefereed polytope theorem.
- **Part II's Research question 1 ("Extend the matching-support interpretation") is answered.** The interpretation extends to every finite preorder with ordered transport pairs in place of matchable vertex sets; for posets of height at most two the two coincide (Section 55.3), so Part II's Theorem 13.2 is the special case. The "step of the lower-zero-set argument" that survives weighted blocks is the zero-set partition itself, taken over *all* zero coordinates rather than a minimal shore (Lemma 50.1); coordinates in nontrivial blocks may exceed one, as the question's remark warns, and the reduction allows that.
- **Part III's Research question 18 ("Transfer beyond height-two posets") is answered in its aim, by a different route:** the combinatorial support model is valid for every preorder, and no stable signed-support method is needed. Whether such methods give a second route is not addressed.
- **Part II's Research question 2 ("A purely combinatorial bridge") remains open:** both manuscripts use the same imported counting identity (Lemma 14.1) and construct no bijection; it is re-posed in support-resolved form as Research question 20.
- **Part I's nonpalindromic reflexive example** (Section 8, subsection 8.5) is an instance of Theorem 48.3: its support counts $(1, 5, 6, 1)$ are $(1, |R|, |\overline{R}|, 1)$ with one missing transitive pair (Example 53.3).
- **Conjecture 5.1(d) (flag realization) and Conjecture 5.3 remain as Parts II and III left them:** Part IV proves no flag realization, and γ -positivity does not imply real-rootedness (Section 60). Palindromicity and unimodality, Conjecture 5.1(a)–(b), follow again, by a route independent of Part I's geometry (Corollary 50.3).

Part IV refutes no statement of Parts I–III, and nothing in them is retracted. Parts I–III gain dated pointers: the title block, abstract, scope box and reading route; Section 10; Part I's reflexive example; Part II's status box, Section 11.3 and Research question 1; Part III's Section 26.3 and Research question 18.

46.4 Where the merge had to choose

- *Placement.* Part IV follows Part III and precedes Part I's appendices, whose letters and numbers are unchanged. The manuscripts' appendices (06: normalization checklist and source ledger; 01: proof certificate and source ledger) are printed as the ordinary Sections 63–65.
- *Order and base.* Manuscript 06's order of sections is kept; manuscript 01's sections are placed beside the matching ones of 06, and its sections without a counterpart are printed as separate sections or subsections marked "(manuscript 01)". Every result of both is printed; a result proved in both is printed once, with both sources named.

- *Duplicated results.* The main theorem, its support-fibre (06) or all-zero-set (01) lemma and its Boolean summation are the same argument in both manuscripts and are printed once, from 06, with manuscript 01's remarks and example. The same holds for the doubled-graph identity and the second derivation through cancellation, the support-resolved counting identity, γ -monotonicity, disjoint unions, γ_0 and γ_1 , the universal bounds, the chain and universal-equivalence formulas, the degree and -1 -multiplicity theorem, and the algorithmic remark on block formulas. Where one statement is sharper (manuscript 01's strict unimodality, strict increase of γ_1 , bounds on h and equality cases), the sharper statement is printed and attributed.
- *Second routes.* Genuinely different proofs are kept: manuscript 01's path-decomposition proof of overlap cancellation (after Corollary 51.2) beside 06's ideal-Hall proof; manuscript 01's profile form of the block formula (Theorem 54.3) beside 06's signed form; manuscript 01's direct initial-segment proof of the chain count and its reflection proof of the Narayana formula.
- *The counting identity.* Both manuscripts prove the demand/support identity by the reduction of Part II's Section 14, step for step (augmentation, the two published inputs at $u = 1$, the clone-and-Hall hypertree bijection), and say so. It is printed once, in Part II; Section 49 states it in Part IV's notation and records the manuscripts' additional remarks.
- *Research questions.* The two manuscripts' twenty questions continue Part III's numbering as Research questions 20–36. Three pairs ask the same question (flag realization; a support-preserving bijection; formal verification) and are printed once each with both formulations, so there are seventeen.
- *Self-citation.* Both manuscripts cite this report as an external repository report (reference "Repo", with their pins). Here those citations become references to Parts I–III, and "the repository('s)" becomes "Part II('s)" or "this report" where it means this report; the pins are kept in Section 47.
- *File names.* The delivered names, such as `code/verify.py`, `results/verification.json` and `data/verification.json`, are replaced by the shipped names in Section 59, because `code/verify.py` in this directory is Part I's verifier; the run commands are replaced by commands that run on a copy or with explicit output paths. Package-content lists that name unshipped files are kept and annotated.
- *Bibliography.* Every external reference of both manuscripts was already in this report's bibliography: 06's and 01's OT is [8], 01's Dai is [2]; nothing is appended.
- *Labels and front matter.* Labels carry the prefix `gam:`, and `gam:gv:` for material only manuscript 01 contains. Part II's Research questions 1 and 2 and Part I's subsection on reflexive relations received the labels `gam:q:extend`, `gam:q:bridge` and `gam:sec:reflexiveex` so that Part IV can cite them. The two title pages, abstracts and status boxes are printed in this section, and manuscript 01's "shortest proof audit" box in Section 47; both tables of contents and manuscript 06's reading-route box, which only lists its own sections, are dropped.

47 The question, its provenance, and the advance

Athanasiadis and Chapoton associate a support polynomial to every finite preorder and ask whether it has nonnegative coefficients in the gamma basis $t^k(1+t)^{n-2k}$ [1, Conjecture 5.2]. The question is stronger than palindromicity and unimodality: positive combinations of these

centered binomial polynomials have both properties, but a symmetric unimodal polynomial need not be gamma-positive. It is weaker than real-rootedness. Their separate Conjecture 5.1(d) asks for a flag simplicial-polytope realization; that is a geometric statement, not a reformulation of gamma-positivity (manuscript 01).

Part II of this report provides the immediate research context. At the snapshot manuscript 06 inspected, Part II proves gamma-positivity for posets of height at most two. Its explicit open boundary includes arbitrary height and height-two quotient posets whose elements have been replaced by nontrivial equivalence classes. The main result below removes both restrictions. It also identifies every gamma coefficient by a finite matching condition. The answer to Conjecture 5.2 is affirmative. The word “supports” is essential: a pair of vertex sets is counted once whenever at least one suitable matching exists, regardless of how many such matchings it has. Likewise, “disjoint” refers to the underlying elements of the preorder, not merely to left and right copies of a bipartite graph (manuscript 01).

47.1 What is inherited, and what is proved here

The matching-support interpretation for augmented bipartite root polytopes is prior work of Ohsugi and Tsuchiya [8, Propositions 3.3–3.4], restated by Davis and Kohl [7, Theorem 3.10]. The needed total-count identity follows with the hypertree interpretation of root-polytope volume [3]; Part II already uses this identity in its height-two argument (Lemma 14.1). None of these facts is claimed as new.

Dai, Hou, Liu, Thawinrak, and Wang identify bipartite demand support polynomials with the corresponding root-polytope numerators and prove preorder duality [2, Theorems 1.2–1.3]. Menon gives a bijective proof of gamma-positivity for octopuses and an extension to lopsided octopuses [6]. These are relevant antecedents, not results that this addition claims to discover. The theorem below includes Menon’s families and all other finite preorders; their explicit special-case formulas are recoveries (Section 55), and univariate duality is recovered and refined rather than claimed (Theorem 55.2).

The additional argument is the universal support-fibre reduction in Lemma 50.1, together with its Boolean summation. It applies to *every* finite preorder. The cancellation theorem gives another explanation, and the endpoint-defect theorem identifies exactly when palindromicity occurs in the balanced perfect-matching setting. The block, extremal, matching-deficiency, and probabilistic formulas make the resulting structure usable beyond the original conjecture.

47.2 Precise scope of the conclusions

The theorem settles the statement of Conjecture 5.2 in the version of [1] consulted here; it also proves its asserted palindromicity and unimodality consequences. Part I’s ordinary simplicial-polytopal realization is not an input. In particular, no unrefereed geometric assertion in ProveIt is needed to complete the proof.

The stronger flag-polytopal realization question is not settled. Nor does gamma-positivity imply real-rootedness: Part II records and extends previously posted counterexamples to the latter conjecture, with priority credited there to Shivam Patel [11]. We do not re-claim those counterexamples. We also do not identify h_τ with the Ehrhart numerator of the *original* preorder polytope; those are different invariants.

The ProveIt tree inspected by manuscript 06 is pinned to commit

9b24a3a8d545af9624f6ac455f5b548be62818b6.

[Merge note: at that commit this report consisted of Parts I and II as printed above; see

Section 46.1.] Its relevant report path and the primary-source details appear in the bibliography and in the accompanying source audit (05-gamma-positivity-SOURCE_AUDIT.md). A targeted search on 29 September 2026 did not locate an earlier general proof of the main formula, but search results were not uniformly informative. No exhaustive priority claim follows from that search.

47.3 The repository frontier seen by manuscript 01

The ProveIt snapshot inspected by manuscript 01 is commit

077672c84f743ff5de82ce6becc3ad6b2ff8787a.

Its report *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, in

SetTheory/Cardinals/docs/reports/enumerative-combinatorics/
preorder-root-polytopes/

contains a height-two theorem in Part II and later cactus results in Part III. [Merge note: that is this report, Parts I–III as printed above.] Part II expresses the gamma coefficients of a height-at-most-two *poset* through perfectly matchable subsets of its bipartite comparability graph. Its further-research section explicitly asks for an extension to greater height and to nontrivial preorder-equivalence blocks (Research question 1). The README at that snapshot still distinguishes this restricted theorem from the unsettled general gamma-positivity question.

The new observation is that one should partition by *all* zero coordinates, not only by zeros on a minimal shore. For a support S with complement Z , excess demand at any element of S can be supplied from comparable elements of Z . Transitivity turns the ideal inequalities into the appropriate Hall inequalities. This construction applies unchanged to arbitrary height and to supports which split equivalence classes.

Part II supplies the useful demand-counting lemma by a reduction to classical root-polytope results. Manuscript 01 reproduces the reduction with its published dependencies identified. Therefore the general theorem does not require accepting Part I’s unrefereed geometric claims.

Item	Status (manuscript 01’s table, as delivered)
General gamma-positivity	Proved for every finite preorder, with an explicit nonnegative formula for each coefficient.
Multivariate and structural laws	Proved: support refinement, complement-duality, Boolean support-pair action, relation-monotonicity, exact degree and -1 -multiplicity, and block compression.
Classical input	The demand-matching-support count, derived from the Kálmán-Postnikov and Ohsugi-Tsuchiya root-polytope identities.
Previously settled statements	Height-two gamma-positivity, special arbor formulas, and univariate duality are credited and recovered.
Not established	Flag simplicial-polytope realization, a natural lattice-point bijection implementing the Boolean action, or a general real-rootedness statement.
Verification	Exhaustive integer tests and explicit proofs; no new proof-assistant verification.

The priority assessment concerns the cited primary papers and the pinned repository report. A search that finds no matching general theorem does not establish that no such theorem exists elsewhere.

Shortest proof audit (manuscript 01). Read the main statement in Section 48, the counting lemma in Section 49 (with its proof in Part II, Section 14), and the zero-set reduction in Section 50. The latter reduction is the step that removes Part II's height-two restriction. Section 51 gives a second reduction sharing the same counting lemma. No unrefereed root-polytope theorem from this report is assumed.

48 Definitions and main theorems

Let E be a finite set of size n . A preorder τ is a reflexive, transitive relation $\leq_\tau$ on E . We do not assume antisymmetry. An *ideal* is a subset I such that $v \in I$ and $u \leq_\tau v$ imply $u \in I$. Write $\mathcal{J}(\tau)$ for all ideals, including $\emptyset$ and E . For $T \subseteq E$, put

$$\downarrow T = \{u \in E : u \leq_\tau v \text{ for some } v \in T\}.$$

Reflexivity gives $T \subseteq \downarrow T$, and transitivity makes $\downarrow T$ an ideal. These are exactly the two properties used in the main reduction.

For $x \in \mathbb{R}^E$ and $I \subseteq E$, write $x(I) = \sum_{i \in I} x_i$. The preorder polytope and its support polynomials are, as in Part I,

$$Q_\tau = \{x \in \mathbb{R}_{\geq 0}^E : x(I) \leq |I| \text{ for every } I \in \mathcal{J}(\tau)\}, \quad (48.1)$$

$$h_\tau(t) = \sum_{x \in Q_\tau \cap \mathbb{Z}^E} t^{|\text{supp}(x)|}, \quad \text{supp}(x) = \{i : x_i > 0\}, \quad (48.2)$$

$$H_\tau(\mathbf{z}) = \sum_{x \in Q_\tau \cap \mathbb{Z}^E} z_{\text{supp}(x)}, \quad z_A = \prod_{i \in A} z_i, \quad h_\tau(t) = H_\tau(t, \dots, t). \quad (48.3)$$

The full-set inequality makes the lattice-point set finite. The all-zero point is the only point of empty support, and the all-one point is the only point of full support, since its n positive integer coordinates already sum to at least n . Thus h_τ has degree n and both endpoint coefficients are one.

The polynomial h_τ is initially a *support enumerator*. It must not be confused with the Ehrhart numerator $h^*(Q_\tau; t)$. An Ehrhart numerator of a different root polytope appears in Section 52 (manuscript 01).

Definition 48.1 (Disjoint transport supports). Let $\mathcal{T}_k(\tau)$ be the set of ordered pairs (U, V) of subsets of E such that

$$U \cap V = \emptyset, \quad |U| = |V| = k,$$

and there exists a bijection $f : U \rightarrow V$ satisfying $u \leq_\tau f(u)$ for every $u \in U$. Set $\mathcal{T}(\tau) = \bigcup_k \mathcal{T}_k(\tau)$. More generally, for subsets $A, B \subseteq E$, write $A \rightsquigarrow_\tau B$ if $|A| = |B|$ and there exists a bijection $f : A \rightarrow B$ satisfying $a \leq_\tau f(a)$ for every $a \in A$ (manuscript 01's "compatible support pairs"; A and B may overlap).

The empty pair is included. A support pair with several possible bijections is counted once. The order of the pair matters, especially inside a nontrivial equivalence class.

Theorem 48.2 (General preorder gamma formula; manuscripts 06 and 01). *For every finite preorder τ on E ,*

$$h_\tau(t) = \sum_{k=0}^{\lfloor n/2 \rfloor} |\mathcal{T}_k(\tau)| t^k (1+t)^{n-2k}. \quad (48.4)$$

More strongly, for commuting variables $\mathbf{z} = (z_i)_{i \in E}$,

$$H_\tau(\mathbf{z}) = \sum_{x \in \mathcal{Q}_\tau \cap \mathbb{Z}^E} z_{\text{supp}(x)} = \sum_{(U,V) \in \mathcal{T}(\tau)} z_V \prod_{i \in E \setminus (U \cup V)} (1 + z_i). \quad (48.5)$$

Consequently $\gamma_k(\tau) = |\mathcal{T}_k(\tau)|$ are nonnegative integers, proving Athanasiadis–Chapoton Conjecture 5.2 for arbitrary finite preorders.

Here and below a gamma expansion is a polynomial identity; no value is assigned to a rational substitution at its pole $t = -1$. The expansion is unique: $t^k(1+t)^{n-2k}$ has lowest degree k with coefficient one, so successive coefficients in increasing degree determine $\gamma_0, \gamma_1, \dots$ (manuscript 01). Define the gamma polynomial

$$\Gamma_\tau(u) = \sum_k \gamma_k(\tau) u^k. \quad (48.6)$$

Then (48.4) can be abbreviated as $h_\tau(t) = (1+t)^n \Gamma_\tau(t/(1+t)^2)$, with the right side interpreted by its polynomial expansion, not by substitution at the undefined quotient $t = -1$. The degree of Γ_τ and the multiplicity of -1 as a root of h_τ are determined exactly in Theorem 56.1, which both manuscripts state (manuscript 01 already in its introduction): with $\text{Comp}(\tau)$ the simple undirected graph on E in which distinct vertices are adjacent precisely when they are comparable in at least one direction, $\deg \Gamma_\tau = \nu(\text{Comp}(\tau))$ and $\gamma_k(\tau) > 0$ exactly for $0 \leq k \leq \nu(\text{Comp}(\tau))$.

There is a second result about a relation which need not be transitive. For a reflexive relation R on E , form the bipartite graph G_R on two copies of E , with edge $i_L j_R$ exactly when $i R j$; for a preorder, G_τ has edge $a_L b_R$ exactly when $a \leq_\tau b$. Define

$$p_R(t) = \sum_{\substack{A, B \subseteq E, |A|=|B| \\ G_R[A_L, B_R] \text{ has a perfect matching}}} t^{|A|}. \quad (48.7)$$

Its degree is n , since all diagonal edges are present. The relation $\bar{R}$ denotes reflexive transitive closure, and $|R|$ counts ordered related pairs, including the n diagonal pairs. [Merge note: G_τ is Part I's G_τ with its shores exchanged, and $p_\tau = p_{G_\tau}$ is the polynomial of Part II's Remark 11.1; see Table 3.]

Theorem 48.3 (Transitivity is the palindromicity criterion; manuscript 06). *For $n \geq 1$ and every reflexive relation R ,*

$$[t]p_R(t) = |R|, \quad [t^{n-1}]p_R(t) = |\bar{R}|. \quad (48.8)$$

The following conditions are equivalent:

- (i) R is transitive;
- (ii) every matchable pair (A, B) remains matchable after removing $A \cap B$;
- (iii) $p_R(t)$ is palindromic of degree n ;
- (iv) $p_R(t)$ is gamma-positive of degree n ;
- (v) $[t^{n-1}]p_R(t) = [t]p_R(t)$.

In particular, the asymmetry between these two coefficients counts exactly the ordered pairs missing from transitive closure:

$$[t^{n-1}]p_R - [t]p_R = |\bar{R} \setminus R|. \quad (48.9)$$

For $n = 0$, all statements except the displayed coefficient indices have the obvious trivial interpretation. The theorem's proof in Sections 51 and 53 is elementary and does not use root polytopes or Theorem 48.2.

49 The established bipartite counting identity

We state the non-elementary enumerative input precisely and record its reduction to the cited classical theorems. This also fixes the distinction between support counts, matching counts, and hypertree counts.

Let $H = (X \sqcup Y, F)$ be a finite bipartite graph with specified shores, and let $N_H(T) \subseteq X$ be the neighborhood of $T \subseteq Y$. Its integral demand set is Part II's

$$\mathcal{R}_H = \{c \in \mathbb{Z}_{\geq 0}^Y : c(T) \leq |N_H(T)| \text{ for all } T \subseteq Y\}. \quad (49.1)$$

A vector in $\mathcal{R}_H$ describes integer requests at right vertices that can be served by distinct unit-capacity left vertices; we count demand *vectors*, not allocations serving them. Write $\mathcal{M}(H)$ for the pairs (U, V) with $U \subseteq X$, $V \subseteq Y$, $|U| = |V|$, such that $H[U, V]$ has a perfect matching, and $p_H(u) = \sum_{(U, V) \in \mathcal{M}(H)} u^{|V|}$, the perfectly matchable set polynomial; the specified shores allow us to write a matchable vertex set uniquely as $U \sqcup V$.

Lemma 49.1 (Demand/support equinumeracy, inherited). *For every finite bipartite graph H ,*

$$|\mathcal{R}_H| = |\mathcal{M}(H)| = p_H(1). \quad (49.2)$$

This is Lemma 14.1 of Part II. Both manuscripts prove it by the reduction printed there: adjoin x_0, y_0 with all edges x_0y, xy_0, x_0y_0 to obtain the connected graph $\widehat{H}$; combine (14.3) (Kálmán–Postnikov [3], evaluated at one; manuscript 01 cites its Theorems 1.1 and 3.10) with (14.4) (Ohsugi–Tsuchiya [8, Propositions 3.3–3.4], in the augmented form of Davis–Kohl [7, Theorem 3.10]) through $h^*(B; 1) = \text{Vol}_{\mathbb{Z}}(B)$, with the full affine integer lattice; and identify the right hypertree vectors with $\mathcal{R}_H$ by the forest count and the clone-and-Hall construction of Section 14. The manuscripts' versions differ from Part II's only in presentation: manuscript 06 bounds the forest on $T \cup N_H(T) \cup \{x_0\}$ directly, and manuscript 01 spells out the augmenting-path proof of the required direction of Hall's criterion (choose a largest matching; absence of an augmenting path from an unmatched clone yields a reached clone set larger than its neighborhood). Both note that empty shores give the single empty matching support, and manuscript 01 that the construction need not choose a unique tree, because the resulting *vector* is unique. Manuscript 01 adds: the augmentation matters; identity (14.4) is about the edge polytope of $\widehat{H}$, not the unaugmented edge polytope of H and not a polytope whose vertices are individual matchings.

Remark 49.2 (The input is not a matching enumeration; manuscript 06). For $H = K_{2,2}$, there are six demand vectors: the nonnegative pairs of total at most two. There are also six matchable support pairs: the empty pair, four single-edge supports, and one full support. There are two perfect matchings on the full support, but it contributes only one. Replacing $|\mathcal{M}(H)|$ by the number of individual matchings would already fail in this example.

49.1 A support-resolved version by Boolean inversion

For later use, the total identity also determines every prescribed right support. This refinement recovers the known support-enumerator identity; it is not a new priority claim for that identity. It requires no additional geometric theorem.

Proposition 49.3 (Prescribed demand support; manuscripts 06 and 01). *For every $S \subseteq Y$,*

$$\#\{c \in \mathcal{R}_H : \text{supp}(c) = S\} = \#\{U \subseteq X : (U, S) \in \mathcal{M}(H)\}. \quad (49.3)$$

Consequently

$$F_H(\mathbf{z}) := \sum_{c \in \mathcal{R}_H} z_{\text{supp}(c)} = \sum_{(U, V) \in \mathcal{M}(H)} z_V. \quad (49.4)$$

Proof. For $T \subseteq Y$, the demands supported inside T are exactly those for $H[X, T]$: inequalities involving outside vertices follow from the ones on T , since outside coordinates are zero and neighborhoods only enlarge. By Lemma 49.1, their number is

$$a(T) = \#\{(U, V) \in \mathcal{M}(H) : V \subseteq T\}.$$

Both sides of (49.3), as functions of S , therefore have the same sums over all $S \subseteq T$. Boolean inversion gives their equality. Explicitly, a fixed $V \subseteq S$ contributes

$$\sum_{V \subseteq T \subseteq S} (-1)^{|S|-|T|} = \begin{cases} 1 & V = S, \\ 0 & V \neq S. \end{cases}$$

Summing (49.3) against z_S proves (49.4). $\square$

Manuscript 01 proves the same refinement by induction on $|T|$ and records the explicit value: both sides of (49.3) equal $\sum_{V \subseteq S} (-1)^{|S|-|V|} p_{H[X \sqcup V]}(1)$.

No explicit canonical bijection between the two sets in (49.3) has been constructed in this argument. The equality is enough for all the polynomial identities below.

Remark 49.4 (What is not constructed; manuscript 01). The lemmas give equality of cardinalities, refined by the receiver support. They do not specify a natural bijection from a demand vector to a matchable support pair. The explicit tree construction is a bijection with hypertree *vectors*, not with matchings or with the pairs counted by p_H . This distinction remains important when we introduce Boolean actions (Section 52).

50 Proof of the general preorder theorem

The decisive step is to apply the bipartite identity to the zeros and the positive coordinates of an arbitrary preorder point, rather than to fixed minimal and maximal levels of a height-two poset.

Lemma 50.1 (Every support fibre is a bipartite demand set; manuscripts 06 and 01). *Fix $S \subseteq E$ and set $Z = E \setminus S$. Let $H_{Z,S}$ be the bipartite graph with shores Z, S and edges zs exactly when $z \leq_\tau s$. The map*

$$c \mapsto x, \quad x_z = 0 \ (z \in Z), \quad x_s = 1 + c_s \ (s \in S) \quad (50.1)$$

is a bijection from $\mathcal{R}_{H_{Z,S}}$ onto the lattice points of Q_τ with support exactly S .

Proof. The map is plainly a bijection between $\mathbb{Z}_{\geq 0}^S$ and all nonnegative integral vectors of support S . We prove the equivalence of its feasibility conditions. For an ideal I , the preorder inequality is

$$c(I \cap S) \leq |I \cap Z|. \quad (50.2)$$

Suppose these inequalities hold. Given $T \subseteq S$, take the ideal $I = \downarrow T$. Then $T \subseteq I \cap S$, and c is nonnegative, so

$$c(T) \leq c((\downarrow T) \cap S) \leq |(\downarrow T) \cap Z| = |N_{H_{Z,S}}(T)|.$$

Thus $c \in \mathcal{R}_{H_{Z,S}}$.

Conversely, suppose all the displayed neighborhood bounds hold. For an arbitrary ideal I , put $T = I \cap S$. Downward closure gives $\downarrow T \subseteq I$, whence

$$c(I \cap S) \leq |(\downarrow(I \cap S)) \cap Z| \leq |I \cap Z|.$$

This is (50.2), so $x \in Q_\tau$. Both implications use only reflexivity, transitivity, and nonnegativity. They do not use a height bound, or require equivalence classes to be singletons. $\square$

Manuscript 01 calls this the all-zero-set reduction and adds: no property of a minimal or maximal layer is used. In particular, the set Z need not be an ideal, the set S need not be a filter, and neither set need be a union of equivalence classes. The only order-theoretic facts used are that $\downarrow T$ is an ideal and that an ideal contains the lower closure of each of its subsets.

Proof of Theorem 48.2. For each $S \subseteq E$, Lemmas 50.1 and 49.1 give

$$\#\{x \in \mathcal{Q}_\tau \cap \mathbb{Z}^E : \text{supp}(x) = S\} = \#\{(U, V) \in \mathcal{T}(\tau) : U \subseteq E \setminus S, V \subseteq S\}. \quad (50.3)$$

Indeed, the pairs on the right are exactly the matchable support pairs in $H_{E \setminus S, S}$. Multiply by z_S and sum over S ; all sums are finite, so we may sum first over the pairs. For a fixed $(U, V) \in \mathcal{T}(\tau)$, the permissible supports are precisely

$$V \subseteq S \subseteq E \setminus U.$$

Every such S is uniquely $V \cup W$ with $W \subseteq E \setminus (U \cup V)$. Its total contribution is

$$\sum_{W \subseteq E \setminus (U \cup V)} z_V z_W = z_V \prod_{i \in E \setminus (U \cup V)} (1 + z_i).$$

This proves (48.5). Specializing all z_i to t gives (48.4), because $|U| = |V| = k$ leaves exactly $n - 2k$ free labels. $\square$

Equation (50.3) is manuscript 01's Corollary "Support-by-support counting formula": the general theorem is not merely an equality after forgetting which coordinates are nonzero.

Example 50.2 (Why equivalence classes cause no exception; manuscript 01). If $E = \{1, 2, 3\}$ is a single equivalence class, the only ideals are $\emptyset$ and E . The lattice point $(3, 0, 0)$ is allowed. Its support $S = \{1\}$ is not a union of equivalence classes, but the reduction still works: the two zero coordinates supply the excess demand $c_1 = 2$ through the two edges of $H_{Z, S}$. Nothing requires an individual coordinate to be bounded by one. In this example $h_\tau = 1 + 9t + 9t^2 + t^3$ and $\Gamma_\tau = 1 + 6u$.

50.1 What the Boolean expansion does and does not provide

Equation (50.3) says that each lattice-point support count is the number of abstract intervals $[V, E \setminus U]$ containing that support, with (U, V) a transport pair. An interval has bottom rank k , top rank $n - k$, and rank enumerator $t^k(1 + t)^{n-2k}$. This is the source of gamma-positivity.

The intervals are an *enumerative model*; they are not asserted to be actual coordinatewise subposets of the lattice-point set. Manuscript 06 constructs no commuting involutions on those lattice points and no canonical partition of them into geometric cubes. Those stronger structures would require a compatible, explicit realization of the counting identity in Lemma 49.1. (Manuscript 01's commuting toggles, Theorem 52.1, act on a support-pair model, not on the lattice points.)

Corollary 50.3 (Palindromicity and strict unimodality; manuscripts 06 and 01). *For every finite preorder, $h_\tau(t) = t^n h_\tau(t^{-1})$, that is, $h_j(\tau) = h_{n-j}(\tau)$, and $h_0(\tau) \leq h_1(\tau) \leq \dots \leq h_{\lfloor n/2 \rfloor}(\tau)$ (manuscript 06). In fact the coefficients are strictly increasing up to the middle (manuscript 01):*

$$[t^j]h_\tau < [t^{j+1}]h_\tau \quad (0 \leq j < \lfloor n/2 \rfloor).$$

When n is odd, the two middle coefficients are equal. The inequalities on the other side follow by symmetry.

Proof. Each polynomial $t^k(1+t)^{n-2k}$ has symmetric coefficients centered at $n/2$, nondecreasing up to that center. A nonnegative sum preserves both properties. There is no appeal to a geometric realization. Moreover $\gamma_0 = 1$, and $(1+t)^n$ has strictly increasing coefficients for the indicated indices. Adding the remaining nonnegative symmetric-unimodal summands preserves these strict inequalities. $\square$

Thus the theorem proves the palindromicity and unimodality assertions of Athanasiadis–Chapoton Conjecture 5.1(a)–(b) as well as Conjecture 5.2. It makes no implication in the reverse direction: a palindromic unimodal polynomial need not be gamma-positive. [Merge note: Part I proposes these properties, and more, through its simplicial polytope S_τ (Theorem 1.1); the proof here is independent of Part I’s unreferenced geometric argument.]

51 Ideal inequalities, cancellation, and a second proof

The gamma coefficients can be recognized without searching through bijections. This also explains why transitivity is the exact structural hypothesis behind the Boolean intervals.

Lemma 51.1 (The ideal Hall criterion; manuscript 06). *For subsets $A, B \subseteq E$ of equal size, there is a bijection $f : A \rightarrow B$ with $a \leq_\tau f(a)$ if and only if*

$$|A \cap I| \geq |B \cap I| \quad \text{for every } I \in \mathcal{J}(\tau). \quad (51.1)$$

Proof. If $b \in B \cap I$, its matched predecessor belongs to $A \cap I$. Distinct b have distinct predecessors, proving necessity. Conversely, let $T \subseteq B$. The predecessors of T inside A are $A \cap \downarrow T$. Since $\downarrow T$ is an ideal and $T \subseteq B \cap \downarrow T$,

$$|T| \leq |B \cap \downarrow T| \leq |A \cap \downarrow T|.$$

Hall’s criterion gives a matching saturating B ; equality of the shore sizes makes it a bijection. $\square$

Corollary 51.2 (Common-support cancellation; manuscripts 06 and 01). *For any $A, B \subseteq E$ with $|A| = |B|$, the pair (A, B) is matchable in G_τ if and only if $(A \setminus B, B \setminus A)$ is matchable; that is,*

$$A \rightsquigarrow_\tau B \iff A \setminus B \rightsquigarrow_\tau B \setminus A. \quad (51.2)$$

Proof. The common intersection contributes the same quantity to both sides of (51.1), so it cancels. Alternatively, the reverse implication extends a matching by the diagonal edges on $A \cap B$. $\square$

Manuscript 01 states (51.2) as its Lemma “Canceling common support” and proves it without the ideal Hall criterion.

Second proof of Corollary 51.2, by path decomposition (manuscript 01). Suppose first that $f : A \rightarrow B$ is a compatible bijection. On the original ground set draw the directed edges $a \rightarrow f(a)$, retaining possible loops. Every vertex has indegree and outdegree at most one. A vertex in $U = A \setminus B$ has outdegree one and indegree zero; one in $V = B \setminus A$ has indegree one and outdegree zero; one in $A \cap B$ has both degrees one. The nontrivial components are therefore directed paths from U to V and directed cycles lying entirely in $A \cap B$.

To justify that a path beginning in U cannot enter a cycle, the first entry vertex of that cycle would receive two incoming edges. Since the graph is finite, the path must instead terminate, and it can terminate only in V . Different starting vertices yield disjoint paths. Along each path, transitivity implies that its starting vertex is below its ending vertex. Replacing every path by this direct relation gives a bijection from U to V . Cycles and loops are discarded.

Conversely, a compatible bijection from U to V extends to one from A to B by fixing every element of $A \cap B$. The additional relations are valid by reflexivity. $\square$

In particular, the truth of $A \rightsquigarrow_\tau B$ depends on the disjoint core $(A \setminus B, B \setminus A)$, not on the choice of common elements (manuscript 01).

51.1 Signed vectors and transport feasibility

For a disjoint pair (U, V) , define $d_i = 1$ on U , $d_i = -1$ on V , and $d_i = 0$ elsewhere. Lemma 51.1 gives a useful equivalent formula:

$$\Gamma_\tau(u) = \sum_k \gamma_k(\tau) u^k = \sum_{\substack{d \in \{-1, 0, 1\}^E \\ d(E)=0 \\ d(I) \geq 0 \ (I \in \mathcal{J}(\tau))}} u^{|\{i: d_i=1\}|}. \quad (51.3)$$

Thus the coefficients count balanced signed labelings whose cumulative sum on every ideal is nonnegative. This is not a count of integer flows with multiplicity: distinct realizations of the same signed labeling still contribute once.

For a chain, these are Dyck words with zero symbols inserted. For a general preorder, ideals replace initial intervals. Equivalent elements are tested only together in the ideal inequalities, but their signs remain individually visible in (51.3). This distinction will produce the block multiplicities in Section 54.

51.2 Recovering the doubled-graph identity

Lemma 51.3 (The doubled graph's demand body; manuscripts 06 and 01). *After identifying right coordinates with E , $\mathcal{Q}_\tau \cap \mathbb{Z}^E = \mathcal{R}_{G_\tau}$. The analogous equality holds for real nonnegative vectors satisfying the same inequalities.*

Proof. The neighborhood of T_R in G_τ is $(\downarrow T)_L$. If $x \in \mathcal{Q}_\tau$, nonnegativity and the ideal inequality for $\downarrow T$ give $x(T) \leq x(\downarrow T) \leq |\downarrow T|$. Conversely, when the demand inequalities hold, take $T = I$ for an ideal; then $\downarrow I = I$, so $x(I) \leq |I|$. $\square$

[Merge note: this is Part I's Lemma 2.1 with the shores of G_τ exchanged.] Proposition 49.3 therefore yields

$$h_\tau(t) = p_\tau(t), \quad H_\tau(\mathbf{z}) = \sum_{x \in \mathcal{Q}_\tau \cap \mathbb{Z}^E} z_{\text{supp}(x)} = \sum_{A \rightsquigarrow_\tau B} z_B, \quad (51.4)$$

where $p_\tau = p_{G_\tau}$ and the sets A and B need not be disjoint as subsets of E (manuscript 01's Proposition "Relation-graph realization"). The univariate identity is already contained in the cited support-enumerator and root-polytope results; it is Part II's Remark 11.1. Here it follows from the weaker total-count input by Boolean inversion.

For a matchable pair, set

$$U = A \setminus B, \quad V = B \setminus A, \quad W = A \cap B.$$

Cancellation says exactly that $(U, V) \in \mathcal{T}(\tau)$, while W is an arbitrary subset of $E \setminus (U \cup V)$. Conversely, every such triple gives $A = U \cup W$, $B = V \cup W$ and a matchable pair. Hence

$$\sum_{A \rightsquigarrow_\tau B} z_B = \sum_{(U, V) \in \mathcal{T}(\tau)} z_V \sum_{W \subseteq E \setminus (U \cup V)} z_W,$$

which is a second derivation of Theorem 48.2. This proof and the support-fibre proof share the classical counting input but not the same preorder reduction; manuscript 01 stresses that the second route is not an independent replacement for the classical input.

51.3 Cancellation characterizes transitivity

Proposition 51.4 (Manuscript 06). *Let R be reflexive on a finite set. Common-support cancellation holds for all equally sized pairs if and only if R is transitive.*

Proof. The transitive case is Corollary 51.2. If R is not transitive, choose aRb and bRc with aRc false. Reflexivity ensures that a, b, c are distinct. The pair

$$A = \{a, b\}, \quad B = \{b, c\}$$

is matched by $a \mapsto b$ and $b \mapsto c$. After deleting the common vertex b , the pair $(\{a\}, \{c\})$ is not matchable. Thus cancellation fails already on a two-edge path. $\square$

52 Boolean orbits and a recovered Ehrhart interpretation

(Manuscript 01 only.)

Theorem 52.1 (Boolean orbits). *Let $\mathcal{M}_\tau$ be the set of compatible ordered pairs (A, B) , $A \rightsquigarrow_\tau B$, graded by $|B|$. For each $v \in E$, define a map T_v as follows: if v belongs to exactly one of A, B , leave the pair unchanged; otherwise toggle its membership in both sets. Then each T_v is a well-defined involution of $\mathcal{M}_\tau$, the involutions commute, and their orbits are indexed by the disjoint compatible pairs (U, V) . The orbit of a core of size $|U| = |V| = k$ is a Boolean family of dimension $n - 2k$, with rank polynomial $t^k(1 + t)^{n-2k}$.*

Proof. Each toggle preserves the disjoint core, so Corollary 51.2 proves that it preserves compatibility. Toggling twice returns the original pair. Different toggles act on different element memberships and commute. For fixed core (U, V) , the common subset C can be chosen arbitrarily in the complement of $U \cup V$; the toggles there add or remove its elements. The resulting orbit consists of all 2^{n-2k} choices, and $|B| = |V| + |C|$. $\square$

This is an explicit combinatorial orbit interpretation of the gamma coefficients. It is an action on the *support-pair model*, not an explicit action on $\mathcal{Q}_\tau \cap \mathbb{Z}^E$. The counting bridge supplies equal support-fiber cardinalities, but transporting the toggles to lattice points naturally would require an additional bijection. No such bijection is claimed here (compare Research question 21).

52.1 A recovered Ehrhart interpretation

Let $B_{\widehat{G}_\tau}$ be the augmented edge polytope of the relation graph. Its dimension is $2n$: its connected graph has $2n + 2$ vertices, and the sums on the two shores each equal one. Combining (14.4) and (51.4) gives

$$h^*(B_{\widehat{G}_\tau}; t) = h_\tau(t). \quad (52.1)$$

This is a geometric consequence of the counting proof, not an identification of h_τ with $h^*(\mathcal{Q}_\tau; t)$. Root-polytope interpretations of support enumerators are already part of the primary literature [2]; the importance here is the additional explicit gamma interpretation. [Merge note: $B_{\widehat{G}_\tau}$ is Part I's B_τ of (2.1) with its two coordinate blocks exchanged, a lattice isomorphism, so (52.1) is Part I's Proposition 2.2.]

In particular,

$$\text{Vol}_{\mathbb{Z}}(B_{\widehat{G_\tau}}) = |Q_\tau \cap \mathbb{Z}^E| = \sum_k \gamma_k(\tau) 2^{n-2k}, \quad (52.2)$$

$$|mB_{\widehat{G_\tau}} \cap \mathbb{Z}^{2n+2}| = \sum_{j=0}^n [t^j] h_\tau(t) \binom{m+2n-j}{2n} \quad (m \in \mathbb{Z}_{\geq 0}). \quad (52.3)$$

The lattice in the second line is restricted automatically to the affine span of the dilation. These are standard consequences of the Ehrhart numerator identity. A $2n$ -dimensional root-polytope realization does not, by itself, settle the flag n -dimensional simplicial-polytope question. (Manuscript 01 writes NVol for the normalized volume, which is Part I's $\text{Vol}_{\mathbb{Z}}$ of (7.4).)

53 An exact classification for balanced bipartite graphs

For reflexive relations, failure of transitivity is visible in two coefficients of the matching-support polynomial. This is stronger than merely exhibiting a nonpalindromic counterexample. This section is manuscript 06's.

Lemma 53.1 (Near-perfect supports detect reachability). *Let R be reflexive on E . For $a, b \in E$, there is a perfect matching from $E \setminus \{a\}$ to $E \setminus \{b\}$ using R if and only if $b \bar{R} a$.*

Proof. When $a = b$, diagonal edges give the matching and reachability is reflexive. Assume $a \neq b$. A matching is a bijection $f : E \setminus \{a\} \rightarrow E \setminus \{b\}$ with $iRf(i)$. Draw its edges as arrows on E . The vertex b has one outgoing arrow and no incoming arrow; a has one incoming arrow and no outgoing arrow; every other vertex has one of each. Starting at b and following arrows must reach a . It cannot enter a directed cycle: the first cycle vertex entered would have two incoming arrows, unless the path started on the cycle, which is impossible at b . Thus there is an R -path from b to a .

Conversely, choose a simple directed path $b = x_0 R x_1 R \cdots R x_\ell = a$. Match x_i to x_{i+1} along the path, and match every vertex outside the path to itself. These edges match precisely $E \setminus \{a\}$ onto $E \setminus \{b\}$. $\square$

Proof of Theorem 48.3. Matchable singleton pairs are the edges of G_R , so $[t]p_R = |R|$. Pairs of size $n-1$ are indexed uniquely by their omitted labels (a, b) . Lemma 53.1 makes their number $|\bar{R}|$, proving (48.8) and (48.9).

Transitivity is equivalent to cancellation by Proposition 51.4. Under transitivity, the decomposition $(A, B) \leftrightarrow (U, V, W)$ in Section 51, used directly on (48.7), gives

$$p_R(t) = \sum_{(U,V) \in \mathcal{T}(R)} t^{|U|} (1+t)^{n-2|U|}.$$

This implication uses no demand or root-polytope identity. Thus transitivity implies gamma-positivity, which implies palindromicity and hence equality of the coefficients of degrees 1 and $n-1$. If those coefficients are equal, then $|\bar{R}| = |R|$. Since $R \subseteq \bar{R}$ and both sets are finite, $R = \bar{R}$; that is, R is transitive. All five conditions are equivalent. $\square$

Corollary 53.2 (Balanced graphs with a perfect matching). *Let $H = (X \sqcup Y, F)$ be a bipartite graph with $|X| = |Y| = n$, admitting a perfect matching. Label the shores $x_1, \dots, x_n$ and $y_1, \dots, y_n$ so that $x_i y_i$ are the edges of a chosen perfect matching, and define iRj by $x_i y_j \in F$. The following are equivalent:*

- (i) *the perfectly matchable set polynomial $p_H(t)$ is palindromic;*

- (ii) $p_H(t)$ is gamma-positive;
- (iii) the relation R is a preorder.

These conditions do not depend on the chosen perfect matching. They are also equivalent to palindromicity, or to gamma-positivity, of the demand support polynomial $F_H(t)$.

Proof. The chosen matching supplies every diagonal relation, so Theorem 48.3 applies. The polynomial p_H is unchanged by relabeling or by a different choice of distinguished matching. Thus the criterion must hold for every such choice whenever it holds for one. Finally, Proposition 49.3 gives $F_H = p_H$. $\square$

This answers the palindromicity/gamma-positivity part of the support-enumerator classification question in [2, Problem 5.3] for the balanced perfect-matching class. It is not a classification for unbalanced graphs or for balanced graphs without perfect matchings, nor is it a real-rootedness criterion. The matching-theoretic ingredients are standard; the explicit endpoint-defect interpretation and its application here are supplied with proofs, without a separate claim of absolute historical priority.

Example 53.3 (A missing transitive edge). Let $E = \{a, b, c\}$, with all loops and just the additional relations aRb and bRc . There are five related pairs, but its transitive closure has six. Directly,

$$p_R(t) = 1 + 5t + 6t^2 + t^3.$$

The defect $6 - 5 = 1$ detects the single missing pair (a, c) . Adding that pair produces the three-element chain, with

$$p_{\overline{R}}(t) = 1 + 6t + 6t^2 + t^3 = (1 + t)^3 + 3t(1 + t).$$

This example illustrates why one cannot simply drop transitivity in the preorder theorem. It also shows why using a Hasse diagram instead of the full order relation changes the polynomial.

[Merge note, not in manuscript 06: this is Part I's nonpalindromic example of subsection 8.5. There D_i is the set of j with jRi , so $D_1 = \{1\}$, $D_2 = \{1, 2\}$, $D_3 = \{2, 3\}$ is the relation above with $(a, b, c) = (1, 2, 3)$, and by Proposition 49.3 its draconian support counts $(1, 5, 6, 1)$ are the coefficients of p_R : $(1, |R|, |\overline{R}|, 1)$. Part I's explanation of what fails (the reduction to ideal differences) is Proposition 51.4 in another form. The endpoint formula (48.8) and the equivalence of palindromicity with transitivity were also checked by the batch-44 dossier on all 4,165 reflexive relations with $1 \leq n \leq 4$; that check is not shipped.]

54 Positive formulas for equivalence-class blowups

A preorder is a poset of equivalence classes, each carrying a positive integer size. The signed-vector formula compresses the gamma polynomial to this quotient without sacrificing positivity.

Let P be the quotient poset of τ , with blocks E_v of sizes $s_v \geq 1$, and let $N = \sum_{v \in P} s_v$. Ideals of τ are exactly unions of blocks whose indices form ideals of P . Define

$$\mathcal{D}(P, \mathbf{s}) = \left\{ d \in \mathbb{Z}^P : \begin{array}{l} |d_v| \leq s_v \text{ for all } v, \quad \sum_v d_v = 0, \\ \sum_{v \in I} d_v \geq 0 \text{ for every } I \in \mathcal{J}(P) \end{array} \right\}, \quad D(d) = \sum_v \max(d_v, 0). \quad (54.1)$$

For $0 \leq r \leq s$, put

$$B_{s,r}(u) = \sum_{j=0}^{\lfloor (s-r)/2 \rfloor} \frac{s!}{j!(j+r)!(s-r-2j)!} u^j. \quad (54.2)$$

All these coefficients are multinomial integers.

Theorem 54.1 (Quotient-block formula; manuscript 06). *For every finite preorder with quotient data $(P, \mathbf{s})$,*

$$\Gamma_\tau(u) = \sum_{d \in \mathcal{D}(P, \mathbf{s})} u^{D(d)} \prod_{v \in P} B_{s_v, |d_v|}(u). \quad (54.3)$$

This applies to arbitrary quotient height and arbitrary nontrivial block sizes.

Proof. For a transport pair (U, V) , put $a_v = |U \cap E_v|$, $b_v = |V \cap E_v|$, and $d_v = a_v - b_v$. Lemma 51.1 says that the pair is matchable exactly when its block differences lie in (54.1): all labels in a block have the same comparisons to every other block, and all ideals are block unions.

For fixed d_v , let $j_v = \min(a_v, b_v)$. The counts in the block are then $j_v + |d_v|$ of one sign, j_v of the other, and $s_v - |d_v| - 2j_v$ zeros. The number of assignments of these signs to the labeled elements is

$$\frac{s_v!}{j_v!(j_v + |d_v|)!(s_v - |d_v| - 2j_v)!}.$$

Every assignment has the same feasibility status, because the ideal conditions depend only on d . The common cardinality of the two supports is

$$|U| = \sum_v (\max(d_v, 0) + j_v) = D(d) + \sum_v j_v.$$

Multiplying the local counts and summing first over j_v , then over d , proves the formula. $\square$

54.1 A companion formula in the ordinary basis

For $0 \leq r \leq s$, define

$$A_{s,r}(t) = \sum_{j=0}^{s-r} \binom{s}{j+r} \binom{s}{j} t^j. \quad (54.4)$$

Extracting the coefficient of z^r in the Laurent identity

$$(1+z)^s (1+t/z)^s = (1+t+z+t/z)^s$$

gives the positive gamma expansion

$$A_{s,r}(t) = \sum_{j=0}^{\lfloor (s-r)/2 \rfloor} \frac{s!}{j!(j+r)!(s-r-2j)!} t^j (1+t)^{s-r-2j}. \quad (54.5)$$

To see the right-hand count explicitly, choose $j+r$ copies of z , j copies of t/z , and $s-r-2j$ copies of $1+t$ in the second expression. The first expression yields (54.4).

Since $\sum_v |d_v| = 2D(d)$, substitution into Theorem 54.1 gives

$$h_\tau(t) = \sum_{d \in \mathcal{D}(P, \mathbf{s})} t^{D(d)} \prod_{v \in P} A_{s_v, |d_v|}(t). \quad (54.6)$$

No rational functions remain in this identity. Each summand has center $N/2$ because its degree after the shift is $N - D(d)$ and its smallest possible exponent is $D(d)$.

Example 54.2 (A nontrivial height-two blowup). Take two equivalence classes of size two, with every member of the first below every member of the second. The feasible differences are $(r, -r)$, $r = 0, 1, 2$. Since

$$B_{2,0} = 1 + 2u, \quad B_{2,1} = 2, \quad B_{2,2} = 1,$$

we get

$$\Gamma_\tau(u) = (1 + 2u)^2 + 4u + u^2 = 1 + 8u + 5u^2,$$

and therefore

$$h_\tau(t) = 1 + 12t + 27t^2 + 12t^3 + t^4.$$

The quotient has height two, but the preorder is not a height-two poset. It lies beyond the scope of Part II's height-two theorem. The example also exposes the importance of counting ordered support pairs inside an equivalence class.

54.2 Algorithmic consequence and its encoding qualification

The empty quotient gives $\Gamma_\tau = 1$. With $r = |P| \geq 1$ fixed, (54.3) computes Γ_τ in a number of integer arithmetic operations polynomial in the expanded size N . Enumerate the ideals of the r -element quotient once. Enumerate $d_1, \dots, d_{r-1}$ and determine d_r from the zero-sum equation, rejecting out-of-range values and failed ideal inequalities. There are at most $(2N + 1)^{r-1}$ candidates. For each accepted candidate multiply the local polynomials, using ordinary polynomial multiplication.

For example, a coarse bound after local precomputation is

$$O(2^r r^2 + (2N + 1)^{r-1}(r2^r + rN^2))$$

arithmetic operations. This is an arithmetic, not bit-operation, estimate. It is polynomial for fixed quotient size when N measures the expanded input. If the block sizes are encoded in binary, the bound is only pseudopolynomial in that compressed input; no stronger complexity assertion is intended.

54.3 The profile form of the block formula (manuscript 01)

Manuscript 01 proves an equivalent block formula with a different parametrization and proof. Let $P = \tau/\sim$ be the quotient poset, where $u \sim v$ means both $u \leq_\tau v$ and $v \leq_\tau u$. Index its equivalence classes by $1, \dots, b$, with positive sizes $m_1, \dots, m_b$. Relations between two classes are uniform: every element of class i lies below every element of class j precisely when $i \leq_P j$. Set $n = \sum_i m_i$.

For a disjoint pair (U, V) , let $\alpha_i = |U \cap E_i|$ and $\beta_i = |V \cap E_i|$. These profiles satisfy

$$\alpha_i, \beta_i \in \mathbb{Z}_{\geq 0}, \quad \alpha_i + \beta_i \leq m_i, \quad \sum_i \alpha_i = \sum_i \beta_i.$$

For $T \subseteq [b]$, set $\downarrow_P T = \{i : i \leq_P j \text{ for some } j \in T\}$.

Theorem 54.3 (Exact block formula; manuscript 01). *Let $\mathcal{A}(P, \mathbf{m})$ consist of the profiles above which also satisfy*

$$\sum_{j \in T} \beta_j \leq \sum_{i \in \downarrow_P T} \alpha_i \quad (T \subseteq [b]). \quad (54.7)$$

Then

$$\Gamma_\tau(u) = \sum_{(\alpha, \beta) \in \mathcal{A}(P, \mathbf{m})} \left(\prod_{i=1}^b \frac{m_i!}{\alpha_i! \beta_i! (m_i - \alpha_i - \beta_i)!} \right) u^{\sum_i \beta_i}. \quad (54.8)$$

Proof. A matching covering the chosen receivers can use exactly the selected sources in predecessor classes. For the set of all selected receivers in classes T , Hall's inequality is (54.7). These inequalities are also sufficient: any subset of selected receivers uses some class set T and has at most $\sum_{j \in T} \beta_j$ members; its neighbors are all selected sources in $\downarrow_P T$. Thus Hall's criterion applies to the individual selected vertices. Equality of the total source and receiver counts makes a receiver-covering matching a bijection.

For a fixed feasible profile, the choices of disjoint sources and receivers in class i number $\binom{m_i}{\alpha_i} \binom{m_i - \alpha_i}{\beta_i}$. Compatibility depends only on the profiles, not on the chosen labels, because class-to-class relations are uniform. Multiplying these counts and summing the feasible profiles gives (54.8). $\square$

This is a closed finite formula, not a claim that the quotient poset alone determines h_τ : the multiplicities m_i are indispensable. Replacing the classes by singletons loses both the profile range and the multinomial weights. [Merge note: grouping the profiles of (54.8) by $d_i = \alpha_i - \beta_i$ and $j_i = \min(\alpha_i, \beta_i)$ gives (54.3); (54.7) is Hall's condition on receivers, whereas (54.1) is the ideal form of Lemma 51.1.]

Algorithmic consequence (manuscript 01). For fixed b , there are at most

$$\prod_{i=1}^b \binom{m_i + 2}{2} = O(n^{2b}) \quad (54.9)$$

profiles before imposing equal totals. Testing a profile by (54.7) takes at most 2^b subset inequalities, each involving at most b additions after the lower closures are precomputed. Therefore (54.8) computes the entire gamma polynomial in polynomial time in the *expanded cardinality* n for a fixed number of classes. This is not a bound polynomial in the bit length of binary-encoded m_i , and it is not a polynomial bound when b varies with n . The weights and their sums use exact integers. The universal gamma bound is at most 3^n , since a disjoint pair assigns each element to source, receiver, or neither, so the output coefficients have $O(n)$ bits. Large block sizes therefore enlarge the enumerated state space, but do not create uncontrolled coefficient bit lengths relative to expanded n .

Example 54.4 (Examples beyond Part II's domain; manuscript 01). For a chain of three quotient classes of sizes $(2, 1, 2)$, the formula gives

$$\Gamma_\tau(u) = 1 + 12u + 17u^2, \quad h_\tau(t) = 1 + 17t + 63t^2 + 63t^3 + 17t^4 + t^5.$$

The quotient has height three and two classes are nontrivial. Both restrictions excluded by Part II's height-two-poset proof fail here.

For two incomparable classes of sizes 2 and 1 below a class of size 3,

$$\Gamma_\tau(u) = 1 + 17u + 36u^2 + 7u^3,$$

$$h_\tau(t) = 1 + 23t + 119t^2 + 201t^3 + 119t^4 + 23t^5 + t^6.$$

Its comparability graph has a perfect matching. Hence $h_\tau(-1) = -7$, agreeing with Corollary 56.2. The companion program checks these values both by direct ideal-inequality lattice enumeration and by the block formula.

55 Monotonicity, duality, extremal coefficients, and basic families

The transport interpretation gives comparisons between different preorders without comparing their lattice points individually.

Theorem 55.1 (Monotonicity and products; manuscripts 06 and 01). *Suppose τ and σ are preorders on the same set and $\leq_\tau \subseteq \leq_\sigma$. Then*

$$\gamma_k(\tau) \leq \gamma_k(\sigma) \quad \text{for every } k. \quad (55.1)$$

More precisely $h_\sigma - h_\tau$ has a nonnegative expansion in the n -centered gamma basis. If the relations differ, then $\gamma_1(\tau) < \gamma_1(\sigma)$, so the two gamma vectors and the two support polynomials differ (manuscript 01). Moreover, for disjoint unions and opposites,

$$\Gamma_{\tau \sqcup \sigma}(u) = \Gamma_\tau(u) \Gamma_\sigma(u), \quad \Gamma_{\tau^*}(u) = \Gamma_\tau(u). \quad (55.2)$$

Proof. Every matching allowed by τ is allowed by σ , so $\mathcal{T}_k(\tau) \subseteq \mathcal{T}_k(\sigma)$. For $k = 1$, these pairs are exactly the distinct-element relations; since both preorders contain all diagonal relations, a proper inclusion adds at least one such pair. Subtract the two expansions in (48.4). In a disjoint union, matchings cannot cross components. Thus each transport support pair splits uniquely into a transport pair in each component, and cardinalities add. Finally, duality sends (U, V) to (V, U) and reverses a witnessing bijection. $\square$

The corresponding duality for h was already proved in [2]; this is a transport-support explanation of it, not a first proof claim. The monotonicity also implies coefficientwise monotonicity of h , since every gamma-basis polynomial has nonnegative coefficients; that weaker statement also follows from $\mathcal{Q}_\tau \subseteq \mathcal{Q}_\sigma$, because every σ -ideal is a τ -ideal. Manuscript 01 notes that the theorem is stronger: the difference remains nonnegative in the more restrictive gamma basis.

Theorem 55.2 (Multivariate complement-duality; manuscript 01). *Let τ^* be the opposite preorder. Then*

$$H_\tau(\mathbf{z}) = z_E H_{\tau^*}(\mathbf{z}^{-1}). \quad (55.3)$$

In particular $\gamma_k(\tau) = \gamma_k(\tau^)$ and $h_\tau = h_{\tau^*}$.*

Proof. The map $(U, V) \mapsto (V, U)$ is a bijection from $\mathcal{T}_k(\tau)$ to $\mathcal{T}_k(\tau^*)$, since a compatible bijection can be inverted. In the term of (48.5) indexed by (V, U) for τ^* , invert all variables and multiply by z_E . With $F = E \setminus (U \cup V)$, the term becomes

$$z_E z_U^{-1} \prod_{w \in F} (1 + z_w^{-1}) = z_V \prod_{w \in F} (1 + z_w),$$

which is exactly the corresponding term for τ . The Laurent expression simplifies to a polynomial. The univariate conclusions also follow directly by swapping the disjoint pairs. $\square$

Equivalently, the number of lattice points of exact support S for τ equals the number of exact support $E \setminus S$ for τ^* . This does *not* assert that H_τ itself is invariant under complementing the support. For the chain $1 < 2$, $H_\tau = 1 + z_1 + 2z_2 + z_1z_2$, which illustrates the distinction. [Merge note: the univariate identity $h_\tau = h_{\tau^*}$ is Conjecture 5.4 of [1], settled by [2, Theorem 1.3]; see Part I, Section 7.]

Proposition 55.3 (Disjoint unions; manuscript 01). *For preorders on disjoint sets with no cross-relations,*

$$H_{\tau \sqcup \sigma} = H_\tau H_\sigma, \quad h_{\tau \sqcup \sigma} = h_\tau h_\sigma, \quad \Gamma_{\tau \sqcup \sigma} = \Gamma_\tau \Gamma_\sigma.$$

Adjoining an isolated element multiplies h by $1 + t$ and leaves Γ unchanged.

Proof. The defining ideal inequalities separate, giving a product of polytopes. Alternatively a compatible pair and its matching separate into the two ground sets, with equal source and receiver cardinalities within each. Disjoint compatible cores therefore multiply as well. $\square$

Corollary 55.4 (The first gamma coefficient; manuscripts 06 and 01). *For every preorder,*

$$\gamma_0(\tau) = 1, \quad \gamma_1(\tau) = \#\{(u, v) : u \neq v, u \leq_\tau v\} = |\tau| - n. \quad (55.4)$$

In particular, $h_1(\tau) = |\tau| = \sum_{v \in E} |\downarrow\{v\}|$. Equivalent distinct elements contribute in both directions to γ_1 .

Proof. The $k = 1$ transport pairs are exactly the non-diagonal related pairs. The coefficient of t in (48.4) is $n + \gamma_1$. Directly (manuscript 01): for a fixed single-element support $\{v\}$, the admissible positive coordinate values are $1, \dots, |\downarrow\{v\}|$, because every ideal containing v contains its principal ideal. [Merge note: this is Part I's formula (1.4).] $\square$

Proposition 55.5 (The second gamma coefficient; manuscript 01). *Let $D_2(\tau)$ count unordered pairs of directed distinct-element relations with four distinct endpoints. Let $R_{2,2}(\tau)$ count ordered disjoint pairs (U, V) with $|U| = |V| = 2$ and every relation $u \leq_\tau v, u \in U, v \in V$, present. Then*

$$\gamma_2(\tau) = D_2(\tau) - R_{2,2}(\tau). \quad (55.5)$$

Proof. Every pair counted by D_2 is a matching witnessing some disjoint (U, V) of size two. Such a support pair has two witnessing bijections exactly when all four possible source-to-receiver relations exist; otherwise it has one. Subtracting the number of doubled pairs gives (55.5). $\square$

[Merge note: for a height-two poset P_G this is Part II's $\gamma_2 = m_2(G) - c_4(G)$ of Corollary 15.2: $D_2 = m_2(G)$, and $R_{2,2}$ counts the four-cycles.]

Theorem 55.6 (Sharp coefficientwise bounds; manuscripts 06 and 01). *For every preorder on n elements and $0 \leq k \leq \lfloor n/2 \rfloor$,*

$$0 \leq \gamma_k(\tau) \leq \frac{n!}{k!^2(n-2k)!} = \binom{n}{2k} \binom{2k}{k}, \quad \binom{n}{j} \leq [t^j] h_\tau(t) \leq \binom{n}{j}^2 \quad (0 \leq j \leq n). \quad (55.6)$$

For an ordinary poset the sharper bound is

$$\gamma_k(\tau) \leq \binom{n}{2k} C_k, \quad C_k = \frac{1}{k+1} \binom{2k}{k}. \quad (55.7)$$

The upper bounds are attained simultaneously for all k by, respectively, the universal equivalence preorder $\mathbf{E}_n$ and a total order L_n . The antichain attains $\gamma_k = 0$ for all $k \geq 1$, and the lower bounds on h . For $n \geq 2$, equality of the complete gamma vector with the upper or lower extremal vector characterizes the corresponding extremal preorder, and equality with the chain vector characterizes a chain among posets (manuscript 01).

Proof. There are $\binom{n}{k} \binom{n-k}{k} = n!/(k!^2(n-2k)!)$ ordered pairs of disjoint k -element sets, which gives the first upper bound. Every such pair is matchable in the universal equivalence preorder. The antichain relation is contained in every preorder, and every preorder is contained in $\mathbf{E}_n$; Theorem 55.1 gives the bounds on h from (55.8) and (55.11) below. Alternatively, the unit cube is contained in $\mathbf{Q}_\tau$, while $\mathbf{Q}_\tau$ is contained in the full simplex $\{x \geq 0 : x(E) \leq n\}$ (manuscript 01).

Every finite poset has a linear extension, obtained recursively by removing a minimal element. Its relation is a subset of that total order. It is therefore enough, by Theorem 55.1, to compute the chain case. Choose the $2k$ nonzero positions in the signed-vector formula (51.3). In their increasing order, the signs must have total zero and nonnegative partial sums. There are C_k such words: among the $\binom{2k}{k}$ balanced sign words, reflection up to the first negative partial sum bijects the bad ones with words having $k+1$ plus signs and $k-1$ minus signs. Hence the

good count is $\binom{2k}{k} - \binom{2k}{k-1} = C_k$. This proves (55.7) and sharpness. In an antichain a disjoint nonempty transport pair cannot be matched, since the only relation edges are loops.

The characterizations follow from γ_1 : zero means that no off-diagonal relations exist, $n(n-1)$ means that all of them exist, and $\binom{n}{2}$ for a poset forces total comparability. $\square$

Second proof of the chain count, by initial segments (manuscript 01). Manuscript 01 states $\gamma_k(L_n) = \binom{n}{2k} C_k$ as its Proposition “The chain gamma vector”. Fix the $2k$ elements in $U \cup V$, and list them in increasing order. A compatible matching exists exactly when every initial segment contains at least as many elements of U as of V . Necessity holds because a target must be matched to an earlier source. For sufficiency, pair the i th source with the i th target; the initial-segment inequalities force the source to precede the target. Among the $\binom{2k}{k}$ words with k source letters and k target letters, the words whose running source-minus-target balance never becomes negative are counted by $\binom{2k}{k} - \binom{2k}{k-1} = C_k$; the subtraction follows by reflecting the prefix ending at the first negative balance. Multiply by the choice of $U \cup V$. Relation-monotonicity under a linear extension gives the bound. $\square$

55.1 Explicit consistency checks from classical families

For an antichain, $\mathcal{Q}_\tau = [0, 1]^n$, so

$$h_\tau(t) = (1+t)^n, \quad \Gamma_\tau(u) = 1. \quad (55.8)$$

For a total order, the preceding proof gives

$$\Gamma_\tau(u) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} C_k u^k. \quad (55.9)$$

This agrees with the classical Narayana specialization for a chain,

$$[t^j] h_{L_n}(t) = \binom{n}{j}^2 - \binom{n}{j-1} \binom{n}{j+1} = \frac{1}{n+1} \binom{n+1}{j} \binom{n+1}{j+1}, \quad (55.10)$$

recorded among the elementary families of [1]; the identity also follows by expanding (55.9). Here a binomial coefficient with lower index outside its usual range is zero. Manuscript 01 gives a direct proof using (51.4): if the two j -subsets are $A = \{a_1 < \dots < a_j\}$ and $B = \{b_1 < \dots < b_j\}$, compatibility is equivalent to $a_i \leq b_i$ for all i , or to $|A \cap [r]| \geq |B \cap [r]|$ for every r . There are $\binom{n}{j}^2$ pairs in total. For a violating pair, exchange the two memberships on the first prefix for which the target count exceeds the source count by one. This is a bijection to a pair of subsets of sizes $j+1$ and $j-1$; the inverse uses the first prefix with source surplus one. This proves the first equality, and factorial simplification proves the second. Thus (55.10) gives sharp ordinary coefficient bounds for posets as well. These classical formulas are checks, not novelty claims.

For the universal equivalence preorder $\mathbf{E}_n$, the only ideals are $\emptyset$ and E , so $\mathcal{Q}_\tau = \{x \geq 0 : x(E) \leq n\}$. Choosing a support of size j and then its positive coordinates gives

$$h_{\mathbf{E}_n}(t) = \sum_{j=0}^n \binom{n}{j}^2 t^j, \quad \Gamma_{\mathbf{E}_n}(u) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{k!^2 (n-2k)!} u^k. \quad (55.11)$$

For the first formula, subtract one on the chosen support; the number of resulting nonnegative j -tuples of total at most $n-j$ is $\binom{n}{j}$. The second follows directly from unrestricted disjoint transport pairs. These are familiar special-case polynomials (Part I’s indiscrete preorder, Section 8); the point is their role as universal coefficientwise bounds.

Table 4: Four-element examples (manuscript 06). Each h is $1 + h_1t + h_2t^2 + h_1t^3 + t^4$, and each Γ is $1 + \gamma_1u + \gamma_2u^2$.

Preorder	h_1	h_2	γ_1	γ_2
Antichain	4	6	0	0
Diamond poset	9	18	5	2
Total order	10	20	6	2
Two ordered blocks of size two	12	27	8	5
Universal equivalence	16	36	12	6

55.2 The octopus formula as an immediate specialization (manuscript 01)

Consider ℓ pairwise incomparable singleton leaves below a root equivalence class of size $n - \ell$. In a nonempty disjoint compatible pair, a receiver cannot be a leaf, because its only predecessor outside the root order is itself and sources and receivers are disjoint. Conversely, once V is any k -subset of the root, every k -subset $U \subseteq E \setminus V$ is compatible with it. Therefore

$$\gamma_k = \binom{n - \ell}{k} \binom{n - k}{k}. \quad (55.12)$$

This recovers the octopus formula attributed to Athanasiadis, Xiao, and Yan and reproved bijectively by Menon [6, Proposition 1.1]. It is not a new formula claimed here. The general theorem also applies when leaf blocks have arbitrary positive size and when the quotient is not a star; Section 54 gives the exact count without any unit-block assumption.

55.3 Recovery of the height-two matching-support formula

If τ is a poset of height at most two, write its strict relations as edges of a bipartite graph $G = (A \sqcup B, F)$ with $a < b$ on each edge. Every nonempty transport pair has $U \subseteq A$ and $V \subseteq B$. Therefore its union $U \cup V$ determines the ordered pair uniquely, and

$$\Gamma_\tau(u) = \sum_{\substack{W \subseteq A \sqcup B \\ G[W] \text{ has a perfect matching}}} u^{|W|/2} = p_G(u).$$

This recovers Part II's height-two result (Theorem 13.2) exactly. Beyond height two, the two signs of a transport pair need not occupy fixed shores of the original poset, which is why a fixed comparability-graph support count is no longer the right general definition. In a preorder with equivalence classes, the union may also admit several orientations; those are genuinely distinct ordered pairs and must not be collapsed. [Merge note: Part II's proof partitions a point by its zeros on the minimal shore A and its support on B (Section 15); Lemma 50.1 partitions by the whole zero set, which is what removes the height restriction.]

56 Matching deficiency and the exact root at minus one

Let $\text{Comp}(\tau)$ denote the *simple comparability graph*: its vertices are E , and distinct u, v are adjacent when $u \leq_\tau v$ or $v \leq_\tau u$. Write $\nu(\text{Comp}(\tau))$ for its maximum matching size.

Theorem 56.1 (Degree and root multiplicity; manuscripts 06 and 01). *For every finite preorder,*

$$\deg \Gamma_\tau = \nu(\text{Comp}(\tau)), \quad \text{ord}_{t=-1} h_\tau(t) = n - 2\nu(\text{Comp}(\tau)). \quad (56.1)$$

Furthermore $\gamma_k(\tau) > 0$ exactly for $0 \leq k \leq \nu(\text{Comp}(\tau))$.

Proof. A transport pair of size k has a witnessing directed matching on $2k$ distinct vertices. Forgetting orientations gives a k -matching in $\text{Comp}(\tau)$, so $\deg \Gamma_\tau \leq \nu(\text{Comp}(\tau))$. Conversely, orient each edge of an undirected matching in a permitted preorder direction. Its tails and heads form a disjoint transport pair. Submatchings supply every smaller size. Hence the degree is exactly $\nu = \nu(\text{Comp}(\tau))$.

Factor the smallest power of $1 + t$ from (48.4):

$$h_\tau(t) = (1 + t)^{n-2\nu} \left(\gamma_\nu t^\nu + \sum_{k < \nu} \gamma_k t^k (1 + t)^{2(\nu-k)} \right). \quad (56.2)$$

The expression in parentheses has value $(-1)^\nu \gamma_\nu \neq 0$ at -1 . Thus the displayed exponent is the exact multiplicity, not just a lower bound. $\square$

This reduces the gamma degree and the entire -1 multiplicity question to a single ordinary maximum-matching problem, even when the remaining coefficients are expensive to count. It uses the full comparability graph, not merely the cover graph. The maximum matching here is an ordinary matching in the *undirected* comparability graph. It is not a maximum matching in G_τ , whose reflexive diagonal already supplies a perfect matching of size n . Confusing these two graphs would destroy the degree formula (manuscript 01). [Merge note: in height two, $\text{Comp}(P_G) = G$, and this is the multiplicity $n - 2\nu(G)$ of Part II's Proposition 15.1.]

For example, a height-two star with m edges has $n = m + 1$ and maximum matching size one. Its polynomial is

$$h_\tau(t) = (1 + t)^{m-1} ((1 + t)^2 + mt),$$

so its exact multiplicity at -1 is $m - 1$. A chain has matching number $\lfloor n/2 \rfloor$, giving multiplicity zero for even n and one for odd n . Equivalent vertices cause no exception: one may choose either allowed direction when both are present.

Corollary 56.2 (The leading Taylor coefficient at -1 ; manuscript 01). *Put $d = n - 2\nu(\text{Comp}(\tau))$. Then*

$$h_\tau^{(j)}(-1) = 0 \quad (0 \leq j < d), \quad h_\tau^{(d)}(-1) = d!(-1)^{\nu(\text{Comp}(\tau))} \gamma_{\nu(\text{Comp}(\tau))}(\tau). \quad (56.3)$$

For even n , the signed value $(-1)^{n/2} h_\tau(-1)$ is positive exactly when $\text{Comp}(\tau)$ has a perfect matching, and is zero otherwise.

Proof. Equation (56.2) identifies the first nonzero coefficient of the Taylor expansion at -1 . If n is even, $d = 0$ is equivalent to $\nu = n/2$. $\square$

57 Induced restrictions and a deletion–differentiation law

(Manuscript 01 only.) For $W \subseteq E$, let $\tau|W$ be the induced preorder, retaining every relation of τ whose endpoints lie in W . A matching between specified subsets of W is compatible in $\tau|W$ exactly when it is compatible in τ . In particular, no choice of an intermediate element outside W is needed: the relation itself is already transitive.

Theorem 57.1 (Induced-deck identities). *For $0 \leq m \leq n$ and $k \geq 0$,*

$$\sum_{\substack{W \subseteq E \\ |W|=m}} \gamma_k(\tau|W) = \binom{n-2k}{m-2k} \gamma_k(\tau), \quad (57.1)$$

where the right side is zero when $2k > m$ or $2k > n$. In particular, $\gamma_k(\tau)$ is the sum of the top gamma coefficients of its induced $2k$ -element subpreorders. If

$$D_\tau(t) = \sum_{v \in E} h_{\tau|(E \setminus \{v\})}(t),$$

then

$$(1-t)D_\tau(t) = nh_\tau(t) - 2th'_\tau(t). \quad (57.2)$$

Proof. A fixed disjoint compatible pair of size k is counted in the induced preorder on W exactly when its $2k$ endpoints lie in W . There are $\binom{n-2k}{m-2k}$ such W , proving (57.1). For one-element deletions this gives

$$D_\tau(t) = \sum_{2k \leq n-1} (n-2k)\gamma_k(\tau)t^k(1+t)^{n-1-2k}.$$

On each gamma-basis term, direct differentiation yields

$$n t^k(1+t)^{n-2k} - 2t \frac{d}{dt} (t^k(1+t)^{n-2k}) = (n-2k)(1-t)t^k(1+t)^{n-1-2k}.$$

When $2k = n$, the left side is zero and that term is omitted on the right; there is no evaluation of a negative exponent. Sum the identities. $\square$

Corollary 57.2 (Odd-cardinality reconstruction). *If n is odd, the multiset of polynomials of the one-element deletions determines h_τ . Explicitly, writing $D_\tau(t) = \sum_j d_j t^j$ with $d_{-1} = d_n = 0$, one has*

$$[t^j]h_\tau(t) = \frac{d_j - d_{j-1}}{n-2j} \quad (0 \leq j \leq n). \quad (57.3)$$

For even n , the same formulas determine every coefficient except the middle coefficient; the differential identity alone leaves one scalar free.

Proof. Compare coefficients in (57.2). For odd n , none of the integers $n-2j$ is zero. For even n , exactly the middle one vanishes. $\square$

The final statement is a limitation of this identity, not a claim that the full combinatorial deletion deck never determines the middle coefficient. This separates polynomial reconstruction from the harder reconstruction of the preorder itself. (Here d_j are coefficients of D_τ , not the deficiency d of Corollary 56.2.)

58 Support size as a mixture of centered binomials

The Boolean expansion provides a probability law without assuming a canonical bijection on the original lattice points. Choose a lattice point uniformly from $Q_\tau \cap \mathbb{Z}^E$, and let L be its support size. All probabilities below refer to this uniform distribution. This section is manuscript 06's.

Theorem 58.1 (Exact mixture and concentration). *Let $\Gamma = \Gamma_\tau$ and $v = \deg \Gamma$. There is a mixing variable $K \in \{0, \dots, v\}$ with*

$$\mathbb{P}(K = k) = \frac{\gamma_k 4^{-k}}{\Gamma(1/4)} \quad (58.1)$$

such that the law of L is the mixture

$$L \mid K = k \stackrel{\text{law}}{=} k + \text{Bin}(n-2k, 1/2). \quad (58.2)$$

Consequently

$$|Q_\tau \cap \mathbb{Z}^E| = 2^n \Gamma(1/4), \quad (58.3)$$

$$\mathbb{E}L = \frac{n}{2}, \quad \text{Var } L = \frac{n}{4} - \frac{\Gamma'(1/4)}{8\Gamma(1/4)}, \quad (58.4)$$

$$\frac{n-2v}{4} \leq \text{Var } L \leq \frac{n}{4}. \quad (58.5)$$

For $n \geq 1$ and $a \geq 0$,

$$\mathbb{P}(|L - n/2| \geq a) \leq 2 \exp(-2a^2/n). \quad (58.6)$$

The upper variance bound is an equality exactly for antichains.

Proof. Evaluation of (48.4) at one gives (58.3). Divide (48.4) by that value. The k th summand has total mass $\gamma_k 2^{n-2k}/h_\tau(1)$, which is (58.1), and its normalized coefficients are those in (58.2). Every component has mean $n/2$ and variance $(n-2k)/4$. Thus the total variance is $n/4 - \mathbb{E}K/2$. Since

$$\mathbb{E}K = \frac{\Gamma'(1/4)}{4\Gamma(1/4)},$$

we obtain (58.4) and (58.5). Equality at the upper bound means all γ_k with $k > 0$ vanish. By Corollary 55.4, this is equivalent to having no non-diagonal comparabilities.

For the tail bound, the centered moment generating function of the k th component is $\cosh(s/2)^{n-2k}$. Since $\cosh x \leq \exp(x^2/2)$, this is at most $\exp(ns^2/8)$ for every real s . The same bound holds after mixing. For $s > 0$, Markov's inequality gives

$$\mathbb{P}(L - n/2 \geq a) \leq \exp(-sa + ns^2/8).$$

Choosing $s = 4a/n$ gives $\exp(-2a^2/n)$, and the lower tail has the same bound. Their sum proves (58.6). The elementary hyperbolic-cosine inequality follows, for example, by integrating $\tanh x \leq x$ for $x \geq 0$ and using evenness. $\square$

The mixture law is not a representation of L as a sum of a fixed family of independent Bernoulli variables. In particular it does not imply real-rootedness. Its value is that it gives exact central statistics and a uniform tail bound directly from the proven gamma expansion. [Merge note, not in manuscript 06: for a bipartite cactus G , Part II's (19.7) gives $\text{Var } S = (n-2v)/4 + \sum_j 2/(4+\lambda_j)$ for the same statistic ($S = L$), from $p_G(u) = \prod_j (1+\lambda_j u)$. Since then $\Gamma'(1/4)/\Gamma(1/4) = \sum_j 4\lambda_j/(4+\lambda_j)$, formula (58.4) gives $n/4 - \sum_j \lambda_j/(2(4+\lambda_j))$, the same number, because $\frac{1}{2} - \lambda_j/(2(4+\lambda_j)) = 2/(4+\lambda_j)$. Here $\Gamma(1/4)$ is Γ_τ evaluated at $1/4$, not Euler's Gamma function.]

59 Exact computations and reproducibility

Both manuscripts supply a standard-library Python program that uses exact integer arithmetic throughout its mathematical checks. Each is designed as a cross-check of several logically different descriptions, not as an unbounded theorem prover. The shipped files are:

Shipped path (delivered path)	Content
code/05-gamma-positivity-verify.py (code/verify.py)	manuscript 06's verifier
data/05-gamma-positivity-preorders.csv (data/preorders.csv)	every labeled preorder through size five with its h - and γ -vectors (CRLF line endings, as delivered)
data/05-gamma-positivity-verification.json, data/05-gamma-positivity-verification.log	the recorded run: detailed results with all larger samples; its console transcript
05-gamma-positivity-PROOF_STATUS.md, 05-gamma-positivity-SOURCE_AUDIT.md	manuscript 06's theorem-dependency and source-audit notes
code/05-gamma-positivity-build.py (build.py)	manuscript 06's three-pass build script (for its own article; do not run here)
code/04-gamma-vector-verify.py (code/verify.py)	manuscript 01's verifier
data/04-gamma-vector-verification.json, data/04-gamma-vector-verification.txt (results/verification.json, results/verification.txt)	manuscript 01's recorded run and its console output
code/04-gamma-vector-build.sh (build.sh)	manuscript 01's two-pass build script (for its own article; do not run here)

59.1 Manuscript 06: exhaustive preorder tests

The program enumerates every reflexive relation on a labeled n -element set and retains exactly those that are transitive. For each resulting preorder it independently enumerates all nonnegative integer vectors of total at most n and tests every ideal inequality. It also enumerates ordered disjoint support pairs and decides matchability by recursive bipartite matching, without using the ideal Hall criterion in that matching routine. The resulting counts are compared in the following ways:

1. Each individual support fibre is compared with the Boolean-interval count in (50.3); this tests the multivariate identity, not only a specialization.
2. Gamma expansion, the quotient-block formula, and both sharp upper bounds are checked. Matchability is also compared with the ideal Hall test.
3. All equal-size support pairs in the doubled graph are tested for common-support cancellation, and its polynomial is compared with h_τ . An independent undirected matching recursion checks $\deg \Gamma_\tau = \nu(\text{Comp}(\tau))$.

Table 5: Manuscript 06's recorded exhaustive run. A support fibre is specified by a preorder and a subset of its labeled ground set.

n	Preorders	Support fibres	Equal-size doubled pairs
0	1	1	1
1	1	2	2
2	4	16	24
3	29	232	580
4	355	5,680	24,850
5	6,942	222,144	1,749,384
Total	7,332	228,075	1,774,841

All comparisons passed. These are labeled preorders, including nontrivial equivalence

classes; the totals do not count only unlabeled posets. The accompanying CSV contains every tested preorder and both coefficient vectors, not merely summary totals.

Additional independent scopes. The demand/support identity, including its prescribed-support refinement, was tested on all 689 bipartite graphs with each labeled shore of size at most three, including empty shores. The run counts 5,880 feasible demand vectors across those graphs.

The full transitivity/cancellation/palindromicity characterization and both endpoint coefficient formulas were tested on all 4,166 reflexive relations through size four. This includes nontransitive relations, so it tests the negative direction rather than assuming the input is a preorder. [Merge note: 4,166 counts the empty relation at $n = 0$; there are 4,165 with $1 \leq n \leq 4$.]

A further 140 distinct deterministic samples have sizes six, seven, and eight: respectively 46, 47, and 47 examples. They were generated with seed 20260929 by taking transitive closures of both arbitrary directed relations and acyclic ones. This is not uniform sampling from preorders. Every sampled relation is listed in the JSON output. The direct support counts, gamma formula, block formula, ideal Hall criterion, extremal bounds, and matching-number identity were checked on these samples. Exhaustive all-pairs doubled cancellation was reserved for the small-size run.

Finally, the binomial and Narayana specializations were compared with their gamma expansions through $n = 20$. These last tests are identities between explicit formulas, not exhaustive lattice-point enumerations in dimension 20. No root approximations, floating-point certificates, or randomized claims of proof are used.

Files and execution. The delivered command, run from the package root, was `python code/verify.py -max-n 5 -out data`. In this directory, pass a new output directory, so that the shipped record is preserved (the default `-out data` is relative to the working directory and would write unprefix files):

```
py code/05-gamma-positivity-verify.py -max-n 5 -out /path/to/scratch
```

Python 3.10 or later is required. No external Python packages are needed. The program refuses to run under `python -O`, because it uses assertions. It deliberately rejects exhaustive sizes above five, whose relation search grows too rapidly for its intended role. The article's production algorithm can instead enumerate signed pairs and use an augmenting-path matching test, requiring at most $O(3^n n^3)$ elementary operations and polynomial workspace; the verifier uses simple recursive matching because its small-instance independence is more important than asymptotic speed.

The package also contains `README.md`, `SOURCE_AUDIT.md`, `PROOF_STATUS.md`, a build script, and the compiled PDF. The theorem proofs are in this article, not in the test log. [Merge note: the two notes and the build script are shipped under the prefixed names above; the delivery README and PDF are not shipped.]

59.2 Manuscript 01: independent representations and completed suites

Manuscript 01's program performs several logically distinct finite checks rather than evaluating the same formula twice. The direct enumerator lists all nonnegative n -vectors with total at most n , computes every lower ideal from the input relation, and tests the original inequalities (48.1). It records the number of lattice points *for each exact support*. This part does not use Hall's condition, matching supports, or a gamma expansion.

A separate recursive matching checker tests bijections between selected source and receiver masks. It does not use the ideal inequalities or the Hall inequalities. Disjoint compatible pairs are accumulated to predict both the gamma coefficients and every multivariate coefficient

in (48.5). Independently, compatible pairs with overlapping sets are accumulated to check (51.4). The program also checks cancellation of overlaps, complement-duality, the maximum-matching degree law, and the deletion–differentiation identity using directly enumerated deleted preorders.

Finally, for fixed bipartite shores, the program enumerates integer demands by their defining subset inequalities and compares their exact support counts against matching-support counts. This directly tests the classical bridge and its support refinement on thousands of separate graphs.

Every reflexive relation on a fixed labeled n -element set is generated, and exactly the transitive ones are retained. Thus the following is an exhaustive test of *labeled preorders*, not merely of naturally labeled posets and not merely of random closures:

n	0	1	2	3	4	5
Preorders checked	1	1	4	29	355	6,942

All 7,332 cases passed. Additional completed checks comprise all 512 bipartite graphs on fixed shores of sizes (3,3), all 4,096 on shores (3,4), all 64 on shores (2,3), and the empty-shore boundary cases. A seeded generation of 120 preorder closures on six or seven elements produced 119 distinct relations, all of which passed. These latter cases are explicitly *not* exhaustive.

The block formula was checked on 87 cases: every preorder on three indexed blocks, with block-size vectors (1, 1, 1), (2, 1, 2), and (2, 2, 2). Allowing equivalent block indices is harmless for the capacity formula, although the canonical equivalence-class quotient in Theorem 54.3 is a poset. The five named examples below are also included in the saved output.

Preorder	Gamma vector	Lattice points	ord ₋₁ h
Three-element chain	(1, 3)	14	1
One equivalence class of size 3	(1, 6)	20	1
Diamond poset, size 4	(1, 5, 2)	38	0
Block chain (2, 1, 2)	(1, 12, 17)	162	1
Two lower blocks (2, 1) below size 3	(1, 17, 36, 7)	487	0

To reproduce the full finite run, manuscript 01 gave, from the extracted package directory, `python3 code/verify.py -max-n 5 -output results/verification.json`. In this directory pass an explicit output path (the default is relative to the working directory):

```
py code/04-gamma-vector-verify.py -max-n 5 \
  -output /path/to/scratch/verification.json
```

Python 3.10 or later is required. No external Python packages, network access, computer algebra system, or proof-assistant installation is used. The JSON records the interpreter version, random seed, case counts, examples, and elapsed times. The recorded times are properties of this run, not performance guarantees.

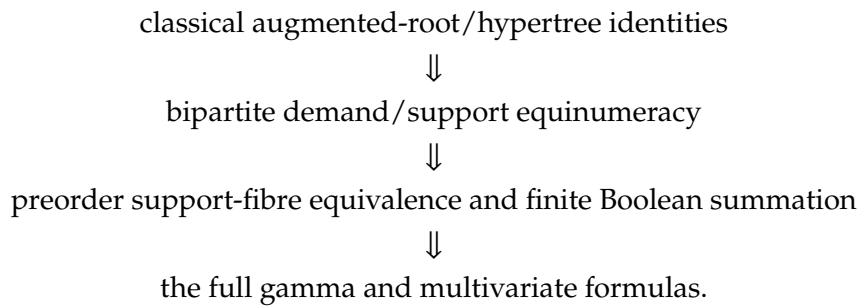
What the checks establish (manuscript 01). Exhaustive agreement detects normalization mistakes, reversed relations, incorrect treatment of equivalence classes, confusion between matchings and supports, and implementation errors in small cases. It does not prove an all-cardinality assertion. The infinite statements are justified by the arguments in Sections 49–58, with the classical inputs (14.3)–(14.4) explicitly imported.

59.3 Reruns and cross-checks in the write phase

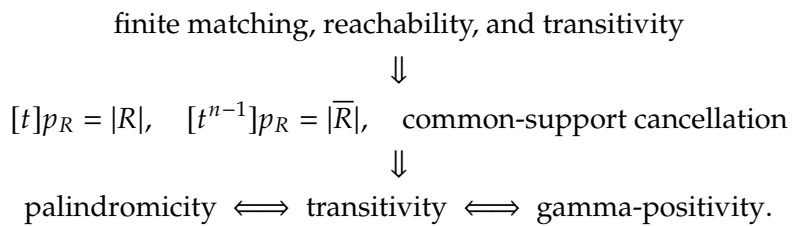
Both programs were rerun on 29 September 2026 (Python 3.14.4, Windows), from a scratch copy with the shipped names and explicit output paths as above. Both passed. Manuscript 06's regenerated CSV is byte-identical to `data/05-gamma-positivity-preorders.csv`, and both JSON records differ from the shipped ones only in timings, the Python version (the recorded runs used 3.13.5) and, for 06, the platform string. Independently, the batch-44 placement dossier compared, for all 390 labeled preorders with $n \leq 4$, a direct count of lattice points by support size with $\sum_k \gamma_k t^k (1+t)^{n-2k}$, and checked the endpoint formula and the transitivity criterion of Theorem 48.3 on all 4,165 reflexive relations with $1 \leq n \leq 4$ (Part I's reflexive example gives $(1, 5, 6, 1)$); there were no mismatches. The write phase recomputed the numerical examples of Sections 50–56 (Examples 50.2, 54.2 and 54.4, Table 4, the star with three edges and an octopus) by direct lattice-point and transport-pair enumeration, and checked the chain and Θ_3 identities by hand; all agree. These scratch checks are not shipped. None of the programs is a proof.

60 Proof dependencies, audit, and a formalization route

The mathematical dependency chain for the main theorem is short (manuscript 06):

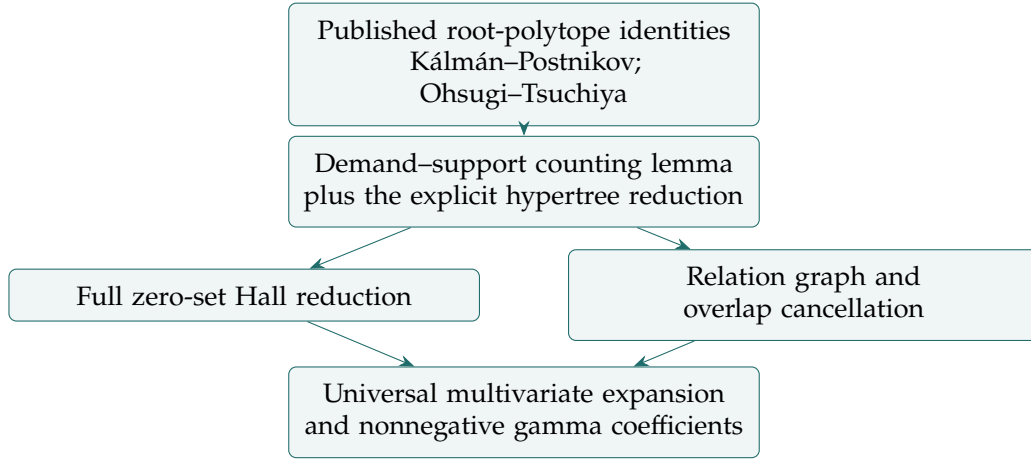


The other principal chain is independent:



A verification of the second chain would not by itself formalize the preorder-polytope theorem: the bridge from lattice points to matching supports must also be checked.

Manuscript 01 displays the central dependencies without any hidden appeal to this report's unrefereed results:



The two lower branches are different reductions but share the same counting bridge. The degree, deletion, monotonicity, and block laws then use only finite combinatorics and polynomial algebra.

60.1 Three easy ways to state a false variant (manuscript 01)

Using only covering relations. For the chain $1 < 2 < 3$, the relation $1 < 3$ must appear in the matching graph. Using just the two Hasse edges would give $\gamma_1 = 2$, whereas the correct value is 3 and $h = 1 + 6t + 6t^2 + t^3$. The lower ideals are unchanged by taking transitive closure, but the compatible-pair formula is not unchanged unless the closure is taken as well. The second proof of Corollary 51.2 explains why: paths must be compressible to edges of the relation.

Testing only principal ideals. For the poset $a < b$ and $a < c$, the vector $(x_a, x_b, x_c) = (0, 2, 2)$ satisfies the inequalities for the three principal ideals. It violates the inequality for the full ideal, because its total is four while the ground set has size three. The polytope in (48.1) uses *all* lower ideals.

Counting witnesses rather than supports. With two sources and two targets and every source below every target, there are two compatible bijections but only one ordered support pair with those specified sources and targets. That pair contributes one, not two. This distinction is the reason the gamma polynomial is not in general the ordinary matching generating polynomial.

60.2 Real-rootedness and flagness remain separate

The universal theorem does not imply real-rootedness. Part II’s height-two analysis already contains counterexamples, with the earlier small example credited to Shivam Patel [11]. To see the distinction in an exact instance (manuscript 01), let Θ_3 be the bipartite graph consisting of three internally disjoint paths of length three with common endpoints. Its height-two poset has gamma polynomial

$$p(u) = 1 + 9u + 24u^2 + 16u^3 + u^4,$$

as counted in Part II (the case $m = 3$ of (13.4)). The identity

$$p(-x) = (1 - x)(1 - 4x)^2 + x^4$$

shows that p is positive on $[-1, 0]$. Its nonnegative coefficients exclude nonnegative roots. If all its roots were real, all four would therefore be strictly below -1 , contradicting the fact that

their product is one. A nonreal zero u_0 of p lifts through $t = u_0(1+t)^2$ to nonreal zeros of $h_\tau(t) = (1+t)^8 p(t/(1+t)^2)$: the quotient is real for real $t \neq -1$, and $t = -1$ cannot solve this equation. Thus h_τ is not real-rooted. This example and its discovery are not claimed anew. [Merge note: this is a short second proof that P_{Θ_3} is not real-rooted; Part II's Theorem 13.3 counts its real roots exactly, and Proposition 15.1 is the lifting argument.]

Similarly, Boolean orbits of an auxiliary finite set are not a flag simplicial sphere. Theorem 52.1 supplies a nonnegative gamma expansion, not the face-incidence structure demanded by Conjecture 5.1(d). The root-polytope formula in (52.1) does not remove that distinction.

60.3 Suggested Lean development order (manuscript 06)

A useful first module would treat finite relations, disjoint pairs of finite sets, and relation-respecting bijections. It should prove the near-perfect support/reachability lemma and the coefficient defect before introducing any geometry. A second module would establish the ideal Hall criterion and common-support cancellation, then perform the finite-sum reindexing that gives the gamma expansion of p_τ .

A third module can express the demand/support identity as an explicit hypothesis and prove the support-fibre reduction and Theorem 48.2 conditional on that hypothesis. Such a theorem must be labeled conditional; it would not be a completed formal proof of the conjecture. Discharging the hypothesis requires formalizing the classical root-polytope route or developing a direct finite combinatorial proof. The quotient-block and deficiency formulas can then be proved by finite counting and elementary polynomial algebra.

The elementary graph classification is therefore a realistic independent formalization target, while the main theorem has one substantial library boundary. No Lean source or successful Lean compilation is claimed in this package. Likewise, the external mathematical theorems used in the paper are ordinary cited inputs, not assertions that they already exist in the repository's trusted kernel boundary.

60.4 A staged proof-assistant project (manuscript 01)

A practical formalization can separate the new order-theoretic work from the existing geometric input. First define finite preorders, lower ideals, and the support-fiber excess map. The equivalence in Lemma 50.1 is a finite-sum inequality argument and should be a small, independent target. Next prove the Boolean summation: this gives a formal implication from the explicit counting-bridge statement to the universal gamma theorem.

A second finite target is overlap cancellation. One may formalize the partial-permutation graph and its path decomposition, or iterate a bijection from a source until it reaches a target. Finiteness and injectivity exclude entry into a cycle. The resulting compatible core supports the commuting toggles and every restriction-counting identity.

To obtain a fully closed formal proof, the counting bridge must itself be proved rather than silently postulated. The supplied ordinary proof reduces it to Hall's theorem and the two published root-polytope results; a natural bijective proof of the bridge could yield a substantially smaller formal library dependency. A Lean theorem with an unproved bridge hypothesis is a valuable intermediate artifact, but is not a complete formal verification of Conjecture 5.2. No uncompiled Lean file or such conditional proof is presented as verified in this package.

[Merge note: neither plan has been started. No Lean or Rocq declaration in ProveIt formalizes any statement of Part IV.]

61 Further research questions and topics

The following questions are proposed research directions. Except where an existing conjecture is named, no claim is made that the formulation is new or that the literature has never addressed a related version (manuscript 06). Manuscript 01 adds: the flag question is the remaining named conjecture in the cited source; the others are proposed directions arising from the proof, whose worldwide open status has not been exhaustively surveyed, and they should be viewed as specific research tasks, not as assertions that the literature contains no related results. Research questions 20–29 are manuscript 06’s, in its order, with manuscript 01’s formulation added where it asks the same question; Research questions 30–36 are the rest of manuscript 01’s.

Research question 20 (A direct demand/support bijection; manuscripts 06 and 01). Construct an explicit bijection in Lemma 49.1 that is compatible with right-support restriction as in Proposition 49.3. A mere lexicographic matching of two equally sized lists is not the target. Can an augmenting-path or activity construction explain the correspondence without importing Ehrhart theory? This would remove the largest formalization dependency of the main theorem. (Manuscript 01: construct an explicit bijection from $\mathcal{R}_H$ to perfectly matchable support pairs that preserves the exact receiver support. Can it respect induced subgraphs or a controlled choice of vertex ordering?)

Proposition 49.3 proves the required cardinality identities, but the geometric proof does not identify a natural map. Such a construction would make the Boolean action on actual preorder lattice points explicit and would simplify a complete formalization (manuscript 01). [Merge note: this is Part II’s Research question 2, asked again with the receiver support preserved.]

Research question 21 (Boolean orbits on actual lattice points). Can one realize the terms in (48.4) as orbits of a natural system of involutions on $\mathcal{Q}_\tau \cap \mathbb{Z}^E$, with a k th orbit containing one point for every subset of $n - 2k$ free labels? Compatibility with relabeling and restriction is desirable. The present support-count identity does not supply such an action.

[Merge note: manuscript 01’s Theorem 52.1 realizes such orbits on the support-pair model $\mathcal{M}_\tau$, not on $\mathcal{Q}_\tau \cap \mathbb{Z}^E$; the question stays open as stated.]

Research question 22 (Flag-polytopal realization; manuscripts 06 and 01). Does every h_τ occur as the h -polynomial of an n -dimensional flag simplicial polytope, as asked in [1, Conjecture 5.1(d)]? The gamma formula gives nonnegative coefficients but no flag face complex. A useful intermediate objective is a simplicial model whose face numbers are the transport-support counts; it must account for the fact that deleting an arbitrary single signed endpoint need not preserve balance. (Manuscript 01: can the disjoint-support model be used to construct a flag simplicial n -polytope whose h -polynomial is h_τ ?)

The existing Boolean families provide ranks but no face-gluing rules. A successful construction must address intersections and sphere topology, not merely reproduce the gamma coefficients. It need not use the same canonical lattice triangulation considered in Part I (manuscript 01). [Merge note: compare Part II’s Research questions 6 and 7 and its crown obstruction, Theorem 13.6.]

Research question 23 (Classification beyond a perfect matching). Extend Corollary 53.2 to arbitrary bipartite graphs, including unequal shores and graphs whose maximum matching does not saturate either shore. Is there a useful decomposition of the palindromic cases into balanced preorder blocks and residual components? Palindromicity is to be measured in the actual polynomial degree. The endpoint argument here does not directly apply when that degree is smaller than a shore size.

Research question 24 (Coefficient shape beyond gamma-positivity). Which preorders have log-concave, unimodal, or real-rooted gamma polynomials? The general theorem answers none of these stronger questions for Γ_τ . For real-rootedness, the already documented height-two counterexamples rule out an unrestricted assertion. Relations to quotient width and induced comparability configurations are concrete targets.

Research question 25 (Extremal profiles with prescribed structure). Determine the maximal and minimal γ_k when block sizes, quotient width, or matching number are fixed. Theorem 55.6 gives sharp unrestricted bounds, but fixing these structural parameters may make the extremizers quite different from chains or universal equivalence classes. The positive block formula offers a way to separate local multiplicities from quotient feasibility.

Research question 26 (Parameterized exact algorithms). Can (51.3) be evaluated efficiently for bounded-width quotient posets, or when the *comparability graph* has bounded treewidth? A cover-graph parameter is a separate question because transitive closure can enlarge neighborhoods and graph width. Establish precise arithmetic and bit-complexity bounds, and determine the counting complexity without parameter restrictions.

Research question 27 (A geometric interpretation of the gamma coefficients). The signed-vector formula counts balanced points of the cube $\{-1, 0, 1\}^E$ satisfying ideal inequalities. Can these points be realized as a natural collection of faces, lattice points in a canonical section, or standard monomials of a graded algebra? Such an interpretation might connect positivity to Part I's root-polytope model, but would require more than the enumerative identity established here.

Research question 28 (Limit laws for support size). For growing families of preorders, what conditions on Γ_τ or the quotient structure imply a central limit theorem for L ? The mixture in Theorem 58.1 reduces the question to the fluctuations of its mixing variable and the conditional binomial size. Can the variance and tail bounds be sharpened using comparability density or matching structure, rather than n alone?

Research question 29 (Independent formal and mathematical review; manuscripts 06 and 01). Formalize first the elementary endpoint-defect and cancellation theorems, and independently review the support-fibre equivalence. Then discharge the bipartite counting bridge in a proof assistant. In parallel, compare the balanced-graph characterization with existing Gorenstein and matching-decomposition criteria to determine which parts are already implicit in the literature and sharpen the priority assessment. (Manuscript 01: formalize the universal theorem with no unproved assumptions, and develop an independent proof of the counting bridge that avoids the root-polytope machinery.)

The finite-order part is short enough to audit separately. Closing the bridge dependency, while retaining the multivariate support interpretation, would turn the current mathematical proof into a compact and reusable formal combinatorial library result (manuscript 01).

Research question 30 (Which gamma vectors are face vectors?; manuscript 01). Do the numbers $\gamma_k(\tau)$ arise as the face numbers of a useful simplicial or balanced complex, with a construction natural under induced restriction?

Simply treating compatible pairs as faces is inadequate: removing one source without a receiver destroys equal cardinality. The locality formula (57.1) suggests studying endpoint-supported structures or quotients of matching complexes with carefully controlled identifications.

Research question 31 (The even-cardinality reconstruction defect; manuscript 01). What additional deletion, marking, or contraction data determines the top gamma coefficient when n is even? Can one reconstruct it from a suitably enriched polynomial deck without reconstructing the entire preorder?

Equation (57.2) reconstructs every other coefficient but annihilates the middle monomial. Corollary 56.2 identifies the missing scalar as a signed value at -1 , counted by full endpoint cores. This gives a precise target for richer reconstruction data.

Research question 32 (Exact counting complexity; manuscript 01). For which classes of quotient posets can the profile formula be evaluated more efficiently than $O(n^{2b}2^b)$? What is the complexity of computing a specified γ_k , or the whole gamma polynomial, for unrestricted preorders?

A single pair's feasibility is a matching problem; counting distinct pairs is a different task. The fixed-block algorithm gives one tractable regime, but does not supply a general efficient counting algorithm or a hardness classification. Tree decompositions of the quotient and repeated neighborhood classes are natural structural parameters.

Research question 33 (Local changes of the relation; manuscript 01). Find effective formulas for the gamma increment when one adds a relation and then takes transitive closure. Can the newly compatible cores be partitioned without overcounting multiple new witnessing matchings?

Theorem 55.1 already proves nonnegativity of every increment. An exact local rule would be stronger and could lead to constructive recurrences, particularly for modular or series-parallel quotient operations. The closure operation must be included in the input model.

Research question 34 (Root geometry and negative dependence; manuscript 01). Identify structural classes of preorders for which Γ_τ is real-rooted or admits a useful multivariate stability refinement. Which operations preserve these properties in the presence of nontrivial blocks?

General positivity is now separated from these stronger analytic properties. The arbitrary height-two bipartite subclass already contains nonreal-root examples, so bounded height alone is insufficient. Part III's cactus results offer positive comparison cases, not a classification of all preorders. [Merge note: compare Part II's Research question 3 and Part III's Research question 11.]

Research question 35 (Refining the lattice-point statistic; manuscript 01). What counterpart of the Boolean expansion survives if one records both support and coordinate sum, or weights the individual excess coordinates?

The present proof deliberately forgets the total demand in the classical bridge. A two-parameter identity would require extra information about its hypertree or matching-support correspondence. It cannot be obtained by inserting arbitrary weights into an equality of unweighted cardinalities.

Research question 36 (Beyond unit ideal budgets; manuscript 01). Which integral budget functions $b(I)$ admit an analogue for $\{x \geq 0 : x(I) \leq b(I)\}$? What replaces the centered gamma basis when the all-one vector is no longer the unique full-support lattice point?

The excess substitution changes the right-hand side to $b(I) - |I \cap S|$. The special cancellation into the number of zero coordinates is lost. Any extension must identify new supply capacities or a different symmetry center, rather than assuming the same formula persists.

62 Conclusion

The main formula treats the gamma coefficients of a preorder not as mysterious alternating combinations of lattice-point counts, but as counts of disjoint feasible transport supports.

Fixing a lattice point's zeros exposes a bipartite demand problem; summing those problems exposes Boolean intervals. This proves gamma-positivity for every finite preorder, including all equivalence-class blowups and all heights.

The matching viewpoint also gives a structural converse. In a balanced bipartite graph with a perfect matching, a palindromic support polynomial forces the relation obtained from that matching to be transitive. The failure is measured exactly by missing reachability pairs. Positive block formulas, coefficientwise monotonicity, sharp bounds, and the exact root multiplicity at -1 then follow from the same finite support model.

The unresolved step toward a more conceptual and more readily formalized proof is not another positivity inequality. It is a natural combinatorial realization of the inherited demand/support equinumeracy. The proofs and checks supplied here isolate that issue while giving an explicit answer to the stated gamma-positivity conjecture.

Manuscript 01 concludes in the same sense. The general preorder problem admits a uniform solution once zero coordinates are allowed on the entire ground set. The ideal inequalities then become a bipartite supply–demand system, and the classical counting bridge turns each support fiber into a family of matchable pairs. Summing over the free coordinates produces the gamma basis with no cancellation of signs. The resulting coefficients count ordered disjoint compatible source and receiver sets. This proves the full gamma-positivity conjecture, explains why arbitrary height and arbitrary equivalence blocks cause no exception, and supplies an exact matching-theoretic meaning for the vanishing at -1 . The same finite model gives relation-monotonicity, complement-duality, induced-deck identities, and a block-capacity formula. The remaining work is not to infer positivity from more examples, but to understand natural bijections, geometric realizations, stronger root properties, and complete formal verification of the classical bridge and its new consequences.

63 Normalization and independent-review checklist

(Appendix A of manuscript 06.)

63.1 The three different cardinalities

A perfect matching is a set of edges. A matchable support pair (U, V) is a pair of vertex sets admitting at least one perfect matching. A hypertree is a degree-minus-one vector admitting at least one realizing spanning tree. The proofs count the latter two without realization multiplicity. The $K_{2,2}$ example in Remark 49.2 is a minimal check against conflating these notions.

63.2 Three polynomial models, with different dimensions

The original preorder polytope Q_τ lies in $\mathbb{R}^n$, and h_τ counts its points by support. The doubled graph G_τ has $2n$ vertices and p_{G_τ} has degree n . The augmented edge polytope $B_{\widehat{G_\tau}}$ has dimension $2n$ and its Ehrhart numerator is the same polynomial, with higher coefficients zero. These dimension statements are consistent because the degree of an Ehrhart numerator can be strictly smaller than the polytope dimension.

For a general graph H with shores of sizes p, q , its augmented graph has $p + q + 2$ vertices, is connected and bipartite, and its edge polytope has dimension $p + q$. The two affine equations say that the coordinate sum on each augmented shore is one. The natural affine lattice is the intersection of that affine space with the ambient integer lattice. No factor of a lattice index is inserted in (14.3).

63.3 Orientation and equality cases

A relation edge is always directed from a source in the lower direction to a target in the upper direction. Hence a matching must satisfy $|A \cap I| \geq |B \cap I|$ for ideals I , not the reverse inequality. In the near-perfect-support lemma, omitting a on the source side and b on the target side detects a path from b to a . The count $|\bar{R}|$ is invariant under swapping their dummy names, but the proof is not.

The signed-vector formula uses signs on individual elements, not just blocks. Once block differences are fixed, the multinomial coefficients in (54.2) recover precisely the lost labeling information. The formula would be wrong without these factors.

63.4 A short audit path

An independent reviewer can check the main theorem by verifying the two implications of Lemma 50.1, the normalization in the established input Lemma 49.1, and the finite sum in (50.3). The second principal theorem can be checked separately from the directed-path construction in Lemma 53.1 and the cancellation counterexample in Proposition 51.4. None of those steps depends on the finite verification data.

64 A compact proof certificate

(Appendix A of manuscript 01.) This section records the complete logical skeleton for a reader checking the generalization rather than its consequences.

Let $S \subseteq E$, $Z = E \setminus S$, and $H_{Z,S}$ have edge zs when $z \leq_\tau s$. Write $x = \mathbf{1}_S + c$ on S , with zero coordinates on Z . For $T \subseteq S$,

$$x \in \mathcal{Q}_\tau \implies c(T) \leq c(S \cap \downarrow T) \leq |Z \cap \downarrow T| = |N_{H_{Z,S}}(T)|.$$

Conversely, for a lower ideal I ,

$$c(I \cap S) \leq |Z \cap \downarrow(I \cap S)| \leq |Z \cap I| \implies x(I) \leq |I|.$$

Thus the exact-support fiber has size $|\mathcal{R}_{H_{Z,S}}|$. The classical bridge says that this is

$$\#\{(U, V) : U \subseteq Z, V \subseteq S, U \rightsquigarrow_\tau V\}.$$

For any fixed disjoint compatible (U, V) , its allowed supports are exactly $S = V \sqcup C$, $C \subseteq E \setminus (U \cup V)$. Hence

$$H_\tau(\mathbf{z}) = \sum_{(U,V)} \sum_C z_{V \cup C} = \sum_{(U,V)} z_V \prod_{w \notin U \cup V} (1 + z_w).$$

This proves the theorem. The only imported count is justified by the explicit demand-to-hypertree-vector correspondence and the published identities (14.3)–(14.4). No statement about preorder gamma-positivity is used in those inputs.

65 Provenance and source-use ledgers

(Appendix B of manuscript 06 and Appendix B of manuscript 01.)

Source	Role in manuscript 06
Athanasiadis–Chapoton [1]	Definitions and the precise target, Conjecture 5.2; flag realization is not settled here.
Kálmán–Postnikov [3]	Root-polytope volume equals the number of hypertree vectors, with ordinary lattice normalization.
Ohsugi–Tsuchiya [8]; Davis–Kohl [7]	The matching-support interpretation of the augmented root-polytope numerator.
Dai et al. [2]	Prior support-enumerator identity and preorder duality; Problem 5.3 motivates the balanced-graph classification.
Menon [6]	A prior special-family gamma-positivity result.
ProveIt (Parts I–II)	The height-two theorem, inherited counting lemma, explicit unresolved boundary, and existing nonreal-root examples.
Manuscript 06	The arbitrary-preorder fibre reduction, general gamma formula, graph classification proof, and the stated derived refinements.

The ledger is a dependency and scope statement, not a certification that every derived refinement is unprecedented. The root-polytope identities are standard mathematical inputs; this report’s unrefereed geometric results are not used as axioms. The delivered proof-status file (`05-gamma-positivity-PROOF-STATUS.md`) repeats this boundary so that computational output is not mistaken for kernel checking.

Source	Use and boundary in manuscript 01
Athanasiadis–Chapoton, v1	Definition of h_τ and precise target Conjecture 5.2; flag realization is kept separate.
ProveIt, pinned commit	Identifies the height-two predecessor and its explicit extension question. Not used as an unproved axiom.
Kálmán–Postnikov	Published normalized-volume/hypertree consequence, with its original h -polynomial convention evaluated at one.
Ohsugi–Tsuchiya, v4	Published augmented-graph matching-support identity; counts covered vertex sets rather than individual matchings.
Davis–Kohl, v1	Theorem 3.10 states the precise augmentation and Ehrhart identity used in the bridge.
Dai et al., v2	Prior univariate duality and root-polytope context; no new claim of having first proved duality.
Menon, v1	Prior octopus and lopsided-octopus results; the special formula is recovered and credited.

All cited online versions were inspected for manuscript 01 on 29 September 2026. The citation labels and conjecture numbers refer to these versions, not to an assumed future revision. The supplied package does not redistribute the source papers or any font files. [Merge note: manuscript 01 cites Davis–Kohl as [arXiv:2207.14759v1](https://arxiv.org/abs/2207.14759v1), and manuscript 06 cites the inspected versions v1 of [1] and [6], v4 of [8] and v2 of [2], as this report’s bibliography does.]

A A hypertree proof of the known root identity

This appendix reconstructs Proposition 2.2 without using the support-enumerator theorem of [2]. It is included for conceptual clarity and as an independent route to the same normalization. Its one substantial imported result is the Kálmán–Postnikov theorem [3].

A.1 Hypertrees and internal inactivity

For a connected bipartite graph with left shore L , a left hypertree is a vector $a \in \mathbb{Z}_{\geq 0}^L$ realized by a spanning tree T through $a_i = \deg_T(i_L) - 1$. Distinct trees realizing the same vector contribute only one hypertree. Order the left shore. A coordinate i is internally inactive when there is a smaller coordinate j such that replacing a by $a - \mathbf{e}_i + \mathbf{e}_j$ gives another hypertree. The interior polynomial is the sum of $t^{\text{number of internally inactive coordinates}}$ over hypertrees.

The Kálmán–Postnikov theorem identifies this interior polynomial with the Ehrhart numerator of the bipartite root polytope. Their paper uses a reversed convention for the triangulation h -polynomial in its Theorem 1.1; the Ehrhart-basis formulation immediately following that theorem, equation (1.2), identifies the coefficients with the usual h^* -numerator used here. Thus there is no additional reversal of t in the statement we apply.

Lemma A.1. *The left hypertrees of $\widehat{G}_\tau$ are exactly*

$$a = (n - x(E), x_1, \dots, x_n), \quad x \in P_\tau. \quad (\text{A.1})$$

Proof. A spanning tree of $\widehat{G}_\tau$ has $2n + 1$ edges, so the sum of its left degree-minus-one vector is $(2n + 1) - (n + 1) = n$. For a nonempty $A \subseteq [n]$, the tree edges incident with A_L form a forest on $A_L \cup N(A)$. Their number is $a(A) + |A|$, and a forest on this nonempty vertex set has at most $|A| + |N(A)| - 1$ edges. Hence

$$a(A) \leq |N(A)| - 1 = |\downarrow A|.$$

Lemma 2.1 now shows that $x = (a_1, \dots, a_n)$ belongs to P_τ , and the sum condition determines a_0 .

Conversely, let $x \in P_\tau$. Replace the left element i by x_i distinct clones, each adjacent to the right set $\downarrow i \subseteq [n]$. A set of clones whose labels lie in A has at most $x(A) \leq |\downarrow A|$ members. Thus Hall's matching condition holds, and all clones can be matched to distinct right vertices. This is also a constructive statement: if an augmenting-path matching leaves a clone unmatched, the alternating paths reachable from it produce a set of clones with fewer neighbors, contradicting Hall's condition.

Build a tree by first taking all edges $i_L 0_R$ for $0 \leq i \leq n$. Each right vertex j_R , $j > 0$, is then attached to the left label of its matched clone, or to 0_L if it is unmatched. The first edges form a star; every subsequent edge attaches a new leaf. Hence the result is a spanning tree. Its left degrees are $x_i + 1$ for $i > 0$ and $n - x(E) + 1$ for $i = 0$, exactly as required. $\square$

Lemma A.2. *Order the left vertices with 0 first. For the hypertree in (A.1), internal inactivity is $|\text{supp}(x)|$.*

Proof. The first coordinate is never internally inactive. A zero coordinate cannot transfer a unit, so it is also not internally inactive. If $x_i > 0$, then $x - \mathbf{e}_i \in P_\tau$ because all the defining inequalities are upper bounds and the coordinates remain nonnegative. By Lemma A.1, the vector obtained by transferring one unit from coordinate i to coordinate 0 is another hypertree. Thus every positive nonzero-index coordinate is internally inactive, and there are no others. $\square$

Summing over (A.1) gives an interior polynomial equal to $h_\tau(t)$. The Kálmán–Postnikov theorem identifies it with $h^*(B_\tau, t)$, proving Proposition 2.2 by this second route. Notice that the argument counts hypertree *vectors*, not spanning trees; counting all realizing trees would introduce incorrect multiplicities.

B Proof dependencies and normalization audit

Claim or input	Where it comes from
$h^*(B_\tau) = h_\tau$	Known: [2, Theorem 1.2], specialized using Lemma 2.1. Alternatively Appendix A plus [3].
Ideal Hall presentation	Proposition 3.2; transitivity is used explicitly.
Integer fiber factorization	Theorem 4.1; all coordinates and inverse parameters are displayed.
$h^*(C_\tau) = h_\tau$	Corollary 4.2; a consequence of the fiber factorization and the known root identity.
Unimodularity	Leaf elimination in Lemma 5.1, followed by Proposition 5.3.
Polytopal pulling boundary	The established pulling lemma [4, Lemma 2.1].
Palindromicity, unimodality	Dehn–Sommerville and the g -theorem applied to the constructed simplicial polytope; an independent reciprocity proof of palindromicity is in Section 6.
Alternative polytopal route	The explicit special simplex and [4, Theorem 3.5].

There are several easy places to make an otherwise plausible argument wrong. The support polynomial of Q_τ is not asserted to be $h^*(Q_\tau)$. It is the numerator of the different polytopes B_τ and C_τ . The larger denominator has exponent $2n + 1$ and the smaller one exponent $n + 1$. Both auxiliary vertices, and the edge between them, are required. The edge orientation convention is $j \leq_\tau i \iff i_L j_R$ is present. The integer-fiber parameters consist of $n + 1$ nonnegative entries; the exponent n in (4.9) results only after dividing by the additional slack factor in the Ehrhart series of C_τ .

No argument substitutes a rational equivalence for an integral one. The fiber base is integral because $a_\tau(y)$ is an integer for integral y . The determinant statements refer to the full relevant lattices. Finally, regularity or shellability alone is not used to claim that a sphere is polytopal: the specific pulling-boundary theorem supplies that step.

C Reproducing the accompanying computations

The archive contains `article.tex`, `article.pdf`, a build script, `code/verify.py`, the full CSV enumeration, a JSON result report, and status notes. From the extracted directory, run

```
python3 code/verify.py --output data
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verifier requires Python 3.10 or later and no third-party Python packages. Do not use Python’s `-O` option, since assertions are part of the checks. The LaTeX document uses standard TeX Live packages, including `newpxtext`, `newpxmath`, `tcolorbox`, and `hyperref`.

The CSV uses zero-based bit positions: the row value 5 represents the set $\{0, 2\}$. Row i records the principal ideal of element i , not its filter. A polynomial is stored as coefficients from

degree zero upward. The JSON report also records explicit edge sets for matching certificates and complete Ehrhart counts for the independently tested instances.

[Added 28 September 2026, batch 39: this appendix describes Part I’s package; Part II’s programs and how to rerun them are in Section 22.4.]

The calculations in the supplied report completed without failed assertions. Their runtime is recorded separately as an observation from this environment, not as a performance guarantee. No claim of formal verification is made.

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